

A self-study textbook · after EECS 598 “Prediction, Learning, and Games”, University of Michigan, Fall 2013

Regret and Rebalancing

Online learning from first principles, with applications to portfolios, option pricing and trading. Sixteen chapters, worked proofs, finance sidebars, and exercises with solutions.

$$\underbrace{\sum_{t=1}^T \ell_t(x_t)}_{\text{what you paid}} - \underbrace{\min_{x \in \mathcal{K}} \sum_{t=1}^T \ell_t(x)}_{\text{best fixed choice in hindsight}} = o(T) \quad \text{on every sequence.}$$

Source course

Jacob Abernethy, EECS 598-006,
24 lectures

Companion texts

Cesa-Bianchi & Lugosi (2006);
Shafer & Vovk (2001)

Prerequisites

Probability, calculus, basic
convex analysis

Online edition with interactive demos: regret-and-rebalancing.html

Contents

How to use this book	vi
I Prediction with Expert Advice	1
1 The online learning game	2
1.1 Why study learning as a game?	2
1.2 Prediction with expert advice	3
1.3 The realizable case: Halving	3
Exercises	5
2 Weighted Majority	6
2.1 Convexity in one page	6
2.2 The Weighted Majority Algorithm	7
2.3 Tuning a learning rate	8
Exercises	8
3 Exponential weights and Hedge	10
3.1 From mistakes to losses	10
3.2 The Exponential Weights Algorithm	10
3.3 The softmin potential	11
3.4 The action setting (Hedge)	12
3.5 Priors over experts	13
Exercises	14
4 Tracking a moving target	16
4.1 When the best expert changes	16
4.2 A tempting idea that fails	16
4.3 A Markov prior	17
4.4 Fixed Share: the efficient implementation	17
Exercises	19
5 Lower bounds	20
5.1 Minimax regret	20
5.2 Two experts	20
5.3 N experts	21

5.4 Regret as a game	21
Exercises	22
II Games, Equilibria and Their Uses	23
6 Zero-sum games and minimax	24
6.1 Games in normal form	24
6.2 Nash's theorem	25
6.3 Von Neumann's minimax theorem	25
6.4 Minimax and linear programming	26
Exercises	27
7 Linear programs and boosting	29
7.1 Solving linear programs with Hedge	29
7.2 Boosting	30
Exercises	32
8 The Perceptron	33
8.1 Online linear classification	33
8.2 The Perceptron as gradient descent	34
8.3 Feasibility via the Perceptron	34
Exercises	35
III Online Convex Optimization	36
9 Online gradient descent	37
9.1 The protocol	37
9.2 Follow the Leader fails again	38
9.3 Online gradient descent	38
9.4 Curvature helps: logarithmic regret	39
9.5 From regret to optimisation and to statistics	39
Exercises	40
10 Follow the Regularized Leader	42
10.1 Be the Leader	42
10.2 Adding a regularizer	42
10.3 Bregman divergences	44
10.4 Follow the Perturbed Leader	44
Exercises	45

IV Finance	47
11 Universal portfolios	48
11.1 Betting on horses: why log-wealth	48
11.2 Constant-rebalanced portfolios	48
11.3 Cover's universal portfolio	49
11.4 Faster algorithms	50
Exercises	52
12 Probability as a game, options	53
12.1 Pricing by replication	53
12.2 Game-theoretic probability	54
12.3 A strong law of large numbers by betting	55
12.4 Regret bounds are option-price bounds	56
Exercises	57
V Bandits	58
13 Stochastic bandits	59
13.1 Bandit feedback	59
13.2 Explore then commit	60
13.3 Upper confidence bounds	60
Exercises	62
14 Adversarial bandits	63
14.1 Estimating what you did not see	63
14.2 A first attempt: separate exploration	63
14.3 EXP3	64
14.4 Bandits on combinatorial and convex sets	65
Exercises	66
VI Approachability, Calibration and Equilibrium	68
15 Blackwell approachability	69
15.1 Vector payoffs	69
15.2 Halfspaces are enough	69
15.3 The theorem and Blackwell's strategy	70
15.4 No-regret learning is approachability	71
15.5 Approachability from OCO	71

Exercises	72
16 Calibration and correlated equilibria	74
16.1 Calibrated forecasts	74
16.2 Calibration implies equilibrium	75
16.3 Correlated equilibria and internal regret	76
16.4 The map of the course	77
Exercises	78
VII Toolkit, Reading and Projects	79
A Mathematical toolkit	80
A.1 Hoeffding's lemma and inequality	80
A.2 Norms, duality and strong convexity	80
A.3 Entropy and KL divergence	81
A.4 Regret bounds at a glance	81
B Reading and projects	82
B.1 Books	82
B.2 Key papers by chapter	82
B.3 Project ideas with a finance angle	83
B.4 About the source course	84

How to use this book

Each chapter follows one or more lectures from the course and links to the original scribe notes. Chapters open with what you will learn and close with key takeaways and exercises; solutions are folded under each exercise so you can try first. Boxes are colour-coded: **blue** for theorems and lemmas, **green** for definitions and protocols, **amber** for finance applications, **red** for warnings and pitfalls. Exercise tags mark difficulty: **warm-up** **core** **challenge** **finance**.

Suggested paths. For the core theory, read Chapters 1–3, 5, 6, 9 and 10. For finance, add Chapters 4, 11 and 12, then 13–14 for trading decisions under partial feedback. Part VI (Chapters 15–16) ties everything together and is the most abstract.

Part I

Prediction with Expert Advice

*How to learn from a crowd of forecasters when nobody promises
the future looks like the past.*

CHAPTER 1

The Online Learning Game

Based on Lecture 1 (Sept 4, 2013) · [scribe notes](#)

In this chapter

- The protocol of sequential prediction, and why we analyse it against an adversary.
- Regret: the yardstick that makes adversarial analysis meaningful.
- The Halving algorithm and its $\log_2 N$ mistake bound.

1.1 Why study learning as a game?

Classical statistics assumes data are drawn i.i.d. from a fixed distribution, then asks how well a model fitted on the past generalises to the future. Financial markets, spam filters, ad auctions and opponents in a game all violate that assumption: the data react to you, drift over time, or are chosen by someone with their own objective.

Online learning drops the probabilistic assumption entirely. Data arrive one at a time; after each, the learner must commit to a decision, then sees the outcome and pays a loss. The outcomes can be arbitrary, even chosen by an adversary who knows our algorithm. It sounds hopeless to promise anything in this setting, and yet the central discovery of the field (Hannan, 1957; Blackwell, 1956; Littlestone and Warmuth, 1989) is that we can guarantee to perform nearly as well as the best of a fixed set of strategies, *in hindsight*, on every sequence.

Year	Who	Idea
1956	Blackwell	Approachability: a vector-valued minimax theorem
1957	Hannan	Sequential decisions with vanishing regret, no stochastic assumptions
1957	Rosenblatt	The Perceptron
1989–94	Littlestone, Warmuth; Vovk	Weighted Majority; aggregating algorithms
1991	Cover	Universal portfolios
1997	Freund, Schapire	Hedge and AdaBoost
2003	Zinkevich	Online convex optimization
2005	Kalai, Vempala	Follow the Perturbed Leader

The same machinery turns up across this book: computing equilibria of zero-sum games (Chapter 6), boosting weak classifiers (Chapter 7), stochastic gradient descent (Chapter 9),

portfolio selection and option pricing (Chapters 11 and 12), bandits (Chapters 13 and 14), and calibrated forecasting (Chapter 16).

1.2 Prediction with expert advice

The simplest version: every morning N pundits predict whether the market will close up (1) or down (0). We want to aggregate them.

Protocol 1.1 (binary prediction with experts)

For rounds $t = 1, 2, \dots, T$:

1. Each expert $i \in [N]$ announces advice $f_{i,t} \in \{0, 1\}$.
2. The learner (the “master algorithm”, MA) predicts $\hat{y}_t \in \{0, 1\}$.
3. Nature reveals the outcome $y_t \in \{0, 1\}$.

The learner’s performance is its number of mistakes, $M_T(\text{MA}) = \sum_{t=1}^T \mathbf{1}[\hat{y}_t \neq y_t]$. Expert i ’s mistakes $M_T(i)$ are defined the same way.

Nature is allowed to be adversarial: y_t may depend on everything that happened before, including the learner’s algorithm. We therefore cannot hope to bound $M_T(\text{MA})$ in absolute terms. An adversary can make any deterministic learner wrong on every round by setting $y_t = 1 - \hat{y}_t$. What we can do is compare to the experts.

Definition 1.2 (regret)

Given losses $\ell_t(\cdot)$, the regret of the learner with respect to a class of comparators $\mathcal{C}$ is

$$\text{Regret}_T = \sum_{t=1}^T \ell_t(\text{learner}) - \min_{c \in \mathcal{C}} \sum_{t=1}^T \ell_t(c).$$

An algorithm is **no-regret** (or Hannan consistent) if $\text{Regret}_T = o(T)$ for every sequence, so that its average loss per round is asymptotically no worse than that of the best comparator in hindsight.

Regret is the idea that makes the adversarial setting workable. If every expert is terrible, the adversary has made the problem hard and we are not blamed. If some expert is good, we must eventually do nearly as well as it, without knowing in advance which one it is.

1.3 The realizable case: Halving

Start with a strong assumption: one expert is perfect, $M_T(i^*) = 0$. Then any expert that errs even once can be discarded forever.

Algorithm 1.3 (Halving)

```

C_1 ← {1, ..., N}                                # experts consistent so far
for t = 1, ..., T:
    receive advice f_{i,t} for i ∈ C_t

```

```

predict  $\hat{y}_t \leftarrow$  majority vote of  $C_t$     (ties  $\rightarrow 1$ )
observe  $y_t$ 
 $C_{t+1} \leftarrow \{ i \in C_t : f_{\{i,t\}} = y_t \}$ 

```

Theorem 1.4 (Halving mistake bound)

If at least one expert is perfect, Halving makes at most $\log_2 N$ mistakes, on any sequence.

Proof. We track the size of the version space C_t . Initially $|C_1| = N$. The perfect expert is never removed, so $|C_t| \geq 1$ for all t . Whenever Halving errs, the majority of C_t predicted wrongly and is removed, so $|C_{t+1}| \leq \frac{1}{2}|C_t|$. On other rounds C can only shrink. After M mistakes,

$$1 \leq |C_{T+1}| \leq N 2^{-M},$$

so $2^M \leq N$, i.e. $M \leq \log_2 N$. □

Tightness. Take $N = 2^n$ experts indexed by binary strings, with expert i predicting bit t of i on round $t \leq n$. Each round the survivors split evenly, and the adversary picks the outcome opposite to Halving's prediction. Halving errs on each of the first $n = \log_2 N$ rounds, and exactly one expert (the one matching the adversary's choices) is perfect. No deterministic algorithm can do better on this family, so $\log_2 N$ is optimal.

The proof uses a pattern that recurs throughout the book. We choose a *potential* (here $|C_t|$) that is bounded below because of a good comparator, and that drops by a constant factor whenever the learner suffers. The learner's loss is then bounded by how far the potential can fall.

Example 1.5 (predictive sorting)

K teams have an unknown true ranking $\pi^* \in S_K$. Each round two teams (i_t, j_t) play and we predict the winner; the better-ranked team always wins. Treat each of the $K!$ permutations as an expert. Halving guarantees at most $\log_2 K! = O(K \log K)$ mistakes, the same count as comparisons needed to sort.

The catch is computation. Halving must compare the number of permutations consistent with the observed games that rank i_t above j_t against those that rank it below. That count is the number of linear extensions of a partial order, which is #P-hard to compute (Brightwell and Winkler, 1991). The course posed as a challenge whether a polynomial-time algorithm can achieve the same mistake bound. It is a first example of a tension that runs through the field: the statistically optimal aggregation over an exponentially large expert class may be computationally intractable, unless the class has structure we can exploit (see Exercise 1.4 and Chapter 4).

In finance

Think of the experts as trading signals: a momentum rule, a mean-reversion rule, an analyst's recommendation, and so on. The assumption that one signal is perfect is

absurd for markets, which is why Halving is only a warm-up. The regret framework keeps the right question, though: *can I, with no model of how returns are generated, guarantee performance close to the best signal in my library over the period?* Chapters 2 to 4 answer this for noisy signals, and Chapter 11 answers it for portfolios.

Key takeaways

- Online learning makes no distributional assumption. Performance is measured by regret against a comparator class on every individual sequence.
- If one expert is perfect, majority vote over the consistent experts makes at most $\log_2 N$ mistakes, and this is optimal.
- Potential-function arguments are the main tool.

Exercises

1.1 warm-up Suppose at least K of the N experts are perfect. Show Halving makes at most $\log_2(N/K)$ mistakes.

1.2 warm-up Show that for every deterministic learner there is a sequence on which it makes T mistakes while the best of two experts (one always predicting 0, one always 1) makes at most $T/2$.

1.3 core (Randomized Halving.) Instead of majority vote, predict with a uniformly random member of C_t . Show the expected number of mistakes is at most $\ln N$ (natural log). Hint: if a fraction r_t of C_t is wrong, the expected mistake is r_t and $|C_{t+1}| = (1 - r_t)|C_t|$.

1.4 challenge (From the course.) Is there an algorithm for predictive sorting with $O(K \log K)$ mistakes and running time polynomial in K and T ? Hint: you do not have to run Halving exactly. Think about sampling linear extensions approximately (Karzanov and Khachiyan's Markov chain), or about any rule that guarantees each mistake removes a constant fraction of the consistent permutations.

Solutions

1.1. The K perfect experts are never removed, so $|C_{T+1}| \geq K$. Combined with $|C_{T+1}| \leq N2^{-M}$ this gives $2^M \leq N/K$.

1.2. Let the adversary set $y_t = 1 - \hat{y}_t$, which is possible because $\hat{y}_t$ is a deterministic function of the past. The learner errs every round. The two constant experts together err exactly T times, so one of them errs at most $T/2$ times. This is why later chapters randomize or allow fractional predictions.

1.3. $1 \leq |C_{T+1}| = N \prod_t (1 - r_t) \leq N \exp(-\sum_t r_t)$ using $1 - x \leq e^{-x}$. So $\mathbb{E}[M] = \sum_t r_t \leq \ln N$. Since $\ln N = 0.69 \log_2 N$, randomization improves the constant.

CHAPTER 2

Weighted Majority and a Convexity Toolkit

Based on Lecture 2 (Sept 9, 2013) · [scribe notes](#)

In this chapter

- The handful of inequalities used in almost every proof in this book.
- The Weighted Majority Algorithm, which drops the perfect-expert assumption.
- How to tune a learning rate by balancing two terms.

2.1 Convexity in one page

Definition 2.1 (convex function)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if for all x, y and $\alpha \in [0, 1]$,

$$f((1 - \alpha)x + \alpha y) \leq (1 - \alpha)f(x) + \alpha f(y).$$

Equivalently (Jensen's inequality), $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ for every random vector X with finite mean. For continuous f it suffices to check $\alpha = \frac{1}{2}$.

Proposition 2.2 (first- and second-order tests)

1. If f is twice differentiable and $\nabla^2 f(x) \succeq 0$ everywhere, then f is convex.
2. If f is differentiable, f is convex iff for all x, δ : $f(x + \delta) \geq f(x) + \langle \nabla f(x), \delta \rangle$.

For non-differentiable convex f , any vector g with $f(x + \delta) \geq f(x) + \langle g, \delta \rangle$ for all δ is a *subgradient*; the set of them, $\partial f(x)$, is non-empty on the interior of the domain.

The first-order condition says a convex function lies above all of its tangent planes. It is the single most-used fact in Part III.

Four scalar inequalities

These do most of the work in Part I.

- (I1) $e^x \geq 1 + x$ for all $x \in \mathbb{R}$,
- (I2) $\ln(1 + x) \leq x$ for all $x > -1$,
- (I3) $e^{sx} \leq 1 + (e^s - 1)x$ for $x \in [0, 1]$, $s \in \mathbb{R}$,
- (I4) $-\ln(1 - x) \leq x + x^2$ for $x \in [0, \frac{1}{2}]$.

(I1) is the tangent line of e^x at 0; (I2) is (I1) after taking logs. (I3) holds because

$x \mapsto e^{sx}$ is convex, so on $[0, 1]$ it lies below its chord between $(0, 1)$ and $(1, e^s)$. For (I4), let $g(x) = x + x^2 + \ln(1 - x)$. Then $g(0) = 0$ and $g'(x) = 1 + 2x - \frac{1}{1-x} = \frac{x(1-2x)}{1-x} \geq 0$ on $[0, \frac{1}{2}]$.

2.2 The Weighted Majority Algorithm

Halving throws an expert away after one mistake. When no expert is perfect that is fatal: every expert is eventually discarded. Littlestone and Warmuth's fix is to *shrink* the weight of an expert that errs rather than delete it.

Algorithm 2.3 (Weighted Majority, WMA)

```

input:  $\varepsilon \in (0, 1/2]$ 
 $w_{\{i,1\}} \leftarrow 1$  for all  $i$ 
for  $t = 1, 2, \dots$ :
    receive  $f_{\{i,t\}} \in \{0,1\}$ 
    predict  $\hat{y}_t \leftarrow 1$  if  $\sum_i w_{\{i,t\}} f_{\{i,t\}} \geq \frac{1}{2} \sum_i w_{\{i,t\}}$ , else 0
    observe  $y_t$ 
     $w_{\{i,t+1\}} \leftarrow w_{\{i,t\}} \cdot (1-\varepsilon)^{\mathbb{1}[f_{\{i,t\}} \neq y_t]}$ 

```

Halving is the limit $\varepsilon \rightarrow 1$.

Theorem 2.4 (WMA mistake bound)

For any sequence, any expert i , and $\varepsilon \in (0, \frac{1}{2}]$,

$$M_T(\text{WMA}) \leq \frac{2 \ln N}{\varepsilon} + 2(1 + \varepsilon) M_T(i).$$

Proof. Let $\Phi_t = \sum_i w_{i,t}$ be the total weight, so $\Phi_1 = N$.

Lower bound. Weights are positive, and expert i 's weight has been multiplied by $(1 - \varepsilon)$ once per mistake, so $\Phi_{T+1} \geq w_{i,T+1} = (1 - \varepsilon)^{M_T(i)}$.

Upper bound. If WMA errs at round t , the experts that were wrong held at least half of Φ_t . Write $\rho \geq \frac{1}{2}$ for their share. Their weight is multiplied by $1 - \varepsilon$, so

$$\Phi_{t+1} = \Phi_t(1 - \rho\varepsilon) \leq \Phi_t(1 - \frac{\varepsilon}{2}).$$

On rounds where WMA is correct, $\Phi_{t+1} \leq \Phi_t$. Hence $\Phi_{T+1} \leq N(1 - \varepsilon/2)^{M_T(\text{WMA})}$.

Combine. $(1 - \varepsilon)^{M_T(i)} \leq N(1 - \varepsilon/2)^{M_T(\text{WMA})}$. Take $-\ln$ of both sides:

$$\begin{aligned}
 M_T(\text{WMA}) \cdot \underbrace{\left(-\ln(1 - \frac{\varepsilon}{2})\right)}_{\geq \varepsilon/2 \text{ (I2)}} & \\
 \leq \ln N + M_T(i) \cdot \underbrace{\left(-\ln(1 - \varepsilon)\right)}_{\leq \varepsilon + \varepsilon^2 \text{ (I4)}} &
 \end{aligned}$$

So $\frac{\varepsilon}{2} M_T(\text{WMA}) \leq \ln N + (\varepsilon + \varepsilon^2) M_T(i)$. Multiply by $2/\varepsilon$. □

2.3 Tuning a learning rate

Bounds of the form $A/\varepsilon + B\varepsilon$ appear constantly. The first term is the cost of learning, which large steps pay less of; the second is the cost of overreacting to noise, which small steps pay less of.

Lemma 2.5 (balancing)

For $A, B > 0$, $\min_{\varepsilon > 0} \{A/\varepsilon + B\varepsilon\} = 2\sqrt{AB}$, attained at $\varepsilon^* = \sqrt{A/B}$. At the optimum the two terms are equal.

Apply it to Theorem 2.4 with $A = 2 \ln N$ and $B = 2M_T(i)$ (assuming the resulting $\varepsilon^* \leq \frac{1}{2}$):

$$M_T(\text{WMA}) \leq 2M_T(i) + 4\sqrt{M_T(i) \ln N}.$$

The additive overhead is sublinear, but the factor 2 in front of $M_T(i)$ is not: if the best expert errs on 30% of days, WMA may err on 60%. This factor is unavoidable for *deterministic* predictions with 0/1 loss (Exercise 1.2 already showed why). The next chapter removes it by letting the prediction be a probability, or equivalently by randomizing.

Tuning requires knowing the future

ε^* depends on $M_T(i)$, which we do not know in advance. Two standard fixes are the *doubling trick* (Exercise 2.4) and time-varying rates $\varepsilon_t \propto 1/\sqrt{t}$. Both lose only a constant factor.

Key takeaways

- Multiplicative down-weighting is Halving made robust to noise.
- WMA errs at most $2(1 + \varepsilon)M_T(i) + \frac{2 \ln N}{\varepsilon}$ times.
- Set a learning rate by balancing the A/ε and $B\varepsilon$ terms.

Exercises

2.1 warm-up Prove that the chord definition of convexity implies Jensen's inequality for random variables taking finitely many values.

2.2 warm-up Tune these bounds in $\eta > 0$ (from Homework 1): (a) $T\eta + \frac{\ln N}{\eta}$; (b) $\max(T\eta, \eta^{-2})$; (c) $\frac{T\varepsilon}{\eta} + \frac{N}{\varepsilon} + T\eta$, jointly in ε, η .

2.3 core Show that no deterministic algorithm can guarantee $M_T \leq c M_T(i) + o(T)$ for all sequences with $c < 2$, even with $N = 2$.

2.4 core (Doubling trick.) Suppose an algorithm with a known horizon T guarantees regret $\leq c\sqrt{T}$. Run it on epochs of length $1, 2, 4, \dots$, restarting each time. Show the total regret on any horizon T is at most $\frac{\sqrt{2}}{\sqrt{2}-1} c\sqrt{T} \approx 3.41 c\sqrt{T}$.

Solutions

2.1. Induct on the number of atoms k . For $k = 2$ it is the definition. For X taking value x_j with probability p_j , write $\mathbb{E}X = p_1x_1 + (1 - p_1)\bar{x}$ where $\bar{x} = \sum_{j \geq 2} \frac{p_j}{1-p_1} x_j$. Then $f(\mathbb{E}X) \leq p_1f(x_1) + (1 - p_1)f(\bar{x})$, and apply the induction hypothesis to $f(\bar{x})$.

2.2. (a) $\eta = \sqrt{\ln N / T}$, value $2\sqrt{T \ln N}$. (b) Equalize: $T\eta = \eta^{-2}$ gives $\eta = T^{-1/3}$ and value $T^{2/3}$. (c) For fixed η , the first two terms balance at $\varepsilon = \sqrt{N\eta/T}$, giving $2\sqrt{NT/\eta} + T\eta$. Balancing in η : set derivative $-\sqrt{NT}\eta^{-3/2} + T = 0$, so $\eta = (N/T)^{1/3}$, value $3N^{1/3}T^{2/3}$. This $T^{2/3}$ rate is typical of problems with limited feedback (Chapter 13).

2.3. Use the two constant experts and the adversary $y_t = 1 - \hat{y}_t$. The learner errs T times while $\min_i M_T(i) \leq T/2$. A bound with $c < 2$ would give $T \leq cT/2 + o(T)$, a contradiction for large T .

2.4. Epoch m has length 2^m and regret at most $c2^{m/2}$. Regret against the best expert overall is at most the sum of per-epoch regrets, because the sum of per-epoch minima is at most the overall minimum. If T falls in epoch K , $2^K \leq T$ and the sum is $c \sum_{m=0}^K 2^{m/2} = c \frac{2^{(K+1)/2} - 1}{\sqrt{2} - 1} \leq \frac{\sqrt{2}}{\sqrt{2} - 1} c\sqrt{T}$.

CHAPTER 3

Exponential Weights and Hedge

Based on Lectures 3–4 (Sept 11 and 16, 2013) · [notes 3](#) · [notes 4](#)

In this chapter

- Exponential Weights for convex losses, with leading constant 1.
- The action (Hedge) setting and the $\sqrt{(T/2) \ln N}$ regret bound.
- Priors over experts, and the softmin potential that underlies everything.

3.1 From mistakes to losses

We now let predictions live in $[0, 1]$ (think of them as probabilities that the market closes up) and measure them with a loss $\ell : [0, 1] \times \{0, 1\} \rightarrow [0, 1]$.

Definition 3.1 (convex loss)

A loss $\ell(\hat{y}, y)$ is *convex* if it is convex in its first argument for each y . Examples: absolute loss $|\hat{y} - y|$, squared loss $(\hat{y} - y)^2$, and log loss $-\hat{y} \ln \hat{y} - (1 - \hat{y}) \ln(1 - \hat{y})$ (convex but unbounded, so it is treated separately in Exercise 3.5).

Squared loss and log loss are also *proper*: if $y \sim \text{Bernoulli}(q)$, the expected loss is minimised by predicting $\hat{y} = q$. Proper losses reward honest probability forecasts, which is the starting point for calibration (Chapter 16).

3.2 The Exponential Weights Algorithm

Algorithm 3.2 (Exponential Weights, EWA)

```
input: learning rate  $\eta > 0$ 
 $w_{\{i,1\}} \leftarrow 1$  for all  $i$ 
for  $t = 1, 2, \dots$ :
    receive  $f_{\{i,t\}} \in [0,1]$ 
    predict  $\hat{y}_t \leftarrow \sum_i w_{\{i,t\}} f_{\{i,t\}} / \sum_i w_{\{i,t\}}$ 
    observe  $y_t$ ;  $\ell_{\{i,t\}} \leftarrow \ell(f_{\{i,t\}}, y_t)$ 
     $w_{\{i,t+1\}} \leftarrow w_{\{i,t\}} \cdot \exp(-\eta \ell_{\{i,t\}})$ 
```

Equivalently $w_{i,t} = \exp(-\eta L_{i,t-1})$ where $L_{i,t} = \sum_{s \leq t} \ell_{i,s}$ is expert i 's cumulative loss.

The key probabilistic lemma controls the log-moment-generating function of a bounded random variable.

Lemma 3.3

If $X \in [0, 1]$ and $s \in \mathbb{R}$, then $\ln \mathbb{E}[e^{sX}] \leq (e^s - 1) \mathbb{E}X$.

Proof. By (I3), $e^{sX} \leq 1 + (e^s - 1)X$ pointwise. Take expectations and then logs, and use (I2): $\ln \mathbb{E}e^{sX} \leq \ln(1 + (e^s - 1)\mathbb{E}X) \leq (e^s - 1)\mathbb{E}X$. $\square$

Theorem 3.4 (EWA loss bound)

For any convex loss with values in $[0, 1]$, any sequence, and any expert i ,

$$L_{MA,T} \leq \frac{\eta L_{i,T} + \ln N}{1 - e^{-\eta}}.$$

Proof. Let $W_t = \sum_i w_{i,t}$ and use the potential $\Phi_t = -\ln W_t$. Then

$$\Phi_{t+1} - \Phi_t = -\ln \frac{\sum_i w_{i,t} e^{-\eta \ell_{i,t}}}{W_t} = -\ln \mathbb{E}[e^{-\eta X}],$$

where X takes the value $\ell_{i,t}$ with probability $p_{i,t} = w_{i,t}/W_t$. Lemma 3.3 with $s = -\eta$ gives

$$\Phi_{t+1} - \Phi_t \geq (1 - e^{-\eta}) \sum_i p_{i,t} \ell(f_{i,t}, y_t) \geq (1 - e^{-\eta}) \ell\left(\sum_i p_{i,t} f_{i,t}, y_t\right) = (1 - e^{-\eta}) \ell(\hat{y}_t, y_t),$$

where the second inequality is Jensen, using convexity of the loss. Summing over t telescopes:

$$(1 - e^{-\eta}) L_{MA,T} \leq \Phi_{T+1} - \Phi_1 = -\ln \sum_j e^{-\eta L_{j,T}} + \ln N \leq \eta L_{i,T} + \ln N.$$

$\square$

Corollary 3.5 (tuned EWA)

If $L^* \geq \min_i L_{i,T}$ is known, choosing $\eta = \ln(1 + \sqrt{2 \ln N / L^*})$ gives

$$L_{MA,T} \leq L^* + \sqrt{2 L^* \ln N} + \ln N.$$

The leading constant is now 1. The three terms have a clean reading: what the best expert paid; the price of not knowing which expert is best; and a Halving-like cost of $\ln N$ that remains even when $L^* = 0$. Because the overhead scales with $\sqrt{L^*}$, the bound is small when some expert is nearly perfect. Bounds of this kind are called *first-order* or *small-loss* bounds.

3.3 The softmin potential

The proof never looked at individual rounds except through Φ , which depends only on the cumulative loss vector $L_t = (L_{1,t}, \dots, L_{N,t})$. Define the softmin

$$\Phi_\eta(L) = -\frac{1}{\eta} \ln \sum_{i=1}^N e^{-\eta L_i}, \quad \min_i L_i - \frac{\ln N}{\eta} \leq \Phi_\eta(L) \leq \max_i L_i.$$

EWA's play is exactly its gradient: $p_t = \nabla \Phi_\eta(L_{t-1})$. The analysis shows “the learner's loss on a round is at most the increase of the potential, up to a small term”. Summing, the learner's loss is about $\Phi_\eta(L_T)$, which is within $\ln N / \eta$ of the best expert's loss. Chapter 10 explains this as Follow the Regularized Leader with entropy, and the view generalises from the simplex to any convex set.

3.4 The action setting (Hedge)

In many problems there is no “prediction” to average: we must choose an *action*, such as which strategy to trade today or which route to drive.

Protocol 3.6 (Hedge setting / decision-theoretic online learning)

For $t = 1, \dots, T$: the learner picks a distribution $p_t \in \Delta_N$; the adversary picks a loss vector $\ell_t \in [0, 1]^N$; the learner pays $\langle p_t, \ell_t \rangle$ (its expected loss if it samples $I_t \sim p_t$). Regret is $\sum_t \langle p_t, \ell_t \rangle - \min_i \sum_t \ell_{i,t}$.

Hedge is EWA with $p_{i,t} \propto \exp(-\eta L_{i,t-1})$. Two facts connect the settings:

- The action setting is harder. With $g(x) = \ell(x, y_t)$ convex, the predictive loss $g(\mathbb{E}X)$ is at most the action loss $\mathbb{E}g(X)$. Any action-setting bound therefore transfers to the predictive setting.
- The action setting handles non-convex losses, including 0/1 loss, as long as we are allowed to randomize. That is how the factor 2 of WMA disappears.

For a bound that is uniform in T we use a second-order version of Lemma 3.3.

Lemma 3.7 (Hoeffding's lemma)

If $X \in [a, b]$ then for every $s \in \mathbb{R}$, $\ln \mathbb{E}e^{sX} \leq s \mathbb{E}X + \frac{s^2(b-a)^2}{8}$. (Proof in [Appendix A](#).)

Theorem 3.8 (Hedge regret)

For losses in $[0, 1]$ and any $\eta > 0$,

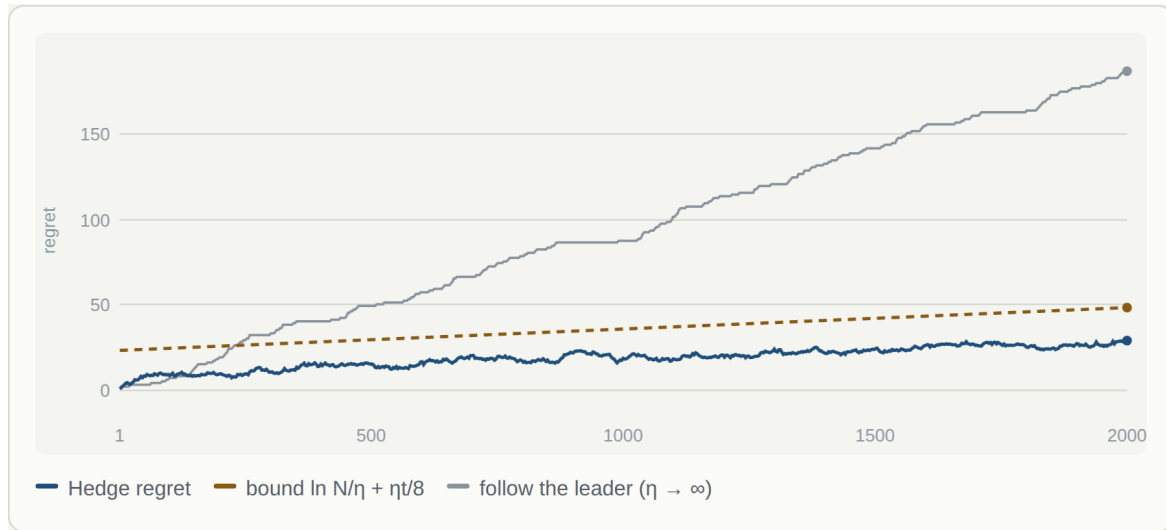
$$\sum_{t=1}^T \langle p_t, \ell_t \rangle - L_{i,T} \leq \frac{\ln N}{\eta} + \frac{\eta T}{8} \quad \text{for all } i.$$

With $\eta = \sqrt{8 \ln N / T}$ the regret is at most $\sqrt{(T/2) \ln N}$.

Proof. As before, $\ln \frac{W_{t+1}}{W_t} = \ln \mathbb{E}_{I \sim p_t} e^{-\eta \ell_{I,t}} \leq -\eta \langle p_t, \ell_t \rangle + \eta^2 / 8$ by Lemma 3.7. Summing, $\ln \frac{W_{T+1}}{W_1} \leq -\eta \sum_t \langle p_t, \ell_t \rangle + \eta^2 T / 8$. Also $\ln \frac{W_{T+1}}{W_1} \geq -\eta L_{i,T} - \ln N$. Rearrange and divide by η . The tuning is Lemma 2.5 with $A = \ln N$, $B = T/8$. $\square$

Two features deserve emphasis. First, regret grows like $\sqrt{T}$, so average regret vanishes: Hedge is no-regret. Second, the dependence on the number of experts is only $\sqrt{\ln N}$. We

can afford exponentially many experts, which is what makes the constructions in Chapters 4, 7 and 11 possible. Chapter 5 shows both dependences are optimal.



Simulated market of N signals over $T = 2000$ days. Each signal's daily loss is Bernoulli with a mean that drifts; one signal is slightly better on average, and the environment periodically punishes whoever currently leads. Try a very large η to see Follow the Leader get whipsawed, and a tiny η to see learning stall. (A static snapshot at the default settings; the HTML edition has live sliders.)

3.5 Priors over experts

Nothing forces uniform initial weights. If we start from $w_{i,1} = \pi_i$ for a prior $\pi \in \Delta_N$, the same proof gives:

Theorem 3.9 (EWA with a prior)

$$L_{MA,T} \leq \frac{\eta L_{i,T} + \ln(1/\pi_i)}{1 - e^{-\eta}} \quad \text{and, for Hedge,} \quad \sum_t \langle p_t, \ell_t \rangle - L_{i,T} \leq \frac{\ln(1/\pi_i)}{\eta} + \frac{\eta T}{8}.$$

The “ $\ln N$ ” was really the description length of the comparator: how many nats it takes to name expert i under the prior. Experts we believe in a priori cost less to compete with. This view has an information-theoretic flavour: regret is the cost of encoding the best expert. It is also exactly what makes infinite or structured expert classes tractable.

In finance

A fund-of-funds allocator who rebalances capital across N managers each month by multiplying each allocation by $e^{-\eta \cdot (\text{manager's loss})}$ is running Hedge. Theorem 3.8 says: over T months, the allocator's cumulative loss exceeds that of the best single manager by at most $\sqrt{(T/2) \ln N}$, whatever the markets do, provided monthly losses are scaled into $[0, 1]$. With $N = 20$ managers and $T = 120$ months, the bound is $\sqrt{60 \ln 20} \approx 13.4$ loss units, or about 0.11 per month. The prior version says a manager

you trust more (larger π_i) is cheaper to track. The catch is the scaling: real returns are unbounded, and a single catastrophic month breaks the $[0, 1]$ assumption. Chapter 11 treats multiplicative wealth properly with log-loss.

Key takeaways

- EWA: weights $\propto e^{-\eta \cdot \text{cumulative loss}}$. With convex losses its loss is at most $\frac{\eta L_i + \ln N}{1 - e^{-\eta}}$.
- Hedge: regret $\leq \frac{\ln N}{\eta} + \frac{\eta T}{8} = \sqrt{(T/2) \ln N}$ at the best η .
- With a prior, $\ln N$ becomes $\ln(1/\pi_i)$, the description length of the comparator.

Exercises

3.1 warm-up Verify that $\nabla \Phi_\eta(L)$ is the Hedge distribution and that $\min_i L_i - \frac{\ln N}{\eta} \leq \Phi_\eta(L) \leq \max_i L_i$.

3.2 core (Follow the Leader fails.) Let $N = 2$, $\ell_1 = (\frac{1}{2}, 0)$ and thereafter alternate $(0, 1), (1, 0), (0, 1), \dots$. Show that the algorithm putting all mass on the current leader (ties broken toward expert 1 in round 1) has regret about $T/2$. Which η in Hedge corresponds to FTL?

3.3 core Prove Theorem 3.9.

3.4 core (Subsets as experts, from Homework 1.) The experts are all k -subsets $S \subseteq [N]$, with loss $\sum_{i \in S} \ell_{i,t}$ (say, the loss of a k -stock equally weighted portfolio). Show that EWA's marginal weight $u_i = \sum_{S \ni i} w_S$ can be computed in $O(Nk)$ time, even though there are $\binom{N}{k}$ experts.

3.5 challenge (Mixability of log loss.) For log loss, show that EWA with $\eta = 1$ satisfies $L_{\text{MA}, T} \leq L_{i, T} + \ln N$, with no $\sqrt{T}$ term at all. Hint: $e^{-\ell(\hat{y}, y)} = \hat{y}^y (1 - \hat{y})^{1-y}$ and $\hat{y}_t = \sum_i p_{i,t} f_{i,t}$, so $e^{-\ell(\hat{y}_t, y_t)} = \sum_i p_{i,t} e^{-\ell(f_{i,t}, y_t)}$.

Solutions

3.1. $\partial_i \Phi_\eta = \frac{e^{-\eta L_i}}{\sum_j e^{-\eta L_j}}$. For the bounds: $e^{-\eta \min L} \leq \sum_j e^{-\eta L_j} \leq N e^{-\eta \min L}$; apply $-\frac{1}{\eta} \ln$.

3.2. After round 1 the leader is expert 2 (loss 0), which then takes loss 1. Now expert 1 leads with 0.5 vs 1, and gets loss 1. The leader always takes the loss of 1, so FTL pays about T , while each expert pays about $T/2$. FTL is Hedge with $\eta \rightarrow \infty$. Chapter 10 shows that regularization (finite η) is what stabilises it.

3.3. Now $W_1 = \sum_i \pi_i = 1$ and $W_{T+1} \geq \pi_i e^{-\eta L_{i,T}}$. The per-round inequality is unchanged. So $(1 - e^{-\eta}) L_{\text{MA}} \leq -\ln W_{T+1} + \ln W_1 \leq \eta L_{i,T} + \ln(1/\pi_i)$.

3.4. $w_S = \prod_{i \in S} v_i$ with $v_i = e^{-\eta L_i}$, so $\sum_S w_S = e_k(v)$, the elementary symmetric polynomial. It satisfies the recursion $e_k(v_{1:n}) = e_k(v_{1:n-1}) + v_n e_{k-1}(v_{1:n-1})$, computable by dynamic programming in $O(nk)$. And $u_i = v_i e_{k-1}(v_{-i})$, where v_{-i} drops coordinate i . Given the table $e_0(v), \dots, e_k(v)$, the identity $e_j(v) = e_j(v_{-i}) + v_i e_{j-1}(v_{-i})$ recovers $e_0(v_{-i}), \dots, e_{k-1}(v_{-i})$ in $O(k)$ steps for each i , so all

marginals cost $O(Nk)$. (In floating point, prefer working in log space or sampling S one element at a time from the DP table.)

3.5. From the hint, $\ell(\hat{y}_t, y_t) = -\ln \frac{W_{t+1}}{W_t}$ exactly. Telescoping gives $L_{MA,T} = -\ln \frac{W_{T+1}}{N} \leq L_{i,T} + \ln N$. This is Bayesian model averaging, and it is also the reason Cover's universal portfolio (Chapter 11) has regret logarithmic in T .

Tracking a Moving Target: Hyperexperts and Fixed Share

Based on Lectures 4–5 (Sept 16 and 18, 2013) · [notes 5](#)

In this chapter

- Competing against comparators that switch between experts.
- Why the naive “all sequences” prior learns nothing.
- Fixed Share: a Markov prior over switching sequences, run in $O(N)$ time per round.

4.1 When the best expert changes

Markets have regimes. A momentum signal may dominate for eighteen months, then a value signal takes over after a crash. Measured against the single best expert over the whole period, Hedge’s guarantee is weak in such cases: the best fixed expert may itself be mediocre, and Hedge, having piled weight onto the old leader, may take a long time to move. Indeed, after $T/2$ rounds of expert 1 dominating, expert 2’s weight is about $e^{-\eta T/2}$ times smaller, and Hedge needs on the order of another $T/2$ rounds to switch.

The strongest comparator imaginable is the best expert *on each round*, $\sum_t \min_i \ell_{i,t}$. No algorithm can have sublinear regret against it: with two experts and losses $(X_t, 1 - X_t)$ for random bits X_t , every algorithm pays $T/2$ in expectation while the per-round best pays 0. We need something in between.

Definition 4.1 (hyperexperts and switches)

A *hyperexpert* is a sequence $\mathbf{i} = (i_1, \dots, i_T) \in [N]^T$ that follows expert i_t on round t . Its loss is $L_{\mathbf{i}} = \sum_t \ell_{i_t,t}$. Its number of switches is $\sigma(\mathbf{i}) = |\{t < T : i_{t+1} \neq i_t\}|$. Tracking regret against the class of hyperexperts with at most k switches is $\sum_t \langle p_t, \ell_t \rangle - \min_{\sigma(\mathbf{i}) \leq k} L_{\mathbf{i}}$.

4.2 A tempting idea that fails

There are N^T hyperexperts. Treat each as an expert and run Hedge on them. When we must play a distribution over base experts, we take the marginal: $p_{t,j} \propto v_{t,j} = \sum_{\mathbf{i}: i_t=j} w_{t,\mathbf{i}}$.

Claim 4.2

With uniform initial weights on $[N]^T$, the marginal p_t is uniform on $[N]$ for every t , regardless of the losses.

Proof. $w_{t,i} = \exp(-\eta \sum_{s \leq t} \ell_{i,s})$ depends only on the prefix $(i_1, \dots, i_{t-1})$. Hence $v_{t,j} = \sum_{i_1, \dots, i_{t-1}} \exp(-\eta \sum_{s \leq t} \ell_{i_s, s}) \cdot N^{T-t}$ for every j : the same number. $\square$

Consistent with this, the regret bound $\sqrt{(T/2) \ln N^T} = T \sqrt{(\ln N)/2}$ is linear in T and says nothing. The uniform prior treats a hyperexpert that switches every round as just as plausible as one that never switches, so past data carry no information about the future. The fix is to *put a prior on hyperexperts that prefers few switches*, and use Theorem 3.9.

4.3 A Markov prior

Definition 4.3 (switching prior)

Fix $\alpha \in (0, 1)$. Draw i_1 uniformly. For $t \geq 1$, set $i_{t+1} = i_t$ with probability $1 - \alpha$; otherwise move to each of the other $N - 1$ experts with probability $\frac{\alpha}{N-1}$. Then

$$\Pi(\mathbf{i}) = \frac{1}{N} \left(\frac{\alpha}{N-1} \right)^{\sigma(\mathbf{i})} (1 - \alpha)^{T-1-\sigma(\mathbf{i})}.$$

By Theorem 3.9, Hedge over hyperexperts with prior Π has regret at most $\frac{\ln(1/\Pi(\mathbf{i}))}{\eta} + \frac{\eta T}{8}$ against every $\mathbf{i}$, and

$$\ln \frac{1}{\Pi(\mathbf{i})} = \ln N + \sigma \ln \frac{N-1}{\alpha} + (T-1-\sigma) \ln \frac{1}{1-\alpha}.$$

Theorem 4.4 (tracking regret)

With $\alpha = 1/T$, for every hyperexpert with at most k switches, $\ln(1/\Pi(\mathbf{i})) \leq (k+1) \ln(NT) + 1$. With η tuned for this value,

$$\sum_t \langle p_t, \ell_t \rangle - L_{\mathbf{i}} \leq \sqrt{\frac{T}{2} ((k+1) \ln(NT) + 1)} = O(\sqrt{T k \ln(NT)}).$$

Proof. $\sigma \ln \frac{N-1}{\alpha} \leq k \ln(NT)$. For the last term, $\ln \frac{1}{1-\alpha} \leq \frac{x}{1-x}$, so with $x = 1/T$, $(T-1-\sigma) \ln \frac{1}{1-1/T} \leq (T-1) \cdot \frac{1}{T-1} = 1$. Add $\ln N$ and apply Theorem 3.9 with Lemma 2.5. $\square$

Compare with an oracle that knows where the k switches happen and restarts Hedge on each of the $k+1$ blocks: its regret is $\sum_b \sqrt{(T_b/2) \ln N} \leq \sqrt{(k+1)(T/2) \ln N}$ by Cauchy-Schwarz. Fixed Share pays only an extra $\ln T$ inside the square root for not knowing the switch times.

4.4 Fixed Share: the efficient implementation

Maintaining N^T weights is impossible, but the prior is a Markov chain, and marginals of Markov chains can be propagated forward, as in the forward algorithm for hidden Markov models.

Algorithm 4.5 (Fixed Share, Herbster and Warmuth 1998)

```

input:  $\eta > 0, \alpha \in (0,1)$ 
 $v_{\cdot\{j,1\}} \leftarrow 1/N$  for all  $j$ 
for  $t = 1, 2, \dots$ :
  play  $p_t = v_t / \sum_j v_{\cdot\{j,t\}}$ 
  observe  $\ell_t$ 
   $u_j \leftarrow v_{\cdot\{j,t\}} \cdot \exp(-\eta \ell_{\cdot\{j,t\}})$  # loss update
   $v_{\cdot\{j,t+1\}} \leftarrow (1-\alpha) u_j + (\alpha/(N-1)) \sum_{i \neq j} u_i$  # share update

```

Cost: $O(N)$ per round.

Proposition 4.6

Fixed Share plays exactly the marginal distribution of Hedge over hyperexperts with prior Π . Hence it enjoys Theorem 4.4.

Proof. Let $w_{t,i} = \Pi(i) \exp(-\eta \sum_{s < t} \ell_{i,s})$ and define the prefix weight $\omega_t(i_{1:t}) = \Pi(i_{1:t}) \exp(-\eta \sum_{s < t} \ell_{i_s,s})$, where $\Pi(i_{1:t})$ is the prior's marginal on the first t coordinates. Summing out future coordinates of a Markov chain gives $v_{t,j} = \sum_{i_{1:t}: i_t=j} \omega_t(i_{1:t})$. The Markov property gives the recursion

$$\omega_{t+1}(i_{1:t+1}) = \omega_t(i_{1:t}) e^{-\eta \ell_{i_t,t}} P(i_{t+1} | i_t).$$

Summing over $i_{1:t}$ with $i_{t+1} = j$ yields $v_{t+1,j} = \sum_i v_{t,i} e^{-\eta \ell_{i,t}} P(j | i)$, which is the loss update followed by the share update. $\square$

The share step guarantees every expert keeps weight at least $\frac{\alpha}{N-1} \sum_i u_i$, so an expert that was bad for a long time can be picked up within a few rounds once it starts performing. That is precisely the property plain Hedge lacked.

In finance

Fixed Share is a regime-switching allocator without a regime model. With $N = 5$ strategies over $T = 2520$ trading days (ten years) and a comparator that switches regimes $k = 4$ times, the bound gives roughly $\sqrt{1260(5 \ln 12600 + 1)} \approx 246$ loss units, or 0.098 per day, compared with 0.018 per day for Hedge against a single fixed strategy. The price of tracking is visible and moderate. In practice α is a “forgetting rate”: a larger α adapts faster but pays more in stable periods. Choosing it online is itself an experts problem (run Hedge over a grid of α values).

Key takeaways

- Competing with the per-round best is impossible; competing with k -switch sequences costs $O(\sqrt{Tk \ln(NT)})$.
- Uniform priors over huge classes can erase learning. Priors that encode structure are what make large classes useful.
- A Markov prior lets the exponential-size mixture be computed in $O(N)$ per

round.

Exercises

- 4.1** warm-up Verify that $\sum_{\mathbf{i} \in [N]^T} \Pi(\mathbf{i}) = 1$.
- 4.2** core Show the share update can be written $v_{t+1} = (1 - \beta)\tilde{u} + \beta \frac{1}{N} \sum_i u_i$ with $\tilde{u} = u$, and find β in terms of α and N .
- 4.3** core Optimise α when k is known. Show that $\alpha = k/(T - 1)$ gives $\ln(1/\Pi) \leq \ln N + k \ln(N - 1) + (T - 1)H(\frac{k}{T-1})$, where H is binary entropy in nats, and that this is at most $\ln N + k \ln \frac{e(N-1)(T-1)}{k}$.
- 4.4** challenge (From Homework 3.) In the portfolio setting of Chapter 11, a “forgetful” universal portfolio competes with constant-rebalanced portfolios that change k times. Using the Fixed Share idea, sketch an algorithm and a regret bound of order $(k + 1) d \ln T$ in log-wealth.

Solutions

- 4.1.** Group sequences by switch positions and values: there are $N \binom{T-1}{\sigma} (N-1)^\sigma$ sequences with σ switches, so the sum is $\sum_{\sigma} \binom{T-1}{\sigma} \alpha^\sigma (1 - \alpha)^{T-1-\sigma} = 1$ by the binomial theorem. Or simply note Π is the law of a Markov chain.
- 4.2.** $(1 - \alpha)u_j + \frac{\alpha}{N-1}(U - u_j) = (1 - \alpha - \frac{\alpha}{N-1})u_j + \frac{\alpha}{N-1}U$ where $U = \sum_i u_i$. Matching $\frac{\beta}{N}U$ gives $\beta = \frac{\alpha N}{N-1}$, and indeed $1 - \beta = 1 - \alpha - \frac{\alpha}{N-1}$.
- 4.3.** With $\sigma = k$: $k \ln \frac{1}{\alpha} + (T - 1 - k) \ln \frac{1}{1-\alpha}$ at $\alpha = k/(T - 1)$ is exactly $(T - 1)H(\alpha)$. Use $H(x) \leq x \ln \frac{e}{x}$, so $(T - 1)H \leq k \ln \frac{e(T-1)}{k}$. This is roughly $\ln \binom{T-1}{k}$, the cost of naming the switch times.

CHAPTER 5

Lower Bounds: Why $\sqrt{T \ln N}$ Is the Right Answer

Based on Lecture 6 (Sept 23, 2013) · [scribe notes](#)

In this chapter

- Minimax regret as a game between learner and adversary.
- The probabilistic method: a random adversary is enough to force $\Omega(\sqrt{T \ln N})$ regret.

5.1 Minimax regret

The best guarantee any algorithm can offer is the value of a game in which the learner commits to an algorithm and the adversary then picks the worst loss sequence:

$$V_{T,N} = \inf_{\text{alg}} \sup_{\ell_1, \dots, \ell_T \in [0,1]^N} \left[\sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_i \sum_{t=1}^T \ell_{i,t} \right].$$

Hedge shows $V_{T,N} \leq \sqrt{(T/2) \ln N}$. To show no algorithm does much better, we use the trick that *a maximum is at least an average*: replace the worst-case adversary by a random one, and show that *every* algorithm has large expected regret against it.

5.2 Two experts

Let $X_1, \dots, X_T$ be independent fair coin flips and set $\ell_t = (X_t, 1 - X_t)$.

Argument. 1. Since p_t depends only on $X_{1:t-1}$, and $\mathbb{E}[\ell_t \mid X_{1:t-1}] = (\frac{1}{2}, \frac{1}{2})$, we get $\mathbb{E}\langle p_t, \ell_t \rangle = \frac{1}{2}$ for every algorithm. The learner's expected loss is exactly $T/2$, whatever it does.

2. So $V_{T,2} \geq \mathbb{E}[\frac{T}{2} - \min(S, T - S)]$ with $S = \sum_t X_t$, and $\frac{T}{2} - \min(S, T - S) = |S - \frac{T}{2}|$.

3. With Rademacher signs $Z_t = 2X_t - 1$, this is $\frac{1}{2}\mathbb{E}|\sum_t Z_t|$.

4. By the central limit theorem, $\mathbb{E}|\sum_t Z_t| \sim \sqrt{T} \mathbb{E}|G| = \sqrt{2T/\pi}$ for $G \sim \mathcal{N}(0,1)$.

Hence $V_{T,2} \geq \sqrt{T/(2\pi)} (1 + o(1))$. □

The adversary did nothing clever: pure noise suffices. The learner cannot predict a coin, and in hindsight one of the two experts will have been lucky by about $\sqrt{T}$. Regret measures that luck.

5.3 N experts

For even N , use $N/2$ independent pairs: $\ell_t = (X_t^1, 1 - X_t^1, \dots, X_t^{N/2}, 1 - X_t^{N/2})$. The learner still pays $T/2$ in expectation, and

$$V_{T,N} \geq \frac{1}{2} \mathbb{E} \max_{j \leq N/2} \left| \sum_t Z_t^j \right| \approx \frac{1}{2} \sqrt{T} \mathbb{E} \max_{j \leq N/2} |G_j| \approx \frac{1}{2} \sqrt{T} \sqrt{2 \ln(N/2)}.$$

The last step uses the fact that the maximum of n independent standard Gaussians concentrates around $\sqrt{2 \ln n}$. A sharper version, using N independent (not paired) experts, gives the exact constant.

Theorem 5.1 (Cesa-Bianchi and Lugosi 2006, Thm 3.7)

$$\lim_{N \rightarrow \infty} \lim_{T \rightarrow \infty} \frac{V_{T,N}}{\sqrt{(T/2) \ln N}} = 1.$$

So Hedge with a tuned learning rate is *asymptotically optimal including the constant*, a rare situation in learning theory.

5.4 Regret as a game

This chapter used the adversary's randomness to lower-bound a min-max. Exchanging the order of play is the subject of the minimax theorem in the next chapter, and in Part VI we will see that the learner's side of the game can also be treated geometrically. It is worth noting now that lower bounds in online learning often come from a stochastic adversary, while upper bounds hold for adversarial sequences. When the two match, as here, stochastic data are as hard as adversarial data, and the adversarial model costs nothing in the worst case.

In finance

If daily signal performance is indistinguishable from coin flips, which is roughly the efficient-markets null, then the best signal in a library of N will beat the coin-flip average by about $\sqrt{(T/2) \ln N}$ loss units just by luck. This is the online-learning version of the data-snooping problem (White, 2000; Harvey, Liu and Zhu, 2016): the best backtest in a large library is biased upward by that amount, and any strategy that selects signals online will trail that best backtest by roughly the same amount. Theorem 5.1 says no allocation rule can beat this, which is a principled haircut to apply to backtested Sharpe ratios.

Key takeaways

- A random adversary lower-bounds the worst case. Against i.i.d. coin-flip losses every learner pays $T/2$ in expectation.
- The best of N random walks gains about $\sqrt{2T \ln N}$ by luck, which forces regret $\Omega(\sqrt{T \ln N})$.

- Hedge's $\sqrt{(T/2) \ln N}$ is optimal, constant included.

Exercises

5.1 warm-up Show that for n i.i.d. standard Gaussians, $\mathbb{E} \max_j G_j \leq \sqrt{2 \ln n}$. Hint: for $\lambda > 0$, $e^{\lambda \mathbb{E} \max G} \leq \mathbb{E} e^{\lambda \max G} \leq \sum_j \mathbb{E} e^{\lambda G_j} = n e^{\lambda^2/2}$.

5.2 core For $T = 4$, compute $\mathbb{E} |\sum_{t \leq 4} Z_t|$ exactly and compare with $\sqrt{2T/\pi}$.

5.3 challenge Show that the bound of Chapter 4 is also tight up to logarithmic factors: any algorithm has tracking regret $\Omega(\sqrt{Tk \ln N})$ against k -switch comparators. Hint: split time into $k + 1$ blocks and apply the N -expert lower bound in each.

Solutions

5.1. Taking logs, $\mathbb{E} \max G \leq \frac{\ln n}{\lambda} + \frac{\lambda}{2}$. Minimise at $\lambda = \sqrt{2 \ln n}$. This is the same balancing as in Hedge, and it is no coincidence: the proof of Hedge is a deterministic version of this argument.

5.2. $\sum Z \in \{-4, -2, 0, 2, 4\}$ with probabilities $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$. $\mathbb{E} |\cdot| = \frac{2(4 \cdot 1 + 2 \cdot 4)}{16} = 1.5$. $\sqrt{8/\pi} \approx 1.60$.

Part II

Games, Equilibria and Their Uses

No-regret learning proves the minimax theorem, computes equilibria, solves linear programs and boosts weak classifiers.

CHAPTER 6

Zero-Sum Games and the Minimax Theorem

Based on Lectures 6–8 (Sept 23 – 30, 2013) · [notes 7](#) · [notes 8](#)

In this chapter

- Bimatrix games, mixed strategies, best responses and Nash equilibria.
- Nash's existence theorem via Brouwer.
- Von Neumann's minimax theorem, proved by letting two Hedge players play each other.

6.1 Games in normal form

Definition 6.1 (*bimatrix game*)

A two-player game is a pair of $n \times m$ matrices (A, B) . The row player picks $i \in [n]$ and receives A_{ij} ; the column player picks $j \in [m]$ and receives B_{ij} . Mixed strategies $x \in \Delta_n$, $y \in \Delta_m$ give expected payoffs $x^\top Ay$ and $x^\top By$. The game is **zero-sum** if $B = -A$.

Example 6.2 (*Prisoners' Dilemma*)

Actions: confess (1) or stay silent (2). Payoffs are minus years in prison:

$$A = \begin{pmatrix} -8 & 0 \\ -10 & -1 \end{pmatrix}, \quad B = A^\top = \begin{pmatrix} -8 & -10 \\ 0 & -1 \end{pmatrix}.$$

Confessing is better for each player whatever the other does: $-8 > -10$ and $0 > -1$. So (confess, confess) is the unique equilibrium, with payoff $(-8, -8)$, although (silent, silent) gives both players -1 . Individually rational play can be collectively bad.

Example 6.3 (*penalty kicks*)

The kicker (rows) aims left or right and the goalie (columns) dives left or right. The kicker scores unless the goalie guesses right, so with payoff $+1$ for a goal the kicker gets $A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ and $B = -A$. Here, every pure pair has a player who wants to deviate. The only equilibrium is both players mixing $(\frac{1}{2}, \frac{1}{2})$, with value 0.

Definition 6.4 (*best response, Nash equilibrium*)

$BR_1(y) = \arg \max_{x \in \Delta_n} x^\top Ay$ and $BR_2(x) = \arg \max_{y \in \Delta_m} x^\top By$. Because payoffs are linear in one's own strategy, a pure best response always exists. A pair (x^*, y^*)

is a **Nash equilibrium** if $x^* \in \text{BR}_1(y^*)$ and $y^* \in \text{BR}_2(x^*)$; equivalently, neither player regrets their choice against any pure deviation: $x^{*\top} Ay^* \geq e_i^\top Ay^*$ for all i and $x^{*\top} By^* \geq x^{*\top} Be_j$ for all j . It is an ε -**equilibrium** if each inequality holds up to ε .

6.2 Nash's theorem

Theorem 6.5 (Nash 1950)

Every finite bimatrix game has at least one mixed Nash equilibrium.

The proof uses Brouwer's fixed-point theorem: every continuous map from a compact convex set to itself has a fixed point.

Proof. Define the gains from deviating, $c_i(x, y) = \max(0, e_i^\top Ay - x^\top Ay)$ and $d_j(x, y) = \max(0, x^\top Be_j - x^\top By)$, and the map $f(x, y) = (x', y')$ with

$$x'_i = \frac{x_i + c_i}{1 + \sum_k c_k}, \quad y'_j = \frac{y_j + d_j}{1 + \sum_k d_k}.$$

f is continuous and maps $\Delta_n \times \Delta_m$ into itself, so it has a fixed point (x, y) . Some k with $x_k > 0$ has $e_k^\top Ay \leq x^\top Ay$ (an average cannot be strictly below every term in its support), so $c_k = 0$. At the fixed point $x_k = \frac{x_k}{1 + \sum_i c_i}$, which forces $\sum_i c_i = 0$. The same argument gives $\sum_j d_j = 0$. So nobody gains from deviating. $\square$

The proof is not constructive, and indeed computing a Nash equilibrium of a general bimatrix game is PPAD-complete (Chen and Deng 2006; Daskalakis, Goldberg and Papadimitriou 2006). The zero-sum case is different: equilibria are computable efficiently, and no-regret learning finds them.

6.3 Von Neumann's minimax theorem

From here on, $M \in [0, 1]^{n \times m}$ is a loss matrix: the row player chooses $p \in \Delta_n$ to *minimise* $p^\top Mq$, and the column player chooses $q \in \Delta_m$ to *maximise* it.

Theorem 6.6 (von Neumann 1928)

$$\min_{p \in \Delta_n} \max_{q \in \Delta_m} p^\top Mq = \max_{q \in \Delta_m} \min_{p \in \Delta_n} p^\top Mq =: V.$$

The inequality $\min \max \geq \max \min$ is easy: on the left the maximiser moves second and can react to p ; moving second never hurts. The content of the theorem is the reverse: in zero-sum games *announcing your mixed strategy costs nothing*. There is a p^* with $p^{*\top} Mq \leq V$ for every q , and a q^* with $p^\top Mq^* \geq V$ for every p .

Proof via no-regret learning

Let the row player run Hedge on loss vectors $\ell_t = Mq_t$, and let the column player run any no-regret algorithm on gain vectors $g_t = M^\top p_t$ (Hedge on gains: $w_j \leftarrow w_j e^{\eta g_{j,t}}$). Write $\bar{p} = \frac{1}{T} \sum_t p_t$, $\bar{q} = \frac{1}{T} \sum_t q_t$, and $\varepsilon_T = \text{Regret}_T / T = O(\sqrt{\ln(\max(n, m)) / T})$.

Proof. The row player's regret bound gives

$$\frac{1}{T} \sum_t p_t^\top M q_t \leq \min_p \frac{1}{T} \sum_t p^\top M q_t + \varepsilon_T = \min_p p^\top M \bar{q} + \varepsilon_T \leq \max_q \min_p p^\top M q + \varepsilon_T.$$

The column player's regret bound gives

$$\frac{1}{T} \sum_t p_t^\top M q_t \geq \max_q \bar{p}^\top M q - \varepsilon_T \geq \min_p \max_q p^\top M q - \varepsilon_T.$$

Chaining, $\min \max \leq \max \min + 2\varepsilon_T$. Let $T \rightarrow \infty$. □

The proof gives more than the theorem:

Corollary 6.7 (equilibrium computation by self-play)

The average strategies form a $2\varepsilon_T$ -equilibrium: $\max_q \bar{p}^\top M q \leq V + 2\varepsilon_T$ and $\min_p \bar{p}^\top M q \geq V - 2\varepsilon_T$. With Hedge on both sides, $T = O(\ln(nm)/\varepsilon^2)$ rounds suffice for an ε -equilibrium, each costing $O(nm)$ arithmetic. If the column player instead *best-responds*, $q_t \in \arg \max_q p_t^\top M q$, it has non-positive regret and the guarantee improves to ε_T .

Note that it is the *averages* $\bar{p}, \bar{q}$ that converge. The day-to-day strategies p_t, q_t of two Hedge players typically cycle around the equilibrium without settling. More recent “optimistic” variants of Hedge make the last iterate converge too, at rate $O(1/T)$ for the average (Rakhlin and Sridharan 2013; Daskalakis, Deckelbaum and Kim 2011).

Fictitious play

If both players use Follow the Leader (best response to the opponent's empirical average), we get Brown's fictitious play. Robinson (1951) proved it converges in zero-sum games, but with the rate $T^{-1/(n+m-2)}$. Karlin conjectured the rate is really $O(T^{-1/2})$, as simulations suggest. Daskalakis and Pan (2014) disproved the conjecture when ties are broken adversarially; with consistent tie-breaking it remains open. This is Homework 2, Problem 3(b).

6.4 Minimax and linear programming

The value of a zero-sum game is a linear program. Because $\max_q$ of a linear function is attained at a vertex,

$$V = \min_{c \in \mathbb{R}, p \in \Delta_n} c \quad \text{subject to} \quad (p^\top M)_j \leq c \quad \text{for } j = 1, \dots, m.$$

Its LP dual is the max-min problem, so LP strong duality implies the minimax theorem. The converse also holds, in the sense that any LP can be reduced to a zero-sum game (Dantzig 1951; Adler 2013). The link is the Lagrangian: $\max_{x \geq 0} \{d^\top x : Ax \leq b\} = \max_{x \geq 0} \min_{y \geq 0} d^\top x + y^\top (b - Ax)$, and exchanging max and min gives the dual $\min_{y \geq 0} \{b^\top y : A^\top y \geq d\}$. Chapter 7 uses this to solve LPs with Hedge.

In finance

Robust portfolio choice is a zero-sum game against nature: the investor picks weights p , nature picks a scenario j , and the loss is $M_{.j}$. The minimax theorem says the investor's worst-case-optimal mix p^* is also a best response to a *least-favourable prior* q^* over scenarios. That is the mathematical content behind the claim that robust optimisation is Bayesian optimisation under a pessimistic prior. Self-play Hedge computes both p^* and q^* for scenario sets far too large for an off-the-shelf LP solver to handle comfortably, since the iteration count scales only with $\ln m$.

Key takeaways

- Nash equilibria exist (Brouwer) but are hard to compute in general.
- In zero-sum games $\min \max = \max \min$. Two no-regret learners playing each other converge on average to an equilibrium, at rate $2\text{Regret}_T/T$.
- Minimax duality and LP duality are two views of the same fact.

Exercises

6.1 warm-up Find the value and optimal strategies of $M = \begin{pmatrix} 0.2 & 0.7 \\ 0.6 & 0.3 \end{pmatrix}$ (row player minimises).

6.2 core Prove that in a zero-sum game the set of equilibria is a product $P^* \times Q^*$ of convex sets, and all equilibria have the same value. Show this fails for the general-sum game of Example 6.2 modified so that there are two equilibria with different payoffs (for example, a coordination game).

6.3 core (Regularized best responses, from Lecture 8.) Replace best responses by $x' = \arg \max_x \{x^\top Ay - \varepsilon \|x\|^2\}$ and similarly for y . Show this map is continuous, that any fixed point is an ε -equilibrium, and deduce Nash's theorem by letting $\varepsilon \rightarrow 0$.

6.4 project (Homework 2, Problem 3.) For $M = \begin{pmatrix} 5 & 8 & 3 & 1 & 6 \\ 4 & 2 & 6 & 3 & 5 \\ 2 & 4 & 6 & 4 & 1 \\ 1 & 3 & 2 & 5 & 3 \end{pmatrix}$, run for $T = 10^4$ rounds:

(i) Hedge vs Hedge; (ii) Hedge vs best response; (iii) FTPL vs FTPL with fresh perturbations; (iv) fictitious play. Plot the duality gap $\max_j \bar{p}^\top M e_j - \min_i e_i^\top M \bar{q}$ against T on log-log axes. Which method converges fastest in practice?

6.5 challenge (Homework 3, Problem 1.) Let X, Y be convex compact sets and $f(x, y)$ convex in x , concave in y , with bounded gradients. Prove $\inf_x \sup_y f = \sup_y \inf_x f$ by letting both players run online gradient descent (Chapter 9), and show that the averaged iterates form an ε -saddle point after $O(1/\varepsilon^2)$ rounds.

Solutions

6.1. No saddle point, so both mix. The row player equalises columns: $0.2p + 0.6(1 - p) = 0.7p + 0.3(1 - p)$, so $0.6 - 0.4p = 0.3 + 0.4p$, $p = 3/8$. Value = $0.6 - 0.15 = 0.45$. The column player equalises rows: $0.2q + 0.7(1 - q) = 0.6q + 0.3(1 - q)$, so $0.7 - 0.5q = 0.3 + 0.3q$, $q = 1/2$, value 0.45.

6.2. (p, q) is an equilibrium iff $p \in P^* = \{p : \max_q p^\top Mq = V\}$ and $q \in Q^* = \{q : \min_p p^\top Mq = V\}$, by the minimax theorem. Each set is a sublevel/superlevel set of a convex/concave function, hence convex. In the coordination game $A = B = \text{diag}(2, 1)$, both $(1, 1)$ and $(2, 2)$ are equilibria, with payoffs 2 and 1, and mixing them is not an equilibrium.

6.3. The objective is strictly concave, so the maximiser is unique, and by Berge's maximum theorem it varies continuously in y . At a fixed point x maximises $x^\top Ay - \varepsilon \|x\|^2$, and $0 \leq \|x\|^2 \leq 1$ on the simplex, so $x^\top Ay \geq e_i^\top Ay - \varepsilon$. Take $\varepsilon_k \rightarrow 0$; by compactness a subsequence of fixed points converges, and the limit satisfies the equilibrium inequalities by continuity. Using ln-entropy instead of $\|x\|^2$ gives "logit" or quantal-response equilibria, with softmax best responses: Hedge's update rule.

6.5. Player x runs OGD on $f_t(x) = f(x, y_t)$, player y runs OGD on $-f(x_t, y)$. Both regrets are $O(\sqrt{T})$. Exactly as in §6.3, convexity (Jensen) in x and concavity in y give $\sup_y f(\bar{x}, y) \leq \frac{1}{T} \sum_t f(x_t, y_t) + \frac{R_T}{T} \leq \inf_x f(x, \bar{y}) + \frac{2R_T}{T}$.

Applications of Minimax: Linear Programs and Boosting

Based on Lectures 9–10 (Oct 2 and 7, 2013) · [notes 9](#) · [notes 10](#)

In this chapter

- A multiplicative-weights solver for LP feasibility whose iteration count ignores the number of constraints.
- Boosting: why weak learners can always be combined into a strong one (minimax), and how to find the combination (Hedge).

7.1 Solving linear programs with Hedge

Any bounded LP can be put in a normalised feasibility form. First add $\sum_i x_i = 1$ (rescale, adding a slack coordinate if necessary), which makes $x \in \Delta_n$. Then $a_j^\top x \leq b_j$ becomes $(a_j - b_j \mathbf{1})^\top x \leq 0$. An objective $c^\top x \geq d$ becomes one more constraint $(d\mathbf{1} - c)^\top x \leq 0$, and binary search on d recovers optimisation at a cost of a $\log(1/\varepsilon)$ factor.

Problem 7.1 (approximate LP feasibility)

Given vectors $\{a_j\}_{j \in J} \subset [-\rho, \rho]^n$, either find $x \in \Delta_n$ with $a_j^\top x \leq \varepsilon$ for all j , or certify that no $x \in \Delta_n$ has $a_j^\top x \leq 0$ for all j .

View it as a game: the x -player picks a mixture of coordinates, and an adversary answers with the most violated constraint.

Algorithm 7.2 (multiplicative-weights LP)

```

x_1 ← uniform on [n]
for t = 1, ..., T:
    j_t ← argmax_j a_j^\top x_t           # most violated constraint (an "oracle")
    if a_{j_t}^\top x_t ≤ ε: return x_t
    x_{t+1,i} ∝ x_{t,i} · exp(-η a_{j_t,i}) # Hedge on loss vector a_{j_t}
return INFEASIBLE

```

Theorem 7.3

With $T > \frac{2\rho^2 \ln n}{\varepsilon^2}$ and the matching η , Algorithm 7.2 is correct.

Proof. Loss vectors lie in $[-\rho, \rho]^n$, a range of 2ρ , so Hedge's regret is at most $\rho\sqrt{2T \ln n}$ (Theorem 3.8 rescaled). Suppose a feasible x° exists but the algorithm returns INFEASIBLE.

Every round had $a_{j_t}^\top x_t > \varepsilon$, so

$$\begin{aligned} \varepsilon &< \frac{1}{T} \sum_t a_{j_t}^\top x_t \leq \frac{1}{T} \sum_t a_{j_t}^\top x^\circ + \rho \sqrt{\frac{2 \ln n}{T}} \\ &\leq 0 + \rho \sqrt{\frac{2 \ln n}{T}}, \end{aligned}$$

which contradicts the choice of T . $\square$

The iteration count depends on the dimension only through $\ln n$ and not at all on the number of constraints $|J|$, which may be exponential as long as we can find the most violated one. This is the Plotkin–Shmoys–Tardos framework; Arora, Hazan and Kale (2012) survey its many uses, from multicommodity flow to semidefinite programming. The price is the $1/\varepsilon^2$ dependence, which makes it a tool for fast, low-precision solutions.

In finance: finding a pricing measure

Suppose S states of the world, and m traded assets with discounted payoff vectors $\pi_j \in \mathbb{R}^S$ and prices c_j . By the fundamental theorem of asset pricing (in its one-period, finite-state form), there is no arbitrage iff there is a strictly positive state-price vector, i.e. a risk-neutral measure $q \in \Delta_S$ with $q^\top \pi_j = c_j$ for all j . Writing each equality as two inequalities $\pm(\pi_j - c_j \mathbf{1})^\top q \leq 0$ gives exactly Problem 7.1. Algorithm 7.2 either returns an approximately consistent pricing measure over, say, $S = 10^6$ scenario states in $O(\rho^2 \ln S / \varepsilon^2)$ rounds, or its failure certifies (through LP duality) a portfolio that is an ε -arbitrage. The Hedge weights over states behave like a Gibbs or maximum-entropy pricing measure, which connects to the entropy-based calibration of option surfaces used in practice (Avellaneda’s weighted Monte Carlo).

7.2 Boosting

Can a learner that is only slightly better than guessing be turned into one that is accurate? Kearns and Valiant asked this in 1988; Schapire (1990) and Freund and Schapire (1997, AdaBoost) answered yes. The game-theoretic view makes the answer almost immediate.

Setup

Training points $x_1, \dots, x_N$ with labels $c(x_i) \in \{0, 1\}$; a class $\mathcal{H}$ of base hypotheses (for example decision stumps $h_{k,\theta}(x) = \mathbf{1}[x_k \geq \theta]$). Define the $N \times |\mathcal{H}|$ matrix

$$M_{i,h} = \begin{cases} +1 & h(x_i) \neq c(x_i) \\ -1 & h(x_i) = c(x_i). \end{cases}$$

For a distribution p over points, $p^\top M e_h = 2 \Pr_{i \sim p}[h \text{ errs}] - 1$.

Definition 7.4 (weak learning assumption, WLA $_\gamma$)

For every distribution p over the training points there is $h \in \mathcal{H}$ with $\Pr_{i \sim p}[h(x_i) \neq c(x_i)] \leq \frac{1}{2} - \frac{\gamma}{2}$, i.e. $p^\top M e_h \leq -\gamma$.

Theorem 7.5 (weak implies strong)

Under WLA_γ there is a distribution q over $\mathcal{H}$ whose weighted majority vote is correct on every training point, with margin γ : $e_i^\top Mq \leq -\gamma$ for all i .

Proof. WLA_γ says $\max_p \min_q p^\top Mq \leq -\gamma$. By the minimax theorem, $\min_q \max_p p^\top Mq \leq -\gamma$, so some q has $e_i^\top Mq \leq -\gamma$ for every point i . For point i that means the q -weight of hypotheses that err is at most $\frac{1}{2} - \frac{\gamma}{2}$, so the weighted vote is right. $\square$

Existence is not enough; we want to find q with few calls to the weak learner. Let the *data player* run Hedge to find distributions on which the current ensemble is weak, and let the weak learner best-respond.

Algorithm 7.6 (boosting by Hedge)

```

p_1 ← uniform over the N training points
for t = 1, ..., T:
    h_t ← WeakLearner(p_t)          # error under p_t ≤ ½ - γ/2
    for each i: p_{t+1,i} ∝ p_{t,i} · exp(η M_{i,h_t})
                                # up-weight points h_t got wrong, down-weight the rest
output majority vote of h_1, ..., h_T

```

Theorem 7.7

If $T > \frac{2 \ln N}{\gamma^2}$, the majority vote of $h_1, \dots, h_T$ classifies every training point correctly.

Proof. The data player gains $g_t = Me_{h_t} \in [-1, 1]^N$ and has regret at most $\sqrt{2T \ln N}$. So for every i ,

$$\frac{1}{T} \sum_t M_{i,h_t} \leq \frac{1}{T} \sum_t p_t^\top Me_{h_t} + \sqrt{\frac{2 \ln N}{T}} \leq -\gamma + \sqrt{\frac{2 \ln N}{T}} < 0.$$

The left side is (fraction of h_t wrong on x_i) minus (fraction right), so the majority is right on x_i . $\square$

AdaBoost is this algorithm with a round-dependent step $\eta_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$ (where ϵ_t is h_t 's weighted error) and a weighted rather than plain majority vote. It adapts to each round's actual edge, and its training error is at most $\prod_t \sqrt{1 - \gamma_t^2} \leq \exp(-\frac{1}{2} \sum_t \gamma_t^2)$ where $\gamma_t = 1 - 2\epsilon_t$. Generalisation beyond the training set follows from margin theory (Schapire, Freund, Bartlett and Lee 1998): the vote's large margin γ controls test error independently of T .

In finance

Boosted stumps and trees (AdaBoost, gradient boosting, XGBoost) are among the most common models for cross-sectional return prediction and credit scoring. The game view explains one of their failure modes in finance. The data player concentrates weight on the hardest examples, and in return data those are often outliers or mislabelled

events (a stock split read as a crash). Boosting will spend its capacity fitting them. Robust variants cap the weights (the “smooth boosting” of Servedio), which in game terms restricts the data player to smooth distributions.

Key takeaways

- Hedge solves LP feasibility to accuracy ε in $O(\rho^2 \ln n / \varepsilon^2)$ rounds, independent of the number of constraints.
- Weak learnability for every distribution means (by minimax) that one mixture of weak hypotheses is strong.
- Boosting is Hedge on training points against a best-responding weak learner. $O(\ln N / \gamma^2)$ rounds suffice.

Exercises

7.1 core Show that the converse of Theorem 7.5 holds: if some mixture q has margin γ on every point, then WLA_γ holds.

7.2 core Derive AdaBoost’s step size: with weights p_t and weak error ε_t , find the η that minimises the normaliser $Z_t = \sum_i p_{t,i} e^{\eta M_{i,h_t}}$ after flipping signs so that correct points are down-weighted, and show $\min_\eta Z_t = 2\sqrt{\varepsilon_t(1 - \varepsilon_t)}$.

7.3 finance Three states, a bond paying $(1, 1, 1)$ at price 1, a stock paying $(0.8, 1, 1.3)$ at price 1, and a call on the stock struck at 1 paying $(0, 0, 0.3)$. For which call prices c is there no arbitrage? Run two iterations of Algorithm 7.2 by hand with $c = 0.1$.

Solutions

7.1. For any p , $p^\top Mq = \sum_i p_i e_i^\top Mq \leq -\gamma$. Since $p^\top Mq$ is an average of $p^\top M e_h$ over $h \sim q$, some h has $p^\top M e_h \leq -\gamma$. So weak and strong learnability (with margins) are equivalent: this is minimax duality.

7.2. $Z(\eta) = \varepsilon_t e^\eta + (1 - \varepsilon_t) e^{-\eta}$. Setting $Z' = 0$: $e^{2\eta} = \frac{1 - \varepsilon_t}{\varepsilon_t}$, so $\eta = \frac{1}{2} \ln \frac{1 - \varepsilon_t}{\varepsilon_t}$ and $Z = 2\sqrt{\varepsilon_t(1 - \varepsilon_t)} = \sqrt{1 - \gamma_t^2}$. The training error is at most $\prod_t Z_t$ because the exponential loss upper-bounds the 0/1 loss.

7.3. We need $q \in \Delta_3$ with $0.8q_1 + q_2 + 1.3q_3 = 1$ and $0.3q_3 = c$. From the stock equation, $0.3q_3 = 0.2q_1$, so $q_1 = 1.5q_3$ and $q_2 = 1 - 2.5q_3$. Strict positivity needs $0 < q_3 < 0.4$, so $c = 0.3q_3 \in (0, 0.12)$. For $c = 0.1$: $q_3 = 1/3$, $q_1 = 1/2$, $q_2 = 1/6$. Starting from uniform q , the stock constraint $(-0.2, 0, 0.3)^\top q = 0.033 > 0$ is violated, so Hedge moves weight from state 3 to state 1, as the solution requires.

CHAPTER 8

The Perceptron

Based on Lectures 10–11 (Oct 7 and 9, 2013) · [notes 10](#) · [notes 11](#)

In this chapter

- Online linear classification and the margin.
- Two proofs of the $1/\gamma^2$ mistake bound, one of which previews online gradient descent.
- Using the Perceptron to solve feasibility problems.

8.1 Online linear classification

Examples $(x_t, y_t) \in \mathbb{R}^d \times \{-1, +1\}$ arrive one at a time. We predict $\hat{y}_t = \text{sgn}(w_t \cdot x_t)$ and then see y_t . The expert class is now continuous: all hyperplanes through the origin.

Definition 8.1 (margin)

Assume $\|x_t\|_2 \leq 1$. A unit vector w^* separates the data with margin $\gamma > 0$ if $y_t (w^* \cdot x_t) \geq \gamma$ for all t . Geometrically every point lies at distance at least γ from the hyperplane $\{w^* \cdot x = 0\}$, on the correct side.

Algorithm 8.2 (Perceptron, Rosenblatt 1957)

```
w_1 ← 0
for t = 1, ..., T:
    predict  $\hat{y}_t \leftarrow \text{sgn}(w_t \cdot x_t)$ 
    if  $y_t (w_t \cdot x_t) \leq 0$ :           # mistake
        w_{t+1} ← w_t + y_t x_t
    else:
        w_{t+1} ← w_t
```

Theorem 8.3 (Block 1962; Novikoff 1962)

If some unit w^* has margin γ and $\|x_t\| \leq 1$, the Perceptron makes at most $1/\gamma^2$ mistakes, on any sequence of any length.

Proof 1 (angle). Let M_t be the number of mistakes before round t . On a mistake round:

- $w_{t+1} \cdot w^* = w_t \cdot w^* + y_t x_t \cdot w^* \geq w_t \cdot w^* + \gamma$. So $w_t \cdot w^* \geq \gamma M_t$.
- $\|w_{t+1}\|^2 = \|w_t\|^2 + 2y_t w_t \cdot x_t + \|x_t\|^2 \leq \|w_t\|^2 + 1$, because $y_t w_t \cdot x_t \leq 0$ on a mistake. So $\|w_t\|^2 \leq M_t$.

By Cauchy–Schwarz, $\gamma M \leq w \cdot w^* \leq \|w\| \leq \sqrt{M}$, hence $M \leq 1/\gamma^2$. $\square$

Proof 2 (potential). Rescale $u = w^*/\gamma$, so $y_t u \cdot x_t \geq 1$ and $\|u\| = 1/\gamma$. Let $\Phi_t = \|w_t - u\|^2$. On a mistake,

$$\Phi_t - \Phi_{t+1} = 2y_t u \cdot x_t - 2y_t w_t \cdot x_t - \|x_t\|^2 \geq 2 - 0 - 1 = 1.$$

Φ never increases, starts at $\|u\|^2 = 1/\gamma^2$ and stays nonnegative, so there are at most $1/\gamma^2$ mistakes. $\square$

The bound does not depend on the dimension d or on T . It depends only on how “easy” the data are, measured by the margin. This is why the Perceptron works with kernels in infinite-dimensional feature spaces, where only inner products $x_s \cdot x_t = k(x_s, x_t)$ are ever needed.

8.2 The Perceptron as gradient descent

Proof 2 is a distance-to-comparator argument, the template of Chapter 9. Define the perceptron loss $\ell_t(w) = \max(0, -y_t w \cdot x_t)$. It is convex, with subgradient $-y_t x_t$ when $y_t w \cdot x_t < 0$ and 0 when $y_t w \cdot x_t > 0$ (at the kink, 0 and $-y_t x_t$ are both subgradients). The Perceptron is therefore online (sub)gradient descent with step size 1, choosing the subgradient $-y_t x_t$ at the kink $w \cdot x_t = 0$: $w_{t+1} = w_t - \nabla \ell_t(w_t)$. The mistake bound is an early instance of the regret bounds for gradient descent in Part III.

A caution from the lecture: after T rounds, w_{T+1} need not classify all of $x_1, \dots, x_T$ correctly. A late update can break an earlier example. To get a consistent separator from a batch, cycle through the data until a full pass makes no mistakes; the mistake bound guarantees this happens within $1/\gamma^2$ updates.

8.3 Feasibility via the Perceptron

Given $a_1, \dots, a_m \in \mathbb{R}^d$, find x with $a_i \cdot x > 0$ for all i . Let the “wiggle room” be $v^* = \sup_{\|x\|=1} \min_i \frac{a_i \cdot x}{\|a_i\|}$.

Proposition 8.4 (Homework 2, Problem 2)

Repeatedly pick any violated constraint i and update $x \leftarrow x + a_i / \|a_i\|$. If $v^* > 0$, a feasible x is found after at most $1/(v^*)^2$ updates.

This is the Perceptron on examples $(a_i / \|a_i\|, +1)$ with margin v^* . Compare with Chapter 7: multiplicative weights over the simplex give $\ln n / \varepsilon^2$ iterations, while additive updates over the Euclidean ball give $1/v^2$. The difference between the two is the choice of geometry (entropy versus squared norm), which Chapter 10 makes precise. Dunagan and Vempala (2004) turned this idea, plus a rescaling step, into a polynomial-time LP algorithm.

In finance

Perceptron-style online linear classifiers are a classic baseline for directional prediction

(“up or down tomorrow”) from a vector of features such as lagged returns, volume or sentiment scores. The theory also warns about applying them there: the mistake bound needs a margin, and directional daily returns are close to non-separable. For non-separable data the Perceptron’s mistakes are bounded by $\min_u (\|u\|^2 + 2 \sum_t \max(0, 1 - y_t u \cdot x_t))$ up to constants, i.e. by the hinge loss of the best linear rule, which in efficient markets is not much better than chance. The honest use of these methods is as online regressors of *small edges*, analysed with the convex-loss regret bounds of Part III rather than with mistake counts.

Key takeaways

- With margin γ , the Perceptron makes at most $1/\gamma^2$ mistakes, independent of dimension and horizon.
- The potential-function proof is the gradient-descent analysis of Part III in miniature.
- Additive (Euclidean) and multiplicative (entropic) updates are two geometries for the same kind of problem.

Exercises

- 8.1** warm-up Show that the $1/\gamma^2$ bound is tight: for $d \geq 1/\gamma^2$ construct a sequence forcing $1/\gamma^2$ mistakes. Hint: use $x_t = e_t$ and $w^* = \gamma \sum_t e_t$ for $t \leq 1/\gamma^2$.
- 8.2** core Extend Theorem 8.3 to $\|x_t\| \leq R$: show at most $(R/\gamma)^2$ mistakes.
- 8.3** core (Kernel Perceptron.) Show that $w_t = \sum_{s \in \mathcal{M}_t} y_s x_s$, where $\mathcal{M}_t$ is the set of mistake rounds before t , and hence that predictions need only kernel evaluations $k(x_s, x_t)$.
- 8.4** challenge (Non-separable case.) For any u , show the Perceptron’s mistake count M satisfies $M \leq \|u\|^2 + 2 \sum_{t \in \mathcal{M}} \max(0, 1 - y_t u \cdot x_t)$ when $\|x_t\| \leq 1$. Hint: redo Proof 2 without assuming $y_t u \cdot x_t \geq 1$.

Solutions

- 8.1.** With $k = 1/\gamma^2$ orthonormal examples, $w^* = \frac{1}{\sqrt{k}} \sum_{t \leq k} y_t e_t$ is a unit vector with margin $1/\sqrt{k} = \gamma$ for any labels. Before seeing $x_t = e_t$, the Perceptron’s w_t is orthogonal to e_t , so it predicts using $\text{sgn}(0)$ and the adversary chooses y_t to make it wrong. It errs k times.
- 8.2.** The norm step becomes $\|w_{t+1}\|^2 \leq \|w_t\|^2 + R^2$, so $\|w\| \leq R\sqrt{M}$ and $\gamma M \leq R\sqrt{M}$.
- 8.3.** By induction from $w_1 = 0$ and the update rule. Then $w_t \cdot x_t = \sum_{s \in \mathcal{M}_t} y_s x_s \cdot x_t$.
- 8.4.** On a mistake, $\Phi_t - \Phi_{t+1} \geq 2y_t u \cdot x_t - 1 \geq 1 - 2 \max(0, 1 - y_t u \cdot x_t)$. Summing over mistake rounds, $M - 2 \sum_{t \in \mathcal{M}} h_t \leq \Phi_1 = \|u\|^2$.

Part III

Online Convex Optimization

One framework that contains experts, classification and portfolios, and two algorithm families that solve it.

CHAPTER 9

Online Convex Optimization and Gradient Descent

Based on Lectures 12–14 and 17 (Oct 16 – Nov 4, 2013; Lecture 12 by guest lecturer Ambuj Tewari) · [notes 12](#) · [notes 14](#) · [notes 17](#)

In this chapter

- The OCO protocol and how it contains everything so far.
- Online gradient descent and its $DG\sqrt{T}$ regret; $O(\log T)$ with strong convexity.
- From regret to optimisation and to statistical risk (online-to-batch).

9.1 The protocol

“The great watershed in optimization isn’t between linearity and nonlinearity, but convexity and nonconvexity.” (R. T. Rockafellar)

Protocol 9.1 (online convex optimization, Zinkevich 2003)

A convex, closed, bounded decision set $\mathcal{K} \subseteq \mathbb{R}^d$. For $t = 1, \dots, T$: the learner plays $x_t \in \mathcal{K}$; the adversary reveals a convex $f_t : \mathcal{K} \rightarrow \mathbb{R}$; the learner pays $f_t(x_t)$. Regret against a comparator $u \in \mathcal{K}$ is

$$\text{Regret}_T(u) = \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(u), \quad \text{Regret}_T = \max_{u \in \mathcal{K}} \text{Regret}_T(u).$$

“Converging to the optimum” is meaningless here because the objective keeps changing. Regret is the replacement.

Problem	Decision set $\mathcal{K}$	Loss $f_t(x)$
Hedge / experts	Δ_N	$\langle \ell_t, x \rangle, \ell_t \in [0, 1]^N$
Prediction with experts	Δ_N	$\ell(\langle x, f_t \rangle, y_t)$, convex ℓ
Online regression	$\{w : \ w\ \leq W\}$	$\frac{1}{2}(\langle w, z_t \rangle - y_t)^2$
Online classification	$\{w : \ w\ \leq W\}$	hinge $\max(0, 1 - y_t \langle w, z_t \rangle)$ or logistic $\ln(1 + e^{-y_t \langle w, z_t \rangle})$
Portfolio selection	Δ_n	$-\ln \langle x, r_t \rangle$, price relatives $r_t \in \mathbb{R}_{>0}^n$
Zero-sum game	Δ_n	$x^\top M q_t$

A convex loss composed with a linear map is convex, so all rows are OCO problems.

9.2 Follow the Leader fails again

The obvious algorithm plays $x_t \in \arg \min_{x \in \mathcal{K}} \sum_{s \leq t} f_s(x)$. On $\mathcal{K} = [-1, 1]$ with $f_t(x) = c_t x$, $c_1 = -\frac{1}{2}$, then $c_t = +1, -1, +1, \dots$, the cumulative sum alternates between $-\frac{1}{2}$ and $+\frac{1}{2}$, so FTL jumps between $+1$ and -1 and always lands on the wrong side: it pays about T while $x = 0$ pays 0. FTL is unstable when losses are flat. Two cures follow: take small steps (this chapter), or add curvature (Chapter 10).

9.3 Online gradient descent

Algorithm 9.2 (online gradient descent, OGD)

```

input: step size  $\eta > 0$ ,  $x_1 \in K$ 
for  $t = 1, \dots, T$ :
    play  $x_t$ , observe  $f_t$ , compute  $g_t \in \partial f_t(x_t)$ 
     $y_{t+1} \leftarrow x_t - \eta g_t$ 
     $x_{t+1} \leftarrow \Pi_K(y_{t+1})$            # Euclidean projection onto  $K$ 

```

Only a (sub)gradient at one point is needed per round, so OGD is cheap. Projection onto a convex set has the property that makes the analysis work: it never increases distance to points of the set, $\|\Pi_K(y) - u\| \leq \|y - u\|$ for all $u \in \mathcal{K}$.

Theorem 9.3 (Zinkevich 2003)

Suppose $\|g_t\|_2 \leq G$ and $\text{diam}(\mathcal{K}) = \max_{x, y \in \mathcal{K}} \|x - y\|_2 \leq D$. Then for all $u \in \mathcal{K}$,

$$\text{Regret}_T(u) \leq \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}, \quad \text{and with } \eta = \frac{D}{G\sqrt{T}}, \quad \text{Regret}_T \leq DG\sqrt{T}.$$

Proof. Fix u . By non-expansiveness of the projection,

$$\|x_{t+1} - u\|^2 \leq \|x_t - \eta g_t - u\|^2 = \|x_t - u\|^2 - 2\eta \langle g_t, x_t - u \rangle + \eta^2 \|g_t\|^2.$$

Rearranging, $\langle g_t, x_t - u \rangle \leq \frac{\|x_t - u\|^2 - \|x_{t+1} - u\|^2}{2\eta} + \frac{\eta}{2} G^2$. By convexity, $f_t(x_t) - f_t(u) \leq \langle g_t, x_t - u \rangle$. Summing, the distance terms telescope to at most $\frac{\|x_1 - u\|^2}{2\eta} \leq \frac{D^2}{2\eta}$. $\square$

This is the Perceptron's potential argument (Chapter 8, Proof 2) with a general loss. Two remarks. First, the proof only used $\langle g_t, x_t - u \rangle$, so it bounds regret for the *linearised* losses $\langle g_t, x \rangle$, which dominates regret for the convex losses. Linear losses are therefore the hardest case, and much of the theory can assume them. Second, a time-varying step $\eta_t = \frac{D}{G\sqrt{t}}$ gives $\text{Regret}_T \leq \frac{3}{2} DG\sqrt{T}$ without knowing T .

The geometry matters

On the experts problem, $\mathcal{K} = \Delta_N$ has $D = \sqrt{2}$, but a loss vector in $[0, 1]^N$ has Euclidean norm up to $G = \sqrt{N}$. OGD gives $O(\sqrt{NT})$, exponentially worse in N than

Hedge's $O(\sqrt{T \ln N})$. On portfolios, $\nabla(-\ln\langle x, r \rangle) = -r/\langle x, r \rangle$ is unbounded near the boundary of the simplex, so Theorem 9.3 does not apply at all. Chapter 10 fixes the first problem by changing the geometry, and Chapter 11 fixes the second by exploiting curvature.

9.4 Curvature helps: logarithmic regret

Definition 9.4 (strong convexity)

f is α -strongly convex if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$ for all x, y . For twice-differentiable f this means $\nabla^2 f \succeq \alpha I$.

Theorem 9.5 (Hazan, Agarwal and Kale 2007)

If every f_t is α -strongly convex and $\|g_t\| \leq G$, OGD with $\eta_t = \frac{1}{\alpha t}$ has $\text{Regret}_T \leq \frac{G^2}{2\alpha}(1 + \ln T)$.

Proof. Strong convexity gives $f_t(x_t) - f_t(u) \leq \langle g_t, x_t - u \rangle - \frac{\alpha}{2} \|x_t - u\|^2$. Using the same one-step inequality with η_t ,

$$\sum_t (f_t(x_t) - f_t(u)) \leq \sum_t \|x_t - u\|^2 \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} - \frac{\alpha}{2} \right) + \frac{G^2}{2} \sum_t \eta_t,$$

with $1/\eta_0 := 0$. With $\eta_t = 1/(\alpha t)$ the bracket is exactly 0, and $\sum_t \frac{1}{\alpha t} \leq \frac{1 + \ln T}{\alpha}$. $\square$

Example 9.6 (online Gaussian mean estimation, Lecture 15)

Observe $z_t \in \mathbb{R}^d$ with $\|z_t\| \leq D$ and pay log loss under $\mathcal{N}(\mu_t, I)$, i.e. $f_t(\mu) = \frac{1}{2} \|z_t - \mu\|^2$ plus a constant; this is 1-strongly convex. FTL plays the running mean $\mu_t = \frac{1}{t-1} \sum_{s < t} z_s$, which is also OGD with $\eta_t = 1/t$. Theorem 9.5 or a direct calculation (Exercise 9.3) gives regret $O(D^2 \ln T)$ against the best fixed mean in hindsight.

9.5 From regret to optimisation and to statistics

Offline convex optimisation

To minimise a fixed convex F , feed OCO the linearisations $f_t(x) = F(x_t) + \langle \nabla F(x_t), x - x_t \rangle$. Then $f_t(x_t) = F(x_t)$ and $f_t \leq F$ by convexity, so for $\bar{x} = \frac{1}{T} \sum_t x_t$ and $x^* = \arg \min F$:

$$F(\bar{x}) \leq \frac{1}{T} \sum_t F(x_t) = \frac{1}{T} \sum_t f_t(x_t) \leq \frac{1}{T} \sum_t f_t(x^*) + \frac{\text{Regret}_T}{T} \leq F(x^*) + \frac{\text{Regret}_T}{T}.$$

With OGD this is the projected subgradient method with its classical $DG/\sqrt{T}$ rate.

Online-to-batch conversion

In statistical learning, examples $z_1, \dots, z_n$ are i.i.d. from an unknown $\mathcal{D}$, and we want small risk $r(w) = \mathbb{E}_{z \sim \mathcal{D}} \ell(w, z)$. Run OCO on $f_t(w) = \ell(w, z_t)$ and output $\bar{w} = \frac{1}{n} \sum_t w_t$.

Theorem 9.7 (Cesa-Bianchi, Conconi and Gentile 2004)

For every u , $\mathbb{E} r(\bar{w}) \leq r(u) + \mathbb{E} \frac{\text{Regret}_n}{n}$.

Proof. By Jensen, $r(\bar{w}) \leq \frac{1}{n} \sum_t r(w_t)$. Since w_t depends only on $z_{1:t-1}$ and z_t is a fresh draw, $\mathbb{E} r(w_t) = \mathbb{E} \ell(w_t, z_t)$. So $\mathbb{E} r(\bar{w}) \leq \mathbb{E} \frac{1}{n} \sum_t \ell(w_t, z_t) \leq \mathbb{E} \frac{1}{n} \sum_t \ell(u, z_t) + \mathbb{E} \frac{\text{Regret}_n}{n} = r(u) + \mathbb{E} \frac{\text{Regret}_n}{n}$. $\square$

A single pass of OGD over the data (which is stochastic gradient descent) therefore attains excess risk $O(DG/\sqrt{n})$, matching what uniform-convergence arguments give for empirical risk minimisation, with a one-line proof and no capacity arguments.

In finance

Online least squares is the backbone of adaptive hedging. To hedge a position with d instruments, choose hedge ratios w_t each day and pay $f_t(w) = \frac{1}{2}(\text{P\&L}_t^{\text{position}} + \langle w, \text{P\&L}_t^{\text{hedges}} \rangle)^2$. OGD adjusts the hedge ratios by a step against the gradient of yesterday's squared residual, an LMS filter. Theorem 9.3 guarantees that the cumulative squared tracking error is within $O(\sqrt{T})$ of the best fixed hedge ratios in hindsight, with no assumption that returns are stationary. If you add a ridge penalty $\frac{\lambda}{2} \|w\|^2$ to each loss, the losses become λ -strongly convex and Theorem 9.5 improves this to $O(\ln T)$ against the best ridge-penalised hedge.

Key takeaways

- OCO unifies experts, classification, regression, portfolios and games.
- OGD: $\text{Regret}_T \leq DG\sqrt{T}$. With α -strongly convex losses and $\eta_t = 1/(\alpha t)$: $\frac{G^2}{2\alpha}(1 + \ln T)$.
- Regret bounds give optimisation rates and statistical risk bounds for free.
- Euclidean geometry is wrong for the simplex. That is the motivation for Chapter 10.

Exercises

9.1 warm-up Show that for a closed convex $\mathcal{K}$ and $u \in \mathcal{K}$, $\|\Pi_{\mathcal{K}}(y) - u\| \leq \|y - u\|$. Hint: the optimality condition for the projection $p = \Pi_{\mathcal{K}}(y)$ is $\langle y - p, u - p \rangle \leq 0$ for all $u \in \mathcal{K}$.

9.2 core Prove the $\frac{3}{2}DG\sqrt{T}$ bound for OGD with $\eta_t = \frac{D}{G\sqrt{t}}$.

9.3 core For Example 9.6, use Lemma 10.1 (the FTL–BTL lemma in the next chapter) and the identity $z_t - \mu_{t+1} = (1 - \frac{1}{t})(z_t - \mu_t)$ to show $\text{regret} \leq \sum_t \frac{\|z_t - \mu_t\|^2}{t} \leq 4D^2(1 + \ln T)$.

9.4 challenge (Lecture 16 open problem, dynamic regret.) Show that OGD with fixed η satisfies, against any comparator sequence $u_1, \dots, u_T$, $\sum_t f_t(x_t) - f_t(u_t) \leq \frac{7D^2}{4\eta} + \frac{D}{\eta} \sum_t \|u_{t+1} - u_t\| + \frac{\eta G^2 T}{2}$ (Zinkevich 2003). What η should you use if the path length is P ?

Solutions

9.1. $\|y - u\|^2 = \|y - p\|^2 + \|p - u\|^2 + 2\langle y - p, p - u \rangle \geq \|p - u\|^2$, because the cross term is $-2\langle y - p, u - p \rangle \geq 0$.

9.2. Summing the one-step inequality with varying η_t : $\sum_t \frac{\|x_t - u\|^2 - \|x_{t+1} - u\|^2}{2\eta_t} \leq \frac{D^2}{2} \sum_t \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) = \frac{D^2}{2\eta_T} = \frac{DG\sqrt{T}}{2}$ by Abel summation, and $\frac{G^2}{2} \sum_t \eta_t \leq \frac{G^2}{2} \cdot \frac{D}{G} \cdot 2\sqrt{T} = DG\sqrt{T}$.

9.3. $\text{Regret} \leq \sum_t [f_t(\mu_t) - f_t(\mu_{t+1})] = \frac{1}{2} \sum_t \left(1 - \left(1 - \frac{1}{t} \right)^2 \right) \|z_t - \mu_t\|^2 \leq \sum_t \frac{1}{t} \|z_t - \mu_t\|^2$, and $\|z_t - \mu_t\| \leq 2D$.

9.4. Repeat the proof with u_t in place of u ; the telescoping now leaves $\frac{1}{2\eta} \sum_t (\|x_{t+1} - u_t\|^2 - \|x_{t+1} - u_{t+1}\|^2) \leq \frac{1}{2\eta} \sum_t 2D \|u_{t+1} - u_t\|$ plus boundary terms. Balancing gives $\eta \propto \sqrt{D(D+P)/(G^2T)}$ and regret $O(G\sqrt{D(D+P)T})$. Compare Fixed Share (Chapter 4).

Follow the Regularized Leader

Based on Lectures 12, 15–17 (Oct 16 – Nov 4, 2013) · [notes 15](#) · [notes 16](#) · [notes 17](#)

In this chapter

- The Be-the-Leader lemma: why FTL’s regret is controlled by its stability.
- FTRL: stabilise FTL with a strongly convex regularizer. Quadratic gives OGD; entropy gives Hedge.
- Bregman divergences and the general regret bound.
- Follow the Perturbed Leader: regularization by randomness.

10.1 Be the Leader

Consider a “cheating” algorithm that gets to see f_t before choosing: $\hat{x}_t \in \arg \min_x \sum_{s \leq t} f_s(x)$. Note $\hat{x}_t = x_{t+1}$, the next FTL iterate.

Lemma 10.1 (*Be the Leader; Kalai and Vempala 2005*)

- (a) For every $u \in \mathcal{K}$, $\sum_{t=1}^T f_t(x_{t+1}) \leq \sum_{t=1}^T f_t(u)$.
 (b) Hence $\text{Regret}_T(\text{FTL}) \leq \sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1}))$.

Proof. (a) Induct on T . For $T = 1$, x_2 minimises f_1 . Assuming the claim for $T - 1$, apply it with $u = x_{T+1}$ and add $f_T(x_{T+1})$:

$$\sum_{t=1}^T f_t(x_{t+1}) \leq \sum_{t=1}^T f_t(x_{T+1}) = \min_u \sum_{t=1}^T f_t(u).$$

- (b) Subtract (a) from $\sum_t f_t(x_t)$. □

FTL has low regret exactly when it is *stable*, i.e. consecutive leaders are close. On flat (linear) losses the leader can jump across the whole domain in one step; on curved losses it cannot. That was the moral of Example 9.6.

10.2 Adding a regularizer

Algorithm 10.2 (*Follow the Regularized Leader, FTRL*)

Given a convex regularizer $R : \mathcal{K} \rightarrow \mathbb{R}$ and $\eta > 0$, play

$$x_t = \arg \min_{x \in \mathcal{K}} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} R(x) \right\}.$$

FTRL is FTL on the sequence $f_0 = R/\eta, f_1, f_2, \dots$ with a fictitious round-0 loss. Lemma 10.1 applied to that sequence gives the master bound.

Theorem 10.3 (FTRL master bound)

$$\text{Regret}_T(u) \leq \frac{R(u) - R(x_1)}{\eta} + \sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1})).$$

The regularizer costs a *range term* $\frac{R(u) - \min R}{\eta}$ but buys a *stability term*: consecutive iterates of a strongly convex objective cannot move far. By convexity and Hölder's inequality,

$$f_t(x_t) - f_t(x_{t+1}) \leq \langle g_t, x_t - x_{t+1} \rangle \leq \|g_t\|_* \|x_t - x_{t+1}\|,$$

where $\|\cdot\|_*$ is the dual norm. If R is 1-strongly convex with respect to $\|\cdot\|$, one can show $\|x_t - x_{t+1}\| \leq \eta \|g_t\|_*$ (Exercise 10.3). This gives the general theorem:

Theorem 10.4

If R is 1-strongly convex with respect to a norm $\|\cdot\|$ on $\mathcal{K}$, the losses are convex with $\|g_t\|_* \leq G$, and $R(u) - \min_{\mathcal{K}} R \leq B^2$, then

$$\text{Regret}_T \leq \frac{B^2}{\eta} + \eta G^2 T = 2BG\sqrt{T} \quad \text{at } \eta = \frac{B}{G\sqrt{T}}.$$

The whole game is to choose a norm in which the losses are small and the regularizer's range is small. Two choices recover our earlier algorithms.

Regularizer R	Norm pair	Range B^2	Gradient bound G	Algorithm	Regret
$\frac{1}{2}\ x\ _2^2$	ℓ_2 / ℓ_2	$D^2/2$	$\ g\ _2$	lazy OGD	$O(DG_2\sqrt{T})$
$\sum_i x_i \ln x_i$ on Δ_N	ℓ_1 / ℓ_∞	$\ln N$	$\ g\ _\infty \leq 1$	Hedge	$O(\sqrt{T \ln N})$
$-\sum_i \ln x_i$ (log-barrier)	local norms	$O(N \ln T)$ after shrinking	scale-free	used for bandits & portfolios	see Ch. 11, 14

Quadratic regularizer: OGD again

With linear losses $f_t(x) = \langle g_t, x \rangle$ and $R(x) = \frac{1}{2}\|x - x_1\|^2$, FTRL is $x_{t+1} = \Pi_{\mathcal{K}}(x_1 - \eta \sum_{s \leq t} g_s)$. This is the “lazy” projection variant of OGD (also called dual averaging). Unconstrained, it coincides with OGD; with projections the two differ slightly but share the same bound. With linear losses and no constraint, the stability term is exactly $\langle g_t, x_t - x_{t+1} \rangle = \eta \|g_t\|^2$, so $\text{Regret}_T(u) \leq \frac{\|u\|^2}{2\eta} + \eta \sum_t \|g_t\|^2$.

Entropic regularizer: Hedge again

On Δ_N with $f_t(x) = \langle \ell_t, x \rangle$ and $R(x) = \sum_i x_i \ln x_i$, the Lagrangian condition $L_{i,t-1} + \frac{1}{\eta}(1 + \ln x_i) + \lambda = 0$ gives $x_{t,i} \propto e^{-\eta L_{i,t-1}}$: exactly Hedge. Negative entropy is 1-strongly convex with respect to $\|\cdot\|_1$ on the simplex (Pinsker's inequality), its range is $\ln N$, and $\|\ell_t\|_\infty \leq 1$. Theorem 10.4 gives $2\sqrt{T \ln N}$, within a constant of Theorem 3.8. This is the answer to “why did Hedge beat OGD on the simplex?”: the simplex is small in ℓ_1 , and losses are small in ℓ_∞ .

10.3 Bregman divergences

Definition 10.5 (Bregman divergence)

For a differentiable convex R , $D_R(x, y) = R(x) - R(y) - \langle \nabla R(y), x - y \rangle$: the gap between R and its tangent plane at y , evaluated at x .

- $D_R \geq 0$, with equality iff $x = y$ when R is strictly convex. It is generally not symmetric.
- $R = \frac{1}{2}\|x\|^2$: $D_R(x, y) = \frac{1}{2}\|x - y\|^2$.
- $R = \sum_i x_i \ln x_i$ on the simplex: $D_R(x, y) = \text{KL}(x \parallel y) = \sum_i x_i \ln \frac{x_i}{y_i}$.
- Locally $D_R(x, y) \approx \frac{1}{2}(x - y)^\top \nabla^2 R(y)(x - y)$.

Theorem 10.6 (FTRL with linear losses, Bregman form)

For linear losses and unconstrained (or interior) iterates, with $x_1 = \arg \min R$,

$$\begin{aligned} \text{Regret}_T(u) &= \frac{1}{\eta} \left(R(u) - R(x_1) - D_R(u, x_{T+1}) + \sum_{t=1}^T D_R(x_t, x_{t+1}) \right) \\ &\leq \frac{1}{\eta} \left(R(u) - R(x_1) + \sum_{t=1}^T D_R(x_t, x_{t+1}) \right). \end{aligned}$$

The identity form (Exercise 10.4) is illuminating: regret equals the regularizer's range plus the “movement” of the iterates measured in the regularizer's own geometry. It is the online analogue of mirror descent (Nemirovski and Yudin 1983; Beck and Teboulle 2003), and for linear losses the two algorithm families coincide.

10.4 Follow the Perturbed Leader

Instead of a deterministic regularizer, randomise the cumulative losses and follow the leader of the perturbed sequence. This needs only a linear optimisation oracle over $\mathcal{K}$, which is valuable when $\mathcal{K}$ is combinatorial (paths, spanning trees, rankings).

Algorithm 10.7 (FTPL, Kalai and Vempala 2005; Homework 2)

```
draw X uniform on [0, b]^N          # once, at the start
for t = 1, ..., T:
    play p_t ← argmin_{p ∈ Δ_N} p · (X + Σ_{s<t} ℓ_s)
```

Theorem 10.8

Against an oblivious adversary, FTPL with $b = \sqrt{NT}$ has $\mathbb{E} \text{Regret}_T \leq 2\sqrt{NT}$.

- Proof sketch (Homework 2, Problem 1).*
1. The “be the perturbed leader” algorithm $\tilde{p}_t = \arg \min p \cdot (X + \sum_{s \leq t} \ell_s)$ has regret at most b , by Lemma 10.1 applied with $f_0 = X$.
 2. FTPL’s regret is BTPL’s regret plus $\sum_t (p_t - \tilde{p}_t) \cdot \ell_t$.
 3. $X + L_{t-1}$ and $X + L_t$ are uniform on two cubes of side b offset by $\ell_t \in [0, 1]^N$. They overlap in all but a fraction $\leq N/b$ of their volume, and where they overlap the argmin is the same, so $\mathbb{E}[(p_t - \tilde{p}_t) \cdot \ell_t] \leq N/b$.
 4. Total: $b + NT/b$, minimised at $b = \sqrt{NT}$.

□

With better perturbations the dependence on N improves to $\sqrt{T \ln N}$. In fact, perturbing each cumulative loss by an independent Gumbel variable with scale $1/\eta$ makes the leader’s distribution exactly the Hedge distribution (the Gumbel-max trick), so Hedge *is* a Follow the Perturbed Leader algorithm. Randomness and regularization are two ways of producing the same stability.

In finance

The regularizer is a choice of what “a small change in the portfolio” means. Quadratic regularization penalises trades in absolute weight, which is reasonable for long-short books with roughly linear transaction costs. Entropic regularization penalises trades relative to current weights (moving from 1% to 2% costs as much as from 20% to 40%), which suits long-only allocations across many assets and keeps every asset’s weight positive. The log-barrier penalises approaching zero weight very heavily, which suits problems where the loss blows up at the boundary, such as log-wealth. FTRL’s $1/\eta$ is literally a turnover budget: smaller η means less trading and higher range cost; larger η means more trading and more exposure to noise.

Key takeaways

- FTL’s regret is bounded by its instability $\sum_t f_t(x_t) - f_t(x_{t+1})$.
- FTRL trades a range cost $\frac{R(u) - \min R}{\eta}$ for stability. The general bound is $2BG\sqrt{T}$ with the norm pair matched to the problem.
- Quadratic regularizer gives OGD; entropy gives Hedge; the log-barrier underlies bandit and portfolio methods.
- Random perturbations are an alternative regularizer (FTPL), and with Gumbel noise they reproduce Hedge exactly.

Exercises

10.1 warm-up (Lecture 16 exercise.) Show that FTRL with negative-entropy regularizer and linear losses on Δ_N is exactly EWA with rate η .

10.2 warm-up Compute $D_R(x, y)$ for $R(x) = \sum_i x_i \ln x_i - x_i$ on $\mathbb{R}_{>0}^N$ (unnormalised entropy).

10.3 core Prove the stability lemma used for Theorem 10.4: if $F_t(x) = \sum_{s \leq t} f_s(x) + R(x)/\eta$ and R is 1-strongly convex in $\|\cdot\|$, then $\|x_t - x_{t+1}\| \leq \eta \|g_t\|_*$ for linear f_t .

10.4 challenge Prove the identity in Theorem 10.6. Hint: with linear losses, $\nabla R(x_{t+1}) = \nabla R(x_t) - \eta g_t$ in the unconstrained case, and use the three-point identity $D_R(u, x_t) - D_R(u, x_{t+1}) = \langle \nabla R(x_{t+1}) - \nabla R(x_t), u - x_{t+1} \rangle + D_R(x_{t+1}, x_t)$.

10.5 challenge (Homework 2(h).) Show that if $G_1, \dots, G_N$ are i.i.d. standard Gumbel, then $\Pr [i = \arg \min_j (L_j - G_j/\eta)] = \frac{e^{-\eta L_i}}{\sum_j e^{-\eta L_j}}$.

Solutions

10.1. See §10.2. The minimiser is interior because entropy's gradient blows up at the boundary, so only the equality constraint's multiplier λ is needed, and normalisation fixes λ .

10.2. $\nabla R(y) = \ln y$. $D_R(x, y) = \sum_i (x_i \ln \frac{x_i}{y_i} - x_i + y_i)$, the generalised KL divergence. On the simplex the last two terms cancel.

10.3. F_t is $\frac{1}{\eta}$ -strongly convex, minimised at x_t , so $F_t(x_{t+1}) \geq F_t(x_t) + \frac{1}{2\eta} \|x_{t+1} - x_t\|^2$. Likewise $F_{t+1}(x_t) \geq F_{t+1}(x_{t+1}) + \frac{1}{2\eta} \|x_t - x_{t+1}\|^2$. Adding and using $F_{t+1} = F_t + f_t$: $f_t(x_t) - f_t(x_{t+1}) \geq \frac{1}{\eta} \|x_t - x_{t+1}\|^2$. The left side is at most $\|g_t\|_* \|x_t - x_{t+1}\|$. Divide.

10.5. Equivalently $i = \arg \max_j (-\eta L_j + G_j)$. With Gumbel CDF $F(g) = e^{-e^{-g}}$, $\Pr[\arg \max = i] = \int f(g) \prod_{j \neq i} F(g + a_i - a_j) dg$ where $a_j = -\eta L_j$. Substituting $u = e^{-g}$ turns this into $\int_0^\infty e^{-u \sum_j e^{a_j - a_i}} du = \frac{e^{a_i}}{\sum_j e^{a_j}}$.

Part IV

Finance

Portfolios that compete with the best rebalancing rule in hindsight, and prices that come from hedging games rather than from probability models.

Universal Portfolios

Based on Lectures 11 and 13 (Oct 9 and 21, 2013) · [notes 11](#) · [notes 13](#)

In this chapter

- Why maximise log-wealth: the Kelly criterion from betting on horses.
- Constant-rebalanced portfolios, and why the best one can beat every stock.
- Cover's universal portfolio: within a factor T^{n-1} of the best CRP's wealth on every market sequence.
- Faster methods: exponentiated gradient and Online Newton Step.

11.1 Betting on horses: why log-wealth

m horses run a race. Horse i wins with probability p_i and pays odds o_i (a unit bet returns o_i if it wins). We bet a fraction b_i of wealth on each horse, $b \in \Delta_m$. After one race wealth is multiplied by $b_I o_I$, where I is the winner.

Maximising *expected* wealth, $\sum_i p_i o_i b_i$, puts everything on $\arg \max_i p_i o_i$, and after enough races we are bankrupt with probability tending to one. Over many independent races wealth grows like $\exp(n \mathbb{E} \ln(b_I o_I))$, so the long-run growth rate is the expected *log* return:

$$\mathbb{E} \ln(b_I o_I) = \sum_i p_i \ln o_i + \sum_i p_i \ln b_i = \sum_i p_i \ln o_i - H(p) - \text{KL}(p \parallel b).$$

Proposition 11.1 (Kelly 1956)

The growth-optimal bet is $b^* = p$: bet in proportion to the win probabilities, whatever the odds. The growth rate falls short of the optimum by exactly $\text{KL}(p \parallel b)$ when betting b .

Betting is prediction under log loss. The regret of a betting strategy, measured in log-wealth, is the log-loss regret of a forecaster (Exercise 3.5). The rest of the chapter extends this from horses (exactly one asset pays) to stocks (all assets pay something).

11.2 Constant-rebalanced portfolios

Definition 11.2

There are n assets. The *price relative* of asset i on day t is $r_{t,i} = \frac{\text{price}_{t+1}(i)}{\text{price}_t(i)} > 0$. A portfolio $x_t \in \Delta_n$ multiplies wealth by $\langle x_t, r_t \rangle$. A **constant-rebalanced portfolio** CRP(x) trades back to the same weights x each day, so

$$W_T(x) = \prod_{t=1}^T \langle x, r_t \rangle.$$

Log-wealth regret against the best CRP is $\text{Regret}_T = \max_{x \in \Delta_n} \ln W_T(x) - \sum_t \ln \langle x_t, r_t \rangle$. This is OCO with the loss $f_t(x) = -\ln \langle x, r_t \rangle$.

Example 11.3 (volatility pumping)

Cash has $r_{t,1} = 1$. A volatile stock alternates $r_{t,2} = 2, \frac{1}{2}, 2, \frac{1}{2}, \dots$. Each asset alone ends where it started. The $(\frac{1}{2}, \frac{1}{2})$ CRP multiplies wealth by $\frac{1}{2}(1+2) = 1.5$ on up days and $\frac{1}{2}(1+\frac{1}{2}) = 0.75$ on down days: 1.125 every two days, so its wealth grows as $1.125^{T/2}$, exponentially. Rebalancing sells after rises and buys after falls, harvesting volatility. The best CRP in hindsight can therefore beat the best single stock by an exponential factor, which makes it a far more demanding benchmark than “the best expert”.

The best CRP is a natural benchmark for a second reason: if returns are i.i.d., the best CRP is the growth-optimal (log-optimal) portfolio among *all* non-anticipating strategies (Breiman 1961; Algoet and Cover 1988). Matching it on every sequence therefore means matching the Kelly investor who knows the distribution, without knowing it.

11.3 Cover’s universal portfolio

Cover’s idea is buy-and-hold over experts. Split initial wealth uniformly across all CRPs $x \in \Delta_n$, and let each sub-account run on its own with no transfers between them. Total wealth is the average:

$$W_T(\text{UP}) = \frac{\int_{\Delta_n} W_T(x) dx}{\int_{\Delta_n} dx}.$$

Equivalently, on day t hold the wealth-weighted average of all CRPs:

$$x_t = \frac{\int_{\Delta_n} x W_{t-1}(x) dx}{\int_{\Delta_n} W_{t-1}(x) dx}.$$

This is exactly EWA with $\eta = 1$ over a continuum of experts under log loss, i.e. Bayesian averaging, as in Exercise 3.5.

Theorem 11.4 (Cover 1991)

For every sequence of price relatives and every $x^* \in \Delta_n$,

$$\ln W_T(x^*) - \ln W_T(\text{UP}) \leq (n-1) \ln(T+1) + 1.$$

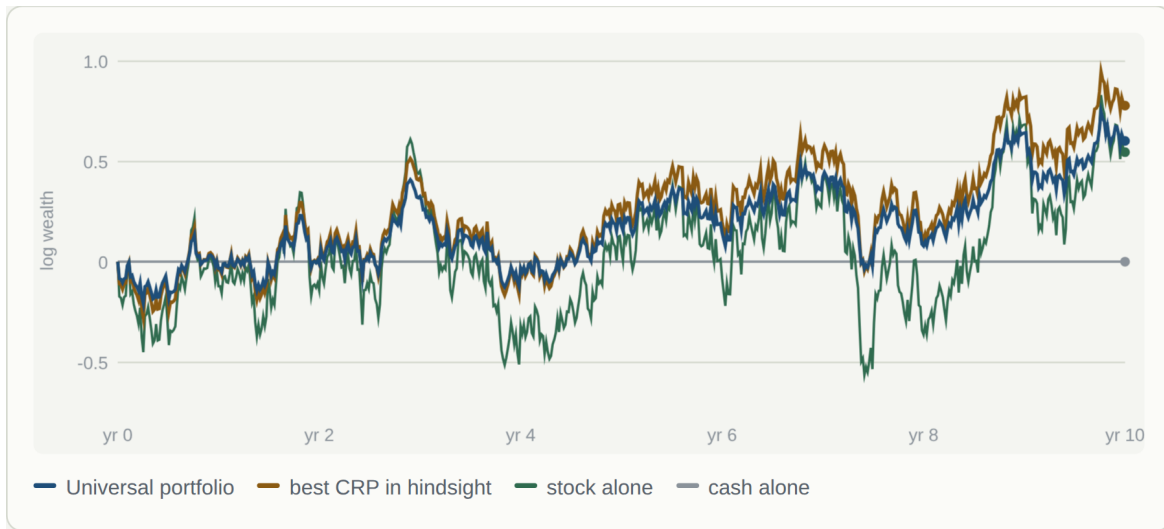
Proof. Fix x^* and $\varepsilon \in (0, 1)$. Consider the shrunken simplex $B_\varepsilon = \{(1-\varepsilon)x^* + \varepsilon v : v \in \Delta_n\}$.

1. *Volume.* B_ε is a copy of Δ_n scaled by ε in $n-1$ dimensions, so $\text{vol}(B_\varepsilon) = \varepsilon^{n-1} \text{vol}(\Delta_n)$.
2. *Wealth.* For $x = (1-\varepsilon)x^* + \varepsilon v \in B_\varepsilon$, $\langle x, r_t \rangle \geq (1-\varepsilon)\langle x^*, r_t \rangle$ because $\langle v, r_t \rangle > 0$. So $W_T(x) \geq (1-\varepsilon)^T W_T(x^*)$.

Hence $W_T(\text{UP}) \geq \frac{1}{\text{vol}(\Delta_n)} \int_{B_\varepsilon} W_T(x) dx \geq \varepsilon^{n-1} (1 - \varepsilon)^T W_T(x^*)$. Take $\varepsilon = \frac{1}{T+1}$. Then $(1 - \varepsilon)^T = (1 + \frac{1}{T})^{-T} \geq e^{-1}$, and taking logs gives the claim. $\square$

Several features stand out:

- No assumption on the returns at all: not bounded, not stationary, not independent. The only thing used is $r_t > 0$.
- No tuning parameter. ε appears only in the analysis.
- Regret is logarithmic in T , much better than $\sqrt{T}$. The reason is the curvature of log loss, which Section 11.4 makes explicit. Per day, the growth rate of UP is within $\frac{(n-1) \ln T}{T} \rightarrow 0$ of the best CRP's.
- Using the Dirichlet($\frac{1}{2}, \dots, \frac{1}{2}$) prior instead of the uniform one improves the constant to $\frac{n-1}{2} \ln T + O(1)$, which is minimax optimal (Cover and Ordentlich 1996).
- The cost is computation: an integral over the simplex each day. Kalai and Vempala (2002) showed it can be approximated in polynomial time by sampling with a random walk. Transaction costs can be handled (Blum and Kalai 1999).



Two assets over ten years of daily data (2,520 days): cash, and a volatile stock whose log-price fluctuates around a trend with the chosen annual growth (it mean-reverts with a half-life of about four months). The universal portfolio is computed exactly by averaging all constant-rebalanced portfolios on a grid of 201 stock weights. With zero trend the stock ends roughly where it started, yet rebalanced portfolios grow steadily: that is volatility pumping, and the universal portfolio captures most of it without knowing the best mix. Press “New market” for other paths. (*A static snapshot at the default settings; the HTML edition has live sliders.*)

11.4 Faster algorithms

Exponentiated gradient

Helmhold, Schapire, Singer and Warmuth (1998) apply the entropic mirror step to the linearised loss. With gradient $\nabla f_t(x_t) = -r_t / \langle x_t, r_t \rangle$ the update is

$$x_{t+1,i} \propto x_{t,i} \exp\left(\eta \frac{r_{t,i}}{\langle x_t, r_t \rangle}\right).$$

It costs $O(n)$ per day. If all price relatives lie in $[c, C]$ with $c > 0$, the gradients are bounded by C/c , and Theorem 10.4 gives regret $O(\frac{C}{c}\sqrt{T \ln n})$. The price for speed is a $\sqrt{T}$ rate and a dependence on how extreme the returns can be.

Exp-concavity and Online Newton Step

Definition 11.5 (exp-concave)

f is α -exp-concave if $e^{-\alpha f}$ is concave. The portfolio loss $f_t(x) = -\ln\langle x, r_t \rangle$ is 1-exp-concave, since $e^{-f_t} = \langle x, r_t \rangle$ is linear.

Exp-concavity is weaker than strong convexity (it gives curvature only in the direction of the gradient) but it is enough for logarithmic regret. Online Newton Step (Hazan, Agarwal and Kale 2007) is OGD preconditioned by the running sum of gradient outer products $A_t = \varepsilon I + \sum_{s \leq t} g_s g_s^\top$:

$$x_{t+1} = \Pi_{\Delta_n}^{A_t} \left(x_t - \frac{1}{\gamma} A_t^{-1} g_t \right),$$

where the projection is in the norm $\|y\|_{A_t} = \sqrt{y^\top A_t y}$. For α -exp-concave losses with bounded gradients it has regret $O((\frac{1}{\alpha} + GD) n \ln T)$ at $O(n^2)$ cost per step, plus the projection. For portfolios it needs a lower bound on price relatives (or a mixing step with the uniform portfolio). Cover's theorem is thus one instance of a general principle: *curvature buys logarithmic regret*.

Algorithm	Regret vs best CRP	Cost per day	Assumptions
Universal portfolio	$(n-1) \ln T + O(1)$	integral over Δ_n	none
Exponentiated gradient	$O(\frac{C}{c} \sqrt{T \ln n})$	$O(n)$	$r_{t,i} \in [c, C]$
Online Newton Step	$O(n \ln T)$ up to constants	$O(n^2)$ plus projection	bounded gradients

Reality check

The best CRP is a benchmark, not a strategy. Rebalancing daily into volatile, uncorrelated assets does pump volatility, but real markets have transaction costs, and the best CRP over real stock data is often concentrated in a few assets that happened to do well. Studies of these algorithms on historical data (e.g. Borodin, El-Yaniv and Gogan 2004; Li and Hoi 2014) find that the universal portfolio typically tracks a diversified CRP closely but rarely beats simple benchmarks after costs. The guarantee is about *robustness*: you never do much worse than the best rebalancing rule, whatever happens.

Key takeaways

- Log-wealth is the right objective for repeated investment; the Kelly bet is $b^* = p$.
- The best constant-rebalanced portfolio can beat every asset exponentially.
- Cover's universal portfolio is EWA over all CRPs. Its regret $(n-1)\ln T + O(1)$ holds on every sequence with no parameters.
- Exp-concavity of $-\ln\langle x, r \rangle$ is the reason for logarithmic regret.

Exercises

11.1 **finance** Three horses with win probabilities $(0.5, 0.3, 0.2)$ and odds $(1.8, 3, 6)$. (a) Is there an arbitrage? (b) Find the Kelly bet and its growth rate. (c) What is the growth rate of betting everything on the horse with the highest expected payoff?

11.2 **finance** Continuing 11.1, allow keeping a fraction c in cash. Show that the optimal strategy bets only on horse 3, with fraction f solving $\max_f 0.2\ln(1-f+6f) + 0.8\ln(1-f)$, and find f .

11.3 **core** For Example 11.3, find the best CRP $x = (1-\theta, \theta)$ in hindsight and its growth per two days.

11.4 **core** Show that $f(x) = -\ln\langle x, r \rangle$ is convex but not strongly convex on Δ_n , and verify that it is 1-exp-concave.

11.5 **challenge** Show that the universal portfolio's wealth equals $\mathbb{E}_{x \sim \text{Unif}(\Delta_n)}[W_T(x)]$, and use Laplace's method to argue that when the best CRP x^* is interior and $\ln W_T(x)$ is strongly concave with curvature $\propto T$ near x^* , the regret is $\frac{n-1}{2} \ln T + O(1)$ rather than $(n-1)\ln T$.

Solutions

11.1. (a) $\sum 1/o_i = 0.556 + 0.333 + 0.167 = 1.056 \geq 1$, so no sure profit from betting on every horse. (b) $b^* = (0.5, 0.3, 0.2)$. Growth rate $= 0.5\ln 0.9 + 0.3\ln 0.9 + 0.2\ln 1.2 = 0.8\ln 0.9 + 0.2\ln 1.2 = -0.0843 + 0.0365 = -0.048$ per race. Negative: with these odds you should not bet at all if you may keep cash. (c) Expected payoffs 0.9, 0.9, 1.2: all on horse 3 has growth rate $-\infty$ (bankruptcy with probability 0.8 each race). Adding a cash option, the Kelly bet becomes a mix of cash and horse 3 only; finding it is Exercise 11.2.

11.2. Horses 1 and 2 have $p_i o_i < 1$, so money on them is better kept in cash. Betting f on horse 3 multiplies wealth by $1+5f$ if it wins and $1-f$ otherwise. Setting the derivative to zero, $\frac{0.2 \cdot 5}{1+5f} = \frac{0.8}{1-f}$, so $1-f = 0.8+4f$ and $f = 0.04$. The Kelly fraction is 4%, matching the classic formula $f = \frac{p \cdot o - 1}{o - 1} = \frac{0.2}{5}$. The growth rate is $0.2\ln 1.2 + 0.8\ln 0.96 \approx 0.0038$ per race: small but positive.

11.3. Maximise $(1+\theta)(1-\theta/2)$: derivative $1-\theta/2 - (1+\theta)/2 = \frac{1}{2} - \theta = 0$, $\theta = \frac{1}{2}$, growth factor 1.125 per two days.

11.4. $\nabla^2 f = \frac{rr^\top}{\langle x, r \rangle^2} \succeq 0$ has rank one, so it is flat in every direction orthogonal to r : convex, not strongly convex. $e^{-f} = \langle x, r \rangle$ is linear, hence concave.

CHAPTER 12

Probability as a Game, and Pricing Options

Based on Lecture 14 (Oct 23, 2013; with Rafael Frongillo) · [scribe notes](#) · Shafer and Vovk, *Probability and Finance: It's Only a Game!* (2001), ch. 1

In this chapter

- No-arbitrage pricing by replication on one- and two-period trees.
- Shafer and Vovk's game-theoretic probability: upper and lower prices defined by hedging, with no probability measure.
- A strong law of large numbers proved by a betting strategy, which turns out to be a universal portfolio.
- Regret bounds as option-price bounds, and Black–Scholes as the value of a game.

The dollar figures in the scribed examples for this lecture did not survive in the PDF. The numbers below are our own; they illustrate the same three phenomena (unique price, price interval, dynamic hedging restoring a unique price).

12.1 Pricing by replication

Interest rates are zero throughout. A stock trades at $S_0 = 100$.

Example 12.1 (one period, two outcomes)

Tomorrow $S_1 \in \{80, 120\}$. A call option struck at 100 pays $(S_1 - 100)^+ \in \{0, 20\}$. Hold Δ shares and c in cash, so tomorrow's value is $c + \Delta(S_1 - 100)$ (buying shares at 100 costs nothing net if financed by borrowing). Matching both outcomes:

$$c + \Delta(80 - 100) = 0, \quad c + \Delta(120 - 100) = 20 \quad \implies \quad \Delta = \frac{1}{2}, \quad c = 10.$$

The option is worth exactly 10. At a price of 12 we would sell it, put 10 into the replicating portfolio and pocket 2 for sure; at 8 we would do the reverse. The price 10 is also $\mathbb{E}_q[(S_1 - 100)^+]$ under the unique "risk-neutral" q with $\mathbb{E}_q[S_1] = S_0$, here $q = (\frac{1}{2}, \frac{1}{2})$. The real-world probability of an up-move plays no role.

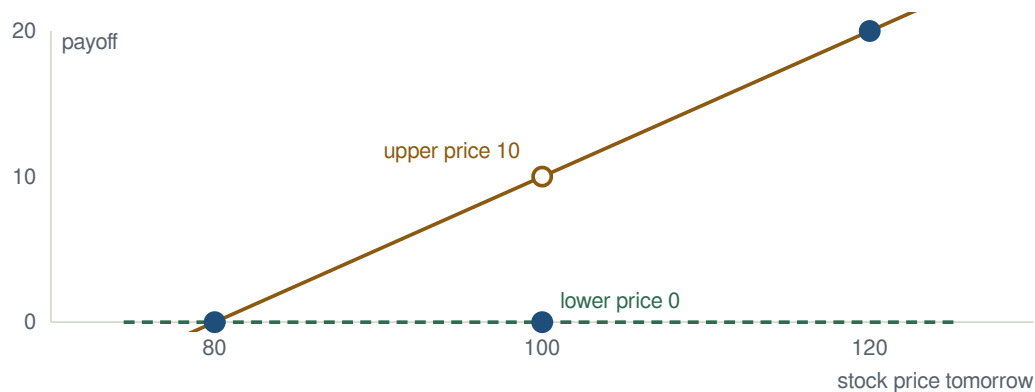
Example 12.2 (one period, three outcomes)

Now $S_1 \in \{80, 100, 120\}$ with payoff $(0, 0, 20)$. Three equations, two unknowns: no exact replication. The cheapest *super-hedge* is the line through $(80, 0)$ and $(120, 20)$, which costs 10 at $S = 100$ and dominates the payoff at every outcome. The most

expensive *sub-hedge* is the line through $(80, 0)$ and $(100, 0)$, worth 0. Any price in $[0, 10]$ is arbitrage-free; outside it there is a sure profit. Probability-free reasoning gives an interval, not a number.

Example 12.3 (two periods)

Split the day: by noon $S \in \{90, 110\}$, by close each moves ± 10 , reaching $\{80, 100, 120\}$ as before. Work backwards. At the 110 node, outcomes $100 \mapsto 0$ and $120 \mapsto 20$ give $\Delta = 1$ and value 10. At the 90 node both outcomes pay 0: value 0, $\Delta = 0$. At the root, the “payoffs” are 0 at 90 and 10 at 110, so $\Delta = \frac{1}{2}$ and the price is 5. Trading at noon restores a unique price, inside the interval of Example 12.2. As trading becomes continuous, this binomial construction converges to Black–Scholes (Cox, Ross and Rubinstein 1979).



Example 12.2 in pictures. Each hedge (cash plus Δ shares) is a line; its intercept at today's price is its cost. The upper price is the lowest line above all payoff points, the lower price the highest line below them. With two outcomes the points are always collinear and the two prices coincide.

12.2 Game-theoretic probability

Shafer and Vovk turn the picture above into a foundation for probability. There is no probability measure, only a game between two players.

Definition 12.4 (the protocol)

- A tree of *situations*. The root is $\square$; the move ω from situation t leads to $t\omega$; leaves are terminal.
- **World** chooses the moves. **Skeptic** chooses a strategy P , a bet $P(t)$ in each situation.
- Skeptic's *capital process* starts at $\mathcal{K}^P(\square) = 0$ and changes linearly with World's move: $\mathcal{K}^P(t\omega) = \mathcal{K}^P(t) + P(t)\omega$. (In Example 12.1, $\omega = S_1 - S_0$ and $P = \Delta$.)
- A *martingale* is $\alpha + \mathcal{K}^P$ for some initial capital α and strategy P .

Definition 12.5 (upper and lower expectation)

For a payoff x on terminal situations,

$$\begin{aligned}\bar{\mathbb{E}}[x] &= \inf\{\alpha : \exists P \forall \text{ terminal } t, \alpha + \mathcal{K}^P(t) \geq x(t)\}, \\ \underline{\mathbb{E}}[x] &= \sup\{\alpha : \exists P \forall t, \alpha + \mathcal{K}^P(t) \leq x(t)\}.\end{aligned}$$

When they agree, the common value is the (game-theoretic) expectation $\mathbb{E}[x]$. For an event E , the upper and lower probabilities are $\bar{\mathbb{P}}[E] = \bar{\mathbb{E}}[\mathbf{1}_E]$ and $\underline{\mathbb{P}}[E] = \underline{\mathbb{E}}[\mathbf{1}_E]$.

The upper expectation is the super-hedging price and the lower expectation the sub-hedging price. Written as

$$\bar{\mathbb{E}}[x] = \inf_P \sup_t (x(t) - \mathcal{K}^P(t)),$$

it is a minimax regret: Skeptic tries to track the payoff x with trading gains, and $\bar{\mathbb{E}}[x]$ is the smallest worst-case shortfall. An event has upper probability zero if Skeptic can start with capital 1, never go into debt, and become arbitrarily rich if the event happens. “Probability zero” means “you could get rich betting against it”.

12.3 A strong law of large numbers by betting

Theorem 12.6 (game-theoretic SLLN, Shafer and Vovk 2001)

In the protocol where, at each round $i = 1, 2, \dots$, Skeptic bets P_i and then World picks $\omega_i \in [-1, 1]$, Skeptic has a strategy that starts with capital 1, never goes negative, and whose capital is unbounded on every path where $\frac{1}{n} \sum_{i \leq n} \omega_i \not\rightarrow 0$. In other words, $\bar{\mathbb{P}}[\frac{1}{n} \sum_{i \leq n} \omega_i \not\rightarrow 0] = 0$.

Proof. One bettor. Bet the fraction $\varepsilon \in [-\frac{1}{2}, \frac{1}{2}]$ of current capital each round, so capital after n rounds is $K_n^\varepsilon = \prod_{i \leq n} (1 + \varepsilon \omega_i) > 0$. Since $\ln(1 + x) \geq x - x^2$ for $|x| \leq \frac{1}{2}$,

$$\ln K_n^\varepsilon \geq \varepsilon S_n - \varepsilon^2 n, \quad S_n = \sum_{i \leq n} \omega_i.$$

Many bettors. Split initial capital: weight 2^{-k-2} each to the bettors with $\varepsilon = +2^{-k}$ and $\varepsilon = -2^{-k}$, $k = 1, 2, \dots$. Each account stays positive, so total capital $K_n \geq 2^{-k-2} K_n^{\pm 2^{-k}}$ for every k .

Conclusion. If $\limsup S_n/n > \delta > 0$, pick k with $2^{-k} < \delta/2$. Along the subsequence where $S_n > \delta n$, $\ln K_n^{2^{-k}} \geq n 2^{-k} (\delta - 2^{-k}) \rightarrow \infty$, so K_n is unbounded. The case $\liminf S_n/n < 0$ uses the negative bettors. $\square$

Look at the structure of that proof: a mixture of constant-fraction betting strategies, with fixed prior weights, compared to the best of them in hindsight, in log-wealth. It is exactly the argument behind Cover’s universal portfolio (Chapter 11) and EWA with a prior (Theorem 3.9). The law of large numbers, seen from the game-theoretic side, is a no-regret statement: a universal bettor gets rich whenever any simple bettor does, so if the averages do not converge, some simple bettor, and hence the universal one, gets rich.

12.4 Regret bounds are option-price bounds

Reverse the roles. Suppose an online algorithm trading a stock and cash, starting with capital 1, guarantees on every price path in some class $\mathcal{S}$ that its final wealth is at least $\beta \max(S_T/S_0, 1)$ for some $\beta \in (0, 1]$. This is a multiplicative no-regret guarantee against the two “experts” hold-the-stock and hold-cash. Consider the claim that pays $\max(S_T/S_0, 1)$, and suppose it is priced above $1/\beta$. Sell it, run the algorithm with capital $1/\beta$, and keep the difference: at maturity the algorithm’s wealth covers the payoff on every path in $\mathcal{S}$. So, with no probability model,

$$\text{price}[\max(S_T/S_0, 1)] \leq \frac{1}{\beta},$$

$$\text{and since } \frac{(S_T - S_0)^+}{S_0} = \max\left(\frac{S_T}{S_0}, 1\right) - 1, \quad \frac{\text{call}_{K=S_0}}{S_0} \leq \frac{1}{\beta} - 1.$$

DeMarzo, Kremer and Mansour (2006) made this precise. Take $\mathcal{S}$ to be paths whose quadratic variation $\sum_t (\ln S_{t+1} - \ln S_t)^2$ is at most Q , the path-wise analogue of $\sigma^2 T$. A Hedge-type algorithm over the two assets has log-wealth regret of order $\sqrt{Q}$, so $\beta = e^{-O(\sqrt{Q})}$ and the at-the-money call is worth at most $S_0(e^{O(\sqrt{Q})} - 1) \approx O(\sqrt{Q}) S_0$. Compare Black–Scholes: the at-the-money call is worth about $0.4 S_0 \sigma \sqrt{T} = 0.4 S_0 \sqrt{Q}$. The $\sqrt{T}$ of regret and the $\sigma \sqrt{T}$ of option prices are the same phenomenon.

The lecture’s co-instructor Rafael Frongillo, with Abernethy and others, took this to its conclusion (Abernethy, Frongillo and Wibisono 2012; Abernethy, Bartlett, Frongillo and Wibisono 2013). Define the minimax price of an option as the value of a hedging game in which Nature moves the price subject to a quadratic-variation budget and a bound on each jump, and the Investor hedges. As the jump bound tends to zero, the value of this game converges to the Black–Scholes price. Black–Scholes then needs no Brownian motion: it is the price of a game against an adversary with a volatility budget.

Why this matters in practice

- Model-free bounds (super- and sub-hedging prices) are what a trader can defend when they do not trust a model. Examples 12.2 and 12.3 show they can be wide or tight depending on what is tradeable.
- The volatility budget Q plays the role of implied volatility. Robust hedging strategies driven by realised quadratic variation, such as variance swaps and “hedging with a volatility band”, are practical relatives of these ideas.
- No-regret algorithms, read backwards, are hedging strategies. Every regret bound in this book has a financial meaning as the capital needed to super-replicate some payoff.

Key takeaways

- Replication gives unique prices when the market is complete, and a price interval otherwise.
- Upper and lower expectations are super- and sub-hedging prices, i.e. minimax

regret problems for Skeptic.

- The strong law of large numbers is a universal-portfolio argument.
- Regret bounds give option-price upper bounds; with a quadratic-variation budget the game's value tends to Black–Scholes.

Exercises

12.1 finance In Example 12.1, price a put struck at 100 by replication, and verify put–call parity $C - P = S_0 - K$.

12.2 finance In Example 12.2, find all risk-neutral measures q on $\{80, 100, 120\}$ (with $\mathbb{E}_q S_1 = 100$) and show that $\{\mathbb{E}_q[\text{payoff}]\}$ is exactly $[0, 10]$ (open or closed, depending on whether q must be strictly positive).

12.3 core Show the martingale property: in a one-step game where the lower and upper expectation of World's move ω are both 0, every capital process $s = \alpha + \mathcal{K}^P$ satisfies $\mathbb{E}[s(t\omega) \mid t] = s(t)$.

12.4 challenge Using Hedge with $N = 2$ experts ("stock" and "cash") on log-returns bounded by δ per step, derive a bound of the form $\ln W_T \geq \max(\ln S_T / S_0, 0) - c\sqrt{Q \ln 2} - O(\delta)$ and deduce an explicit upper bound on the at-the-money call price in terms of Q .

Solutions

12.1. Put payoff $(20, 0)$: $\Delta = -\frac{1}{2}$, $c = 0 - (-\frac{1}{2})(120 - 100) = 10$. So $P = 10$, and $C - P = 0 = S_0 - K$.

12.2. $q = (a, 1 - 2a, a)$ for $a \in [0, \frac{1}{2}]$. $\mathbb{E}_q[\text{payoff}] = 20a \in [0, 10]$. With strictly positive q , $a \in (0, \frac{1}{2})$ and the set is $(0, 10)$. This is the duality between super-hedging and risk-neutral pricing (LP duality again).

12.3. $s(t\omega) = s(t) + P(t)\omega$ is affine in ω with slope $P(t)$. Super-hedge it with initial capital $s(t)$ and bet $P(t)$; sub-hedge it the same way. Both prices equal $s(t)$.

Part V

Bandits

Learning when you only see the outcome of what you chose.

Stochastic Bandits: Explore, Exploit, Be Optimistic

Based on Lectures 18–19 (Nov 6 and 11, 2013) · [notes 18](#) · [notes 19](#)

In this chapter

- The bandit feedback model and pseudo-regret.
- Explore-then-commit and its $O(N \ln(NT) / \Delta^2)$ regret.
- UCB: optimism in the face of uncertainty, $O(\sum_j \ln T / \Delta_j)$ regret.

13.1 Bandit feedback

So far the learner saw the loss of every expert each round. In many problems it sees only the loss of the action it took: the fill price of the venue it routed to, the click-through of the ad it showed, the return of the strategy it funded. This is the *multi-armed bandit* problem, named after slot machines (“one-armed bandits”).

Protocol 13.1 (*bandit*)

N arms. On round t each arm has a loss $X_{i,t} \in [0, 1]$. The learner picks I_t (possibly at random) and observes and pays *only* $X_{I_t,t}$. In the **stochastic** setting, $X_{i,t}$ are i.i.d. over t with mean μ_i ; in the **adversarial** setting (Chapter 14) they are arbitrary. With $i^* = \arg \min_i \mu_i$ and gaps $\Delta_j = \mu_j - \mu_{i^*}$, the (pseudo-)regret is

$$\bar{R}_T = \mathbb{E} \sum_{t=1}^T (\mu_{I_t} - \mu_{i^*}) = \sum_{j \neq i^*} \Delta_j \mathbb{E}[T_j(T)],$$

where $T_j(t)$ is the number of times arm j was played in the first t rounds.

The last identity is the key to every analysis in this chapter: regret is controlled by how often we pull each suboptimal arm. The tension is between *exploration* (pull uncertain arms to learn about them) and *exploitation* (pull the arm that looks best).

Lemma 13.2 (*Hoeffding's inequality*)

If $Z_1, \dots, Z_m$ are i.i.d. in $[0, 1]$ with mean μ , then $\Pr \left[\left| \frac{1}{m} \sum_j Z_j - \mu \right| \geq \epsilon \right] \leq 2e^{-2m\epsilon^2}$, and each one-sided deviation has probability at most $e^{-2m\epsilon^2}$. (Proof in [Appendix A](#).)

13.2 Explore then commit

Algorithm 13.3 (*explore-then-commit, ETC*)

```

input:  $m \geq 1$ 
for each arm  $i$ : play it  $m$  times, let  $\hat{\mu}_i$  be its average loss    #  $t = 1..Nm$ 
for  $t > Nm$ : play  $\hat{i} = \operatorname{argmin}_i \hat{\mu}_i$ 

```

Theorem 13.4

If every suboptimal arm has gap at least $\Delta > 0$, ETC with $m = \lceil \frac{2}{\Delta^2} \ln(2NT) \rceil$ has $\bar{R}_T \leq 1 + Nm \leq 1 + N + \frac{2N \ln(2NT)}{\Delta^2}$.

Proof. Exploration costs at most Nm since losses are in $[0, 1]$. If ETC commits to a wrong arm j , then $\hat{\mu}_j \leq \hat{\mu}_{i^*}$, so $\Delta \leq \mu_j - \mu_{i^*} \leq (\mu_j - \hat{\mu}_j) + (\hat{\mu}_{i^*} - \mu_{i^*}) \leq 2 \max_i |\hat{\mu}_i - \mu_i|$. By Hoeffding and a union bound, $\Pr[\text{wrong commit}] \leq 2Ne^{-m\Delta^2/2} \leq \frac{1}{T}$ for our m . The exploitation phase then costs at most $T \cdot \frac{1}{T} = 1$ in expectation. $\square$

ETC has two weaknesses. It needs to know Δ (or T) to set m , and it spends the same effort on an arm that is obviously bad after two pulls as on one that is hard to distinguish from the best. The $1/\Delta^2$ dependence is the price: without knowing Δ , tuning for the worst case gives $O(T^{2/3})$ regret (Exercise 13.2).

13.3 Upper confidence bounds

UCB resolves the tension with one principle: *act as if the world is as favourable as is plausible given the data*. For losses, “favourable” means the lowest plausible mean. An arm is played either because its estimate is good (exploitation) or because it is uncertain (exploration), and uncertainty shrinks with every pull.

Algorithm 13.5 (*UCB1, Auer, Cesa-Bianchi and Fischer 2002; loss form*)

```

play each arm once
for  $t = N+1, \dots, T$ :
     $I_t \leftarrow \operatorname{argmin}_j [ \hat{\mu}_j - \sqrt{2 \ln t / T_j(t-1)} ]$ 

```

The bonus $\sqrt{2 \ln t / T_j}$ is the width of a Hoeffding confidence interval at confidence level $1 - t^{-4}$.

Theorem 13.6

For every suboptimal arm j , $\mathbb{E}[T_j(T)] \leq \frac{8 \ln T}{\Delta_j^2} + 1 + \frac{\pi^2}{3}$. Hence

$$\bar{R}_T \leq \sum_{j: \Delta_j > 0} \left(\frac{8 \ln T}{\Delta_j} + \left(1 + \frac{\pi^2}{3}\right) \Delta_j \right).$$

Proof. Write $c_{t,s} = \sqrt{2 \ln t / s}$ and let $\ell = \lceil 8 \ln T / \Delta_j^2 \rceil$. If arm j is chosen at round t after it has already been pulled $s \geq \ell$ times, while the optimal arm has been pulled s^* times, then $\hat{\mu}_{j,s} - c_{t,s} \leq \hat{\mu}_{*,s^*} - c_{t,s^*}$. At least one of three things must then hold:

1. $\hat{\mu}_{*,s^*} - c_{t,s^*} > \mu_*$: the optimal arm's lower confidence bound is too high;
2. $\hat{\mu}_{j,s} + c_{t,s} < \mu_j$: arm j 's estimate is too low;
3. $\mu_j - 2c_{t,s} < \mu_*$, i.e. $\Delta_j < 2c_{t,s}$.

(If none holds, $\hat{\mu}_{j,s} - c_{t,s} \geq \mu_j - 2c_{t,s} \geq \mu_* \geq \hat{\mu}_{*,s^*} - c_{t,s^*}$, and ties aside j is not chosen.) Event 3 is impossible when $s \geq \ell$, because then $2c_{t,s} \leq 2\sqrt{2 \ln T / \ell} \leq \Delta_j$. By Hoeffding, events 1 and 2 each have probability at most $e^{-2s \cdot 2 \ln t / s} = t^{-4}$ for fixed s, s^* . So

$$\mathbb{E}[T_j(T)] \leq \ell + \sum_{t=1}^{\infty} \sum_{s=1}^t \sum_{s^*=1}^t 2t^{-4} \leq \ell + \sum_{t \geq 1} 2t^{-2} = \ell + \frac{\pi^2}{3}.$$

□

Compare with ETC: the dependence on each gap is $1/\Delta_j$ rather than $1/\Delta^2$, no parameters need tuning, and each arm's exploration adapts to its own gap. Taking the worst case over gaps gives a distribution-free bound of $O(\sqrt{NT \ln T})$ (Exercise 13.3).

Theorem 13.7 (Lai and Robbins 1985, informal)

For any algorithm that has $o(T^a)$ regret on every instance for every $a > 0$, and Bernoulli arms,

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[T_j(T)]}{\ln T} \geq \frac{1}{\text{KL}(\mu_j \| \mu^*)} \quad \text{for each suboptimal } j,$$

where KL is between Bernoulli distributions. Since $\text{KL}(\mu_j \| \mu^*) \leq \frac{\Delta_j^2}{\mu^*(1-\mu^*)}$, this shows UCB's $\ln T / \Delta_j$ per-arm regret is optimal up to constants. Refinements such as KL-UCB and Thompson sampling achieve the exact constant.

In finance

Order routing is a natural stochastic bandit. A broker splits child orders across N venues or algorithms, and observes the implementation shortfall only for the venue actually used. UCB routes most flow to the venue with the best observed shortfall while guaranteeing that every venue is re-tested at a rate of order $\ln t$, so a venue whose quality improves is eventually noticed. Two caveats from the theory apply. First, the i.i.d. assumption is doubtful: venue quality depends on market conditions and on your own flow. Second, regret $\sum_j \ln T / \Delta_j$ blows up when venues are nearly identical, which is also when choosing among them matters least. When the environment may be adversarial (other traders learning about your routing), use the methods of Chapter 14.

Key takeaways

- Bandit regret equals $\sum_j \Delta_j \mathbb{E}[T_j(T)]$: control how often bad arms are pulled.

- Explore-then-commit: $O(N \ln(NT)/\Delta^2)$, but needs the gap.
- UCB: optimism gives $O(\sum_j \ln T/\Delta_j)$, matching the Lai–Robbins lower bound up to constants.

Exercises

- 13.1** warm-up With $N = 5$, $\Delta = 0.05$ and $T = 10^6$, compute the exploration length of ETC from Theorem 13.4 and compare with UCB's bound on $\mathbb{E}[T_j(T)]$ for an arm with gap 0.05.
- 13.2** core Show that ETC with m chosen without knowledge of Δ can guarantee $\bar{R}_T = O(T^{2/3}(N \ln T)^{1/3})$. Hint: an arm with gap below ε costs at most εT even if chosen.
- 13.3** core Derive UCB's distribution-free bound $\bar{R}_T = O(\sqrt{NT \ln T})$. Hint: split arms into those with $\Delta_j < \delta$ and the rest.
- 13.4** project Simulate ETC, ε -greedy, UCB1 and Thompson sampling (Beta–Bernoulli) on 10 Bernoulli arms with means spaced 0.02 apart. Plot regret against $\ln T$. Which slopes match Theorems 13.6 and 13.7?

Solutions

- 13.1.** ETC: $m = \lceil \frac{2}{0.0025} \ln(10^7) \rceil = \lceil 800 \times 16.118 \rceil = 12,895$ pulls of each arm, total 64,475. UCB: $\frac{8 \ln 10^6}{0.0025} \approx 44,210$ pulls for that arm, plus a constant. The constants in UCB's analysis are loose; in practice it pulls such an arm far fewer times.
- 13.2.** With m pulls per arm, all estimates are within $\sqrt{\ln(2NT)/(2m)}$ w.h.p., so the committed arm has gap at most $2\sqrt{\ln(2NT)/(2m)}$. Regret $\leq Nm + T\sqrt{2\ln(2NT)/m} + 1$. Balancing gives $m \approx (T/N)^{2/3}(\ln(NT))^{1/3}$ and regret $O(N^{1/3}T^{2/3}(\ln NT)^{1/3})$.
- 13.3.** Arms with $\Delta_j < \delta$ contribute at most δT . The rest contribute at most $\sum_j \frac{8 \ln T}{\Delta_j} + O(N) \leq \frac{8N \ln T}{\delta} + O(N)$. Set $\delta = \sqrt{8N \ln T/T}$.

Adversarial Bandits: EXP3 and Beyond

Based on Lectures 19–21 (Nov 11 – 18, 2013) · notes 20 · notes 21

In this chapter

- Importance-weighted loss estimates and why they are enough.
- From $T^{3/4}$ to $T^{2/3}$ to $\sqrt{T}$: ε -exploration versus EXP3.
- The $\Omega(\sqrt{NT})$ lower bound, and bandits over combinatorial and convex sets.

14.1 Estimating what you did not see

With adversarial losses, a deterministic learner is hopeless: the adversary puts loss 1 on whatever arm it will pick and 0 elsewhere. So the learner must randomise, picking $I_t \sim p_t$. It then sees one coordinate of ℓ_t and needs a stand-in for the rest.

Definition 14.1 (importance-weighted estimator)

$$\tilde{\ell}_{t,i} = \frac{\ell_{t,I_t}}{p_{t,I_t}} \mathbf{1}[I_t = i].$$

It is unbiased: $\mathbb{E}_{I_t \sim p_t} \tilde{\ell}_{t,i} = p_{t,i} \cdot \frac{\ell_{t,i}}{p_{t,i}} = \ell_{t,i}$.

Lemma 14.2 (unbiased estimates suffice)

Let $\mathcal{F}_{t-1}$ be the history before round t . If p_t is $\mathcal{F}_{t-1}$ -measurable and $\mathbb{E}[\tilde{\ell}_t \mid \mathcal{F}_{t-1}] = \ell_t$, then for any fixed $u \in \Delta_N$ (oblivious adversary), $\mathbb{E} \sum_t \langle \tilde{\ell}_t, p_t - u \rangle = \mathbb{E} \sum_t \langle \ell_t, p_t - u \rangle$.

So any full-information algorithm run on the estimates inherits its regret bound in expectation, *except* that the bounds of Part I assumed losses in $[0, 1]$, while $\tilde{\ell}_{t,i}$ can be as large as $1/p_{t,i}$. The whole difficulty of adversarial bandits is controlling the size of these estimates.

14.2 A first attempt: separate exploration

With probability ε , explore: play a uniformly random arm and set $\tilde{\ell}_t = \frac{N}{\varepsilon} \ell_{t,I_t} e_{I_t}$. Otherwise exploit: play Hedge on past estimates and set $\tilde{\ell}_t = 0$. The estimate is unbiased and bounded by N/ε . Plugging this range into the Hedge bound gives regret $O(\varepsilon T + \frac{\ln N}{\eta} + \eta T \frac{N^2}{\varepsilon^2}) = O(T^{3/4} \sqrt{N} (\ln N)^{1/4})$. A finer analysis uses the second-moment term $\sum_i p_{t,i} \tilde{\ell}_{t,i}^2$ that appears in the proof of Theorem 14.4 below; its expectation here is at most $\varepsilon \cdot \frac{1}{N} \sum_i p_{t,i} \frac{N^2}{\varepsilon^2} \leq \frac{N}{\varepsilon}$. That gives $O(\varepsilon T + \frac{\ln N}{\eta} + \frac{\eta TN}{\varepsilon}) = O(T^{2/3} (N \ln N)^{1/3})$. Both are worse than $\sqrt{T}$ because

exploration rounds are wasted: they pay loss without exploiting, and exploiting rounds teach nothing.

14.3 EXP3

Algorithm 14.3 (EXP3, Auer, Cesa-Bianchi, Freund and Schapire 2002)

```

input:  $\eta > 0$ ;  $\tilde{L}_{\{i,0\}} \leftarrow 0$ 
for  $t = 1, \dots, T$ :
   $p_{\{t,i\}} \leftarrow \exp(-\eta \tilde{L}_{\{i,t-1\}}) / \sum_j \exp(-\eta \tilde{L}_{\{j,t-1\}})$ 
  draw  $I_t \sim p_t$ ; observe  $\ell_{\{t,I_t\}}$ 
   $\tilde{L}_{\{I_t,t\}} \leftarrow \tilde{L}_{\{I_t,t-1\}} + \ell_{\{t,I_t\}} / p_{\{t,I_t\}}$  # other arms unchanged

```

EXP3 = “exponential weights for exploration and exploitation”. Every round both exploits and gathers information.

Theorem 14.4

Against an oblivious adversary with losses in $[0, 1]$, EXP3 satisfies $\mathbb{E} \text{Regret}_T \leq \frac{\ln N}{\eta} + \frac{\eta NT}{2}$. With $\eta = \sqrt{\frac{2 \ln N}{NT}}$, $\mathbb{E} \text{Regret}_T \leq \sqrt{2NT \ln N}$.

Proof. Use the softmax potential $\Phi_t = -\frac{1}{\eta} \ln \sum_i e^{-\eta \tilde{L}_{i,t-1}}$. As in Chapter 3,

$$\Phi_{t+1} - \Phi_t = -\frac{1}{\eta} \ln \mathbb{E}_{i \sim p_t} e^{-\eta \tilde{\ell}_{t,i}}.$$

Because estimates are nonnegative we can use $e^{-x} \leq 1 - x + \frac{x^2}{2}$ for $x \geq 0$, then $-\ln(1 - y) \geq y$:

$$\Phi_{t+1} - \Phi_t \geq \sum_i p_{t,i} \tilde{\ell}_{t,i} - \frac{\eta}{2} \sum_i p_{t,i} \tilde{\ell}_{t,i}^2.$$

This is the step that avoids needing $\tilde{\ell} \leq 1$: only nonnegativity is used, and the price is a *second-moment* term. Now take expectations over I_t :

$$\begin{aligned} \mathbb{E} \sum_i p_{t,i} \tilde{\ell}_{t,i} &= \langle p_t, \ell_t \rangle, \\ \mathbb{E} \sum_i p_{t,i} \tilde{\ell}_{t,i}^2 &= \sum_i p_{t,i} \cdot p_{t,i} \frac{\ell_{t,i}^2}{p_{t,i}^2} = \sum_i \ell_{t,i}^2 \leq N. \end{aligned}$$

The $1/p$ blow-up is exactly cancelled by weighting with p . Summing over t , $\mathbb{E}[\Phi_{T+1} - \Phi_1] \geq \mathbb{E} \sum_t \langle p_t, \ell_t \rangle - \frac{\eta NT}{2}$. On the other side, $\Phi_1 = -\frac{\ln N}{\eta}$ and $\Phi_{T+1} \leq \tilde{L}_{u,T}$ for any arm u , with $\mathbb{E} \tilde{L}_{u,T} = L_{u,T}$. Rearrange. $\square$

The comparison with full information is instructive: Hedge’s regret is $\sqrt{(T/2) \ln N}$; EXP3 pays an extra $\sqrt{N}$, the price of seeing one coordinate out of N . That price is unavoidable.

Theorem 14.5 (lower bound; Auer et al. 2002; Homework 3)

For every algorithm there is a stochastic instance with $\bar{R}_T \geq c\sqrt{NT}$ for a universal constant $c > 0$.

Idea. All arms are fair coins (loss $\frac{1}{2}$ in expectation) except a secret arm with mean $\frac{1}{2} - \varepsilon$. Identifying a biased coin among N requires about N/ε^2 flips in total, because each arm needs about $1/\varepsilon^2$ samples to be distinguished from fair (the KL divergence per sample is $\Theta(\varepsilon^2)$). An algorithm with regret $\ll \varepsilon T$ must play the secret arm most of the time, so it effectively identifies it, which is impossible if $T \ll N/\varepsilon^2$. Setting $\varepsilon = \sqrt{N/T}$ gives regret $\Omega(\varepsilon T) = \Omega(\sqrt{NT})$. $\square$

Refinements: the $\sqrt{\ln N}$ gap between the bounds is closed by the INF algorithm (Audibert and Bubeck 2009), which is FTRL with the regularizer $-\sum_i \sqrt{x_i}$. EXP3's regret holds only in expectation; its variance is large, and EXP3.P adds explicit exploration to get high-probability bounds.

14.4 Bandits on combinatorial and convex sets

Suppose each round we choose an s - t path in a graph with d edges and pay the sum of edge delays, observing only the total delay of our path. Paths are exponentially many, so EXP3 over paths has regret $\sqrt{(\#\text{paths}) T}$, which is useless. The fix is to work in the convex hull of path indicator vectors (the flow polytope $\mathcal{K} \subset \mathbb{R}^d$) with linear losses $\langle f_t, x \rangle$, and run FTRL on estimated loss vectors:

$$x_t = \arg \min_{x \in \mathcal{K}} \sum_{s < t} \langle \tilde{f}_s, x \rangle + \frac{1}{\eta} R(x), \quad \text{then play a random path } y_t \text{ with } \mathbb{E} y_t = x_t.$$

Since regret is linear in the losses for a fixed comparator, unbiased estimates again suffice in expectation. The problem is again the size of $\tilde{f}_t$: near the boundary of $\mathcal{K}$ we must divide by small numbers.

Abernethy, Hazan and Rakhlin (2008) solved this by choosing R to be a **self-concordant barrier** for $\mathcal{K}$ (for a polytope $\{x : a_k^\top x \leq b_k\}$, the log-barrier $-\sum_k \ln(b_k - a_k^\top x)$) and measuring everything in the local norm $\|v\|_x = \sqrt{v^\top \nabla^2 R(x) v}$. The key geometric fact is that the Dikin ellipsoid $\{y : \|y - x\|_x \leq 1\}$ lies inside $\mathcal{K}$. Their algorithm, SCRiBLE, plays a random endpoint of one of the ellipsoid's principal axes, $y_t = x_t \pm \lambda_i^{-1/2} e_i$ (where λ_i, e_i are eigenpairs of $\nabla^2 R(x_t)$), and sets $\tilde{f}_t = d \langle f_t, y_t \rangle \epsilon_t \lambda_i^{1/2} e_i$ with $\epsilon_t = \pm 1$ the chosen sign. This estimate is unbiased and has local dual norm at most d , and the result is

$$\mathbb{E} \text{Regret}_T = O(d\sqrt{\theta T \ln T}),$$

where θ is the barrier's self-concordance parameter ($\theta = m$ for the log-barrier of m constraints). This was the first efficient $\sqrt{T}$ algorithm for bandit linear optimization.

In finance

Adversarial bandits suit trading problems where your own actions change the environ-

ment, or where competitors adapt. Examples include choosing which of N execution algorithms to use for each parent order (other participants may detect and front-run a predictable choice), or setting quotes as a market maker (you observe only whether your quote was hit, not what would have happened at other prices). EXP3 guarantees average performance within $\sqrt{2N \ln N/T}$ per round of the best fixed choice, against any behaviour of the other side, provided that behaviour does not react to your internal random draws. The randomisation is a feature: an unpredictable router is harder to exploit.

Key takeaways

- Importance weighting $\tilde{\ell}_{t,i} = \ell_{t,i} \mathbf{1}[I_t = i] / p_{t,i}$ gives unbiased estimates; regret bounds transfer in expectation.
- EXP3 is Hedge on these estimates. The second moment $\sum_i p_{t,i} \tilde{\ell}_{t,i}^2$ has expectation at most N , giving $\sqrt{2NT \ln N}$.
- $\Omega(\sqrt{NT})$ is unavoidable: bandit feedback costs a factor $\sqrt{N}$.
- For structured action sets, FTRL with a self-concordant barrier and ellipsoid sampling gives $O(d\sqrt{\theta T \ln T})$.

Exercises

14.1 warm-up Verify $e^{-x} \leq 1 - x + \frac{x^2}{2}$ for $x \geq 0$, and show it fails for $x < 0$. Why does EXP3's proof need losses (not gains)?

14.2 core Carry out the tuning of the ε -exploration algorithm in §14.2 with the improved variance bound, and find the resulting rate.

14.3 core Show that $\mathbb{E}[\tilde{\ell}_{t,i}^2] = \ell_{t,i}^2 / p_{t,i}$, and give an example where the variance of $\tilde{L}_{i,T}$ is of order T^2 for some arm. Why doesn't this ruin Theorem 14.4?

14.4 challenge (Homework 3, Problem 3.) Assume any procedure using fewer than cN/ε^2 coin flips identifies the odd coin with probability at most $\frac{3}{4}$. Show that no bandit algorithm can have regret $O(T^{1/2}N^{1/2-\delta})$ for any $\delta > 0$.

Solutions

14.1. $g(x) = 1 - x + \frac{x^2}{2} - e^{-x}$ has $g(0) = 0$, $g'(x) = -1 + x + e^{-x} \geq 0$ by (I1). For $x < 0$ the third-order term has the wrong sign (e.g. $x = -3$: $e^3 = 20.1 > 8.5$). With gains, the estimates enter the exponent with a positive sign and can be huge, so the quadratic bound fails. The gain version of EXP3 needs explicit exploration to cap $1/p$.

14.2. Minimise $\varepsilon T + \frac{\ln N}{\eta} + \frac{\eta TN}{\varepsilon}$. Over η : $2\sqrt{TN \ln N/\varepsilon}$. Then $\varepsilon T + 2\sqrt{TN \ln N} \varepsilon^{-1/2}$ is minimised at $\varepsilon = (N \ln N/T)^{1/3}$, giving $3 T^{2/3} (N \ln N)^{1/3}$. Compare EXP3: using every round both to exploit and to learn removes the $T^{2/3}$.

14.3. $\mathbb{E}\tilde{\ell}^2 = p \cdot \ell^2/p^2$. If an arm's probability drops to $1/T$ while its loss is 1, each round adds

variance of order T , for a total of order T^2 . The proof never needs the variance of $\tilde{L}_i$ itself, only the p -weighted second moment $\sum_i p_{t,i} \tilde{\ell}_{t,i}^2$, which is small because arms with tiny p get tiny weight.

Part VI

**Approachability, Calibration and
Equilibrium**

*A vector-valued minimax theorem that ties the whole course
together.*

Blackwell's Approachability Theorem

Based on Lectures 21–23 (Nov 18 – Dec 2, 2013) · [notes 21](#) · [notes 22](#)

In this chapter

- Games with vector payoffs, and why the minimax theorem fails for them in one shot.
- Blackwell's theorem: what cannot be guaranteed in one round can be guaranteed on average.
- Blackwell's projection strategy and its $O(1/\sqrt{T})$ rate; no-regret learning as a special case; a second proof via OCO.

15.1 Vector payoffs

Let $r : \Delta_n \times \Delta_m \rightarrow \mathbb{R}^d$ be **biaffine**: $r(p, q) = \sum_{i,j} p_i q_j r(e_i, e_j)$. The learner picks p , the adversary picks q , and the learner wants the payoff to land in a closed convex target set $S \subseteq \mathbb{R}^d$.

When $d = 1$ and $S = (-\infty, c]$, von Neumann says: if for every q some p gets $r(p, q) \leq c$, then a single p works against every q . The analogous statement for vectors is false.

Example 15.1

$n = m = 2$, identify p, q with their first coordinates, $r(p, q) = (p, q) \in [0, 1]^2$, and $S = \{(x, y) : x = y\}$. For every q the response $p = q$ lands in S . But no fixed p lands in S for every q .

Yet in the repeated game the learner can play $p_t = q_{t-1}$, and the average payoff $(\frac{1}{T} \sum_t q_{t-1}, \frac{1}{T} \sum_t q_t)$ is within $O(1/T)$ of the diagonal. Repetition rescues the duality.

15.2 Halfspaces are enough

Lemma 15.2 (three equivalent conditions)

For closed convex S , the following are equivalent:

1. *Response-satisfiability*: $\forall q \exists p : r(p, q) \in S$.
2. *Halfspace-forcing*: for every closed halfspace $H \supseteq S$, $\exists p \forall q : r(p, q) \in H$.
3. For every closed halfspace $H \supseteq S$, $\forall q \exists p : r(p, q) \in H$.

Proof. (2) $\Leftrightarrow$ (3): write $H = \{x : \langle v, x \rangle \leq c\}$. Then $\langle v, r(p, q) \rangle = p^\top M^v q$ with $M_{ij}^v =$

$\langle v, r(e_i, e_j) \rangle$, a scalar zero-sum game, and the minimax theorem swaps the quantifiers. (1) $\Rightarrow$ (3): immediate, since $S \subseteq H$. (3) $\Rightarrow$ (1): if some q_0 had $r(p, q_0) \notin S$ for all p , the compact convex set $\{r(p, q_0) : p \in \Delta_n\}$ would be disjoint from S , and a separating hyperplane would give a halfspace $H \supseteq S$ that no $r(p, q_0)$ enters, contradicting (3). $\square$

15.3 The theorem and Blackwell's strategy

Theorem 15.3 (Blackwell 1956)

If S is closed convex and response-satisfiable, the learner has an adaptive strategy $p_t = f(q_1, \dots, q_{t-1})$ such that for every sequence $q_1, q_2, \dots$,

$$d(\bar{r}_T, S) \rightarrow 0, \quad \bar{r}_T = \frac{1}{T} \sum_{t=1}^T r(p_t, q_t).$$

Quantitatively, if $\|r(p, q) - s\| \leq B$ for all p, q and $s \in S$ near the payoffs, then $d(\bar{r}_T, S) \leq B/\sqrt{T}$.

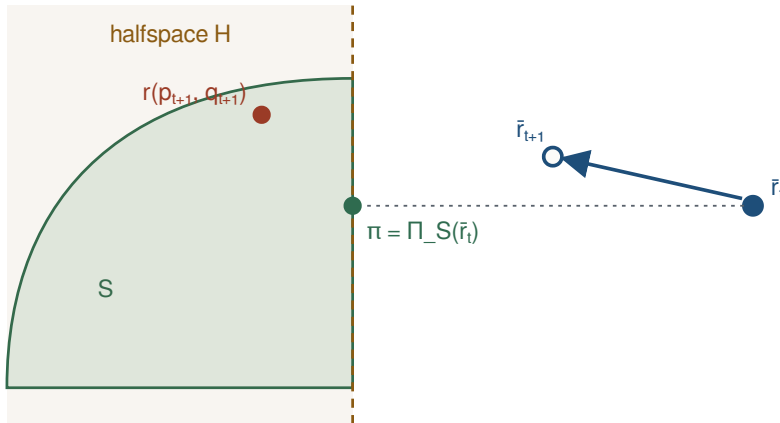
Algorithm 15.4 (Blackwell's projection strategy)

```

for t = 0, 1, 2, ...:
  if  $\bar{r}_t \in S$ : play any  $p_{t+1}$ 
  else:
     $\pi \leftarrow \Pi_S(\bar{r}_t)$                                 # closest point of S
     $v \leftarrow \bar{r}_t - \pi$                                 # outward normal
     $H \leftarrow \{x : \langle v, x - \pi \rangle \leq 0\}$             # halfspace  $\supseteq S$ , tangent at  $\pi$ 
    play  $p_{t+1}$  forcing  $r(p_{t+1}, q) \in H$  for all  $q$  # Lemma 15.2 (2)
    observe  $q_{t+1}$ 

```

Finding p_{t+1} means solving the scalar game $\min_p \max_q \langle v, r(p, q) \rangle$: one small LP per round.



One step of Blackwell's strategy. The new payoff is forced into the halfspace H that supports S at the projection π , so it lies on the far side of the tangent line from $\bar{r}_t$. The new average $\bar{r}_{t+1}$ moves a fraction $\frac{1}{t+1}$ of the way toward it, which makes its distance to S shrink.

Proof of the rate. Let $\pi = \Pi_S(\bar{r}_t)$ and $r = r(p_{t+1}, q_{t+1})$. Since $\bar{r}_{t+1} = \frac{t}{t+1}\bar{r}_t + \frac{1}{t+1}r$,

$$d(\bar{r}_{t+1}, S)^2 \leq \|\bar{r}_{t+1} - \pi\|^2 = \left(\frac{t}{t+1}\right)^2 \|\bar{r}_t - \pi\|^2 + \frac{\|r - \pi\|^2}{(t+1)^2} + \frac{2t}{(t+1)^2} \langle \bar{r}_t - \pi, r - \pi \rangle.$$

The cross term is ≤ 0 because $r \in H$. Multiplying by $(t+1)^2$: $(t+1)^2 d_{t+1}^2 \leq t^2 d_t^2 + B^2$. Telescoping, $T^2 d_T^2 \leq TB^2$. $\square$

15.4 No-regret learning is approachability

For the experts problem, let the learner pick $w \in \Delta_N$, the adversary pick $\ell \in [0, 1]^N$, and set

$$r(w, \ell) = (\langle w, \ell \rangle - \ell_1, \dots, \langle w, \ell \rangle - \ell_N) \in \mathbb{R}^N, \quad S = \mathbb{R}_{\leq 0}^N.$$

Coordinate i of $\bar{r}_T$ is the average regret against expert i , so approaching the negative orthant is exactly vanishing regret. The response condition holds trivially: if you knew ℓ you would pick $e_{\arg \min \ell}$. Blackwell's strategy then gives a no-regret algorithm. Working it out, the projection onto the orthant clips positive coordinates, and the forcing p turns out to be $p_{t+1,i} \propto (R_{t,i})_+$: play each expert in proportion to its positive cumulative regret. This is Hart and Mas-Colell's *regret matching*, the algorithm behind counterfactual regret minimisation for poker. Its regret is $O(\sqrt{NT})$ from the rate $B/\sqrt{T}$ with $B = O(\sqrt{N})$; other potentials give Hedge's $\sqrt{T \ln N}$.

15.5 Approachability from OCO

The converse direction also holds: any no-regret OCO algorithm yields an approachability strategy (Abernethy, Bartlett and Hazan 2011). The key is a dual description of distance.

Lemma 15.5

With support function $h_S(\theta) = \sup_{s \in S} \langle \theta, s \rangle$, for every x : $d(x, S) = \max_{\|\theta\|_2 \leq 1} \{ \langle \theta, x \rangle - h_S(\theta) \}$.

Proof. For $\|\theta\| \leq 1$ and any $s \in S$, $\langle \theta, x \rangle - h_S(\theta) \leq \langle \theta, x - \Pi_S(x) \rangle \leq d(x, S)$. For $x \notin S$ take $\theta = \frac{x - \Pi_S(x)}{d(x, S)}$; the projection optimality condition gives $\langle \theta, s - \Pi_S(x) \rangle \leq 0$ for all $s \in S$, so $h_S(\theta) = \langle \theta, \Pi_S(x) \rangle$ and equality holds. For $x \in S$ take $\theta = 0$. $\square$

Algorithm 15.6 (approachability via OCO)

```

maintain  $\theta_t$  in the unit ball, updated by any OCO algorithm (e.g. OGD)
for  $t = 1, 2, \dots$ :
    play  $p_t$  with  $\langle \theta_t, r(p_t, q) \rangle \leq h_S(\theta_t)$  for all  $q$       # halfspace-forcing
    observe  $q_t$ ; feed the OCO algorithm the convex loss
     $f_t(\theta) = h_S(\theta) - \langle \theta, r(p_t, q_t) \rangle$ 

```

Averaging, $\frac{1}{T} \sum_t f_t(\theta) = h_S(\theta) - \langle \theta, \bar{r}_T \rangle$, so by Lemma 15.5, $\min_{\theta} \frac{1}{T} \sum_t f_t(\theta) = -d(\bar{r}_T, S)$. By the forcing step, $f_t(\theta_t) \geq 0$. Hence $d(\bar{r}_T, S) \leq \frac{1}{T} (\sum_t f_t(\theta_t) - \min_{\theta} \sum_t f_t(\theta)) = \frac{\text{Regret}_T}{T}$. With OGD, $O(1/\sqrt{T})$.

So Blackwell's theorem, no-regret learning and OCO are three presentations of one idea. Blackwell's projection strategy corresponds to running Follow the Leader on the losses f_t ; other OCO algorithms give other approachability strategies with different constants and geometry.

In finance

Many portfolio mandates are sets of linear constraints on *average* outcomes: tracking error to several benchmarks at once, average exposure to sectors within bands, average turnover below a cap. Each is a target set S for a vector of running averages, and the market is the adversary. Blackwell's theorem gives a clean criterion: if for every market scenario there is a portfolio satisfying the mandate against that scenario, then there is an adaptive policy satisfying it on average against every sequence of scenarios, at rate $1/\sqrt{T}$. When the criterion fails, some scenario defeats every portfolio, and no policy can meet the mandate in the worst case. Multi-objective risk budgeting fits this framework directly.

Key takeaways

- Vector minimax fails in one shot but holds on average: response-satisfiable convex sets are approachable.
- Blackwell's strategy forces each new payoff into the supporting halfspace at the projection; distance falls as $B/\sqrt{T}$.
- No-regret learning is approachability of the negative orthant (regret matching), and OCO algorithms give approachability strategies.

Exercises

15.1 warm-up In the regret game of §15.4, show that for $\bar{r}_t \notin S$ with positive part $v = (\bar{r}_t)_+$, the halfspace-forcing condition is $\langle v, r(p, \ell) \rangle \leq 0$ for all ℓ , and that $p \propto v$ satisfies it.

15.2 core Show that a closed convex S that is not response-satisfiable is *excludable*: the adversary can keep $\bar{r}_T$ at distance bounded below from S . (Blackwell's dichotomy for convex sets.)

15.3 core Run Blackwell's strategy on Example 15.1 (the diagonal). Show that it plays $p_{t+1} = 0$ when the learner's running average exceeds the adversary's, and $p_{t+1} = 1$ when it is below. Compare with the strategy $p_t = q_{t-1}$.

Solutions

15.1. $\Pi_S(\bar{r}_t) = \bar{r}_t - v$ and $\langle v, \Pi_S(\bar{r}_t) \rangle = 0$, so $H = \{x : \langle v, x \rangle \leq 0\}$. With $p = v/\|v\|_1$: $\langle v, r(p, \ell) \rangle = \sum_i v_i(\langle p, \ell \rangle - \ell_i) = \|v\|_1 \langle p, \ell \rangle - \langle v, \ell \rangle = 0$.

15.2. Some q_0 has $\{r(p, q_0)\}$ disjoint from S , with a strictly separating halfspace $\{x : \langle v, x \rangle \geq c + \delta\}$ containing it while $S \subseteq \{\langle v, x \rangle \leq c\}$. Playing q_0 forever keeps every payoff, hence the average, in the first halfspace, at distance at least $\delta/\|v\|$ from S .

15.3. Write $\bar{r}_t = (\bar{x}, \bar{y})$. Its projection onto the diagonal is $\pi = (m, m)$ with $m = \frac{\bar{x} + \bar{y}}{2}$, and $v = \bar{r}_t - \pi = (a, -a)$ with $a = \frac{\bar{x} - \bar{y}}{2}$. The forcing condition $\langle v, (p, q) - \pi \rangle = a(p - q) \leq 0$ for all $q \in [0, 1]$ requires $p = 0$ if $a > 0$ and $p = 1$ if $a < 0$. Blackwell's strategy is a bang-bang controller that only looks at the averages, while $p_t = q_{t-1}$ copies the last move. Both approach the diagonal, at rates $O(1/\sqrt{T})$ and $O(1/T)$ respectively.

Calibration, Internal Regret and Correlated Equilibria

Based on Lectures 23–24 (Dec 2 and 4, 2013) · [notes 23](#) · [notes 24](#)

In this chapter

- What it means for probability forecasts to be calibrated, and why a forecaster facing an adversary must randomise.
- Calibration from approachability, and approachability from calibration.
- Internal (swap) regret, and how it makes independent learners converge to correlated equilibria.

16.1 Calibrated forecasts

A weather forecaster says “70% chance of rain”. We cannot judge a single forecast, but we can check the long run: on the days she said about 70%, did it rain on about 70% of them?

Definition 16.1 (ε -calibration)

Forecasts $p_t \in [0, 1]$, outcomes $y_t \in \{0, 1\}$. Fix a grid $q_1, \dots, q_N$ of spacing ε and forecasts on the grid. The forecaster is ε -calibrated if the ℓ_1 calibration score

$$C_T = \sum_{i=1}^N \left| \frac{1}{T} \sum_{t=1}^T (q_i - y_t) \mathbf{1}[p_t = q_i] \right|$$

satisfies $\limsup_T C_T \leq c \varepsilon$ for a constant c . Each term is the frequency of forecast q_i times the gap between q_i and the empirical frequency of $y = 1$ on those days.

Calibration alone is a weak requirement: the forecaster who always announces the long-run base rate is calibrated on a stationary sequence and useless. It becomes interesting because it can be achieved on *every* sequence, and because (as §16.3 shows) it is strong enough to derive equilibria.

Deterministic forecasters cannot be calibrated

If p_t is a deterministic function of the past, Nature sets $y_t = 0$ when $p_t > \frac{1}{2}$ and $y_t = 1$ otherwise. Every forecast is then off by at least $\frac{1}{2}$ on every day it is used, so $C_T \geq \frac{1}{2}$ (Oakes 1985; Dawid 1985). The forecaster must randomise, and Nature must not see the coin flip before choosing y_t .

Theorem 16.2 (Foster and Vohra 1998)

There is a randomised forecaster that is ε -calibrated in expectation against every adversary, for every $\varepsilon > 0$.

Proof via approachability. The forecaster picks $\sigma \in \Delta_N$ and announces q_I with $I \sim \sigma$. Nature picks $\alpha \in [0, 1]$ and $y \sim \text{Bernoulli}(\alpha)$. Define the vector payoff

$$r(\sigma, \alpha) = (\sigma_1(q_1 - \alpha), \dots, \sigma_N(q_N - \alpha)) \in \mathbb{R}^N.$$

It is biaffine, and $\|\bar{r}_T\|_1$ is the expected calibration score. Target $S = \{x : \|x\|_1 \leq \varepsilon\}$, which is convex. Response condition: given α , put all mass on the grid point nearest α ; then $r = (q_{i^*} - \alpha)e_{i^*}$ has ℓ_1 norm at most ε . By Theorem 15.3, S is approachable. $\square$

16.2 Calibration implies equilibrium

Calibration generalises from $[0, 1]$ to any compact convex set Y : partition Y into N cells of diameter ε with representatives q_i , and require that on the rounds when q_i was forecast, the average outcome is within $c\varepsilon$ of q_i . Now play a game by forecasting your opponent and best-responding to the forecast.

Algorithm 16.3 (calibrated best response, zero-sum game M)

```

for t = 1, ..., T:
    forecast the opponent's mixed strategy:  $q_{\{i_t\}} \in \Delta_m$  (calibrated forecaster)
    play  $x_t \leftarrow \arg\min_x x^\top M q_{\{i_t\}}$  # best response to the forecast
    observe  $y_t$ 

```

Theorem 16.4

Against any sequence $y_1, y_2, \dots$, Algorithm 16.3 has average loss at most $\max_y \min_x x^\top M y + O(\varepsilon)$.

Proof. Group rounds by forecast. Let n_i be the number of rounds with forecast q_i , and $\bar{y}_i$ the average of y_t on them. Then

$$\frac{1}{T} \sum_t x_t^\top M y_t = \sum_i \frac{n_i}{T} x(q_i)^\top M \bar{y}_i = \sum_i \frac{n_i}{T} x(q_i)^\top M (q_i + O(\varepsilon)) \leq \sum_i \frac{n_i}{T} \min_x x^\top M q_i + O(\varepsilon),$$

and each $\min_x x^\top M q_i \leq \max_y \min_x x^\top M y$. $\square$

Combined with the trivial direction, this is another proof of the minimax theorem. The same argument with a general vector payoff (best-respond to a calibrated forecast of the opponent's move) proves Blackwell's theorem: the average payoff is a convex combination of points $r(x(q_i), q_i) \in S$, up to $O(\varepsilon)$. So calibration and approachability are equivalent.

16.3 Correlated equilibria and internal regret

In a game with k players, player i picks $j_i \in [M_i]$ and suffers $C_i(j_1, \dots, j_k)$.

Definition 16.5 (correlated equilibrium, Aumann 1974)

A joint distribution μ over action profiles is a correlated equilibrium if, for every player i and every swap $\varphi : [M_i] \rightarrow [M_i]$,

$$\mathbb{E}_{j \sim \mu} C_i(j) \leq \mathbb{E}_{j \sim \mu} C_i(\varphi(j_i), j_{-i}).$$

Interpretation: a mediator draws $j \sim \mu$ and privately tells each player their component; no player gains by deviating from the recommendation, even as a function of it. It is an ε -CE if the inequalities hold up to ε .

Every Nash equilibrium (a product distribution) is a CE, but the set of CEs is a polytope defined by linear inequalities, so CEs can be computed by linear programming. In the game of Chicken, a traffic light that tells one driver “go” and the other “stop” at random is a CE that is fairer than either pure Nash equilibrium and better for both drivers than the mixed one (Exercise 16.2).

Definition 16.6 (internal and swap regret)

External regret compares with always playing one fixed action. **Swap regret** compares with the best modification of one’s own play by a fixed function φ :

$$\text{Regret}_T^{\text{swap}} = \max_{\varphi} \sum_{t=1}^T \sum_i p_{t,i} (\ell_{t,i} - \ell_{t,\varphi(i)}).$$

Internal regret restricts to φ that changes one action $i \mapsto j$ and leaves the others alone; the two notions differ by at most a factor N .

External regret can be small while swap regret is linear: a learner that alternates between two actions, playing A exactly when B would have been better and vice versa, can match the best fixed action on average while the swap “ $A \rightarrow B$, $B \rightarrow A$ ” would have been perfect.

Theorem 16.7 (Foster and Vohra 1997; Hart and Mas-Colell 2000)

If every player uses an algorithm with swap regret $o(T)$, the empirical distribution of joint play $\bar{\mu}_T = \frac{1}{T} \sum_t \mu_t$ converges to the set of correlated equilibria. With swap regret R_T , $\bar{\mu}_T$ is an R_T/T -correlated equilibrium.

Proof. Player i ’s losses on round t are induced by the others’ play. Its swap-regret bound reads $\frac{1}{T} \sum_t (C_i(\mu_t) - C_i^\varphi(\mu_t)) \leq R_T/T$ for every φ . Both $C_i(\mu)$ and $C_i^\varphi(\mu)$ are linear in μ , so this is $C_i(\bar{\mu}_T) \leq C_i^\varphi(\bar{\mu}_T) + R_T/T$. $\square$

Getting low swap regret

Via approachability. Use the N^2 -dimensional payoff $r(w, \ell)_{(i,j)} = w_i(\ell_i - \ell_j)$ with target the negative orthant. Coordinate (i, j) of the average is the regret for the swap $i \mapsto j$. The response condition holds with $w = e_{\arg \min \ell}$, so Blackwell’s theorem gives $o(T)$ internal

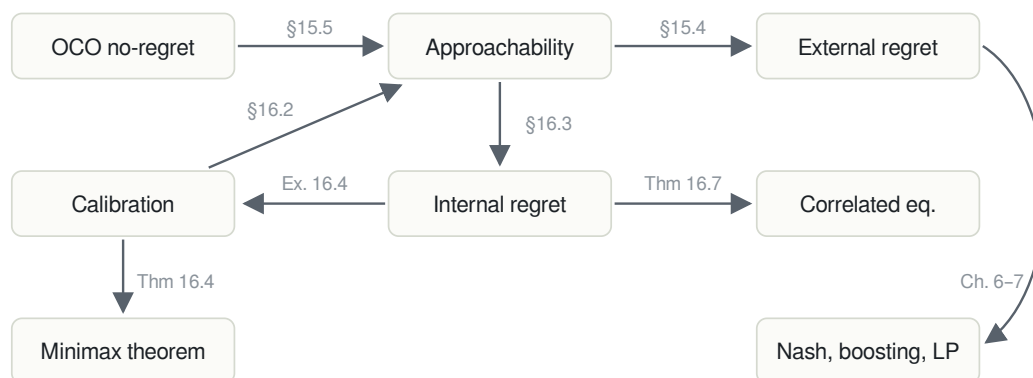
regret.

Via *external regret* (Blum and Mansour 2007). Run N copies $A_1, \dots, A_N$ of Hedge. Each outputs a distribution $q_t^{(i)}$; stack them as the rows of a stochastic matrix Q_t and play its stationary distribution $p_t = p_t Q_t$. Feed copy A_i the loss vector $p_{t,i} \ell_t$. Then

$$\text{Regret}_T^{\text{swap}} \leq \sum_{i=1}^N \text{Regret}_T(A_i) = O(\sqrt{NT \ln N}).$$

The stationarity condition $p_t = p_t Q_t$ is what makes “play p_t ” and “let copy i choose where action i ’s mass goes” the same thing.

16.4 The map of the course



How the main results of the course imply one another. An arrow from A to B means an algorithm for A yields an algorithm or constructive proof for B.

In finance

Calibration is the everyday test of probability forecasts in finance: are 2%-PD obligors defaulting at 2%? Do option-implied probabilities of a 10% move match realised frequencies? Rating agencies and model validators publish exactly the reliability tables of Definition 16.1. Theorem 16.2 has a sobering reading: calibration on its own can be achieved by a forecaster who knows nothing about credit, just by adjusting forecasts to past frequencies with randomisation. So calibration is necessary but far from sufficient, and good forecasts must also be *sharp* (discriminating), as measured by a proper scoring rule such as the Brier or log score (Chapter 3). Prediction markets aggregate beliefs through the same scoring rules: a market maker using a logarithmic market scoring rule (Hanson) is running a cost function equal to the softmin potential of Chapter 3, which connects market making to online learning (Abernethy, Chen and Wortman Vaughan 2013).

Key takeaways

- Calibration is achievable against any adversary, but only by randomised forecasters.
- Calibration $\Leftrightarrow$ approachability, and best-responding to calibrated forecasts proves the minimax theorem.
- Swap (internal) regret is the right notion for multi-player games: when all players have low swap regret, play converges to the set of correlated equilibria.
- Blum–Mansour turns any N external-regret algorithms into one with low swap regret.

Exercises

16.1 warm-up A forecaster announces 0.3 on 100 days (rain on 36) and 0.8 on 50 days (rain on 36). Compute C_T .

16.2 core In Chicken, each driver chooses Go or Stop. Losses: both Go 10; Go vs Stop: 0 for Go and 2 for Stop; both Stop 3. Find the mixed Nash equilibrium, and show the “traffic light” distribution $\mu = \frac{1}{2}(\text{Go}, \text{Stop}) + \frac{1}{2}(\text{Stop}, \text{Go})$ is a correlated equilibrium with lower expected loss.

16.3 core Prove the Blum–Mansour bound. Hint: copy i ’s regret against action j is $\sum_t p_{t,i}(\langle q_t^{(i)}, \ell_t \rangle - \ell_{t,j})$, and stationarity gives $\sum_i p_{t,i} \langle q_t^{(i)}, \ell_t \rangle = \langle p_t Q_t, \ell_t \rangle = \langle p_t, \ell_t \rangle$.

16.4 challenge (Lecture 24.) Show that no-internal-regret implies calibration: treat grid points as experts with loss $(q_i - y_t)^2$, and show that a miscalibrated forecast bin can be profitably swapped to the grid point nearest its empirical frequency.

Solutions

16.1. $T = 150$. Terms: $\frac{100}{150}|0.3 - 0.36| = 0.04$ and $\frac{50}{150}|0.8 - 0.72| = 0.0267$. $C_T \approx 0.067$.

16.2. Mixed Nash: if the other goes with probability g , Go costs $10g$ and Stop costs $2g + 3(1 - g) = 3 - g$. Indifference gives $g = 3/11$ and expected loss $30/11 \approx 2.73$ each. Under μ , a player told Stop knows the other goes: Stop costs 2, Go costs 10, so obey. A player told Go knows the other stops: Go costs 0, Stop costs 3, so obey. Expected loss is $\frac{1}{2}(0) + \frac{1}{2}(2) = 1$ each.

16.3. For any φ , sum copy i ’s regret against $\varphi(i)$ over i : $\sum_i \sum_t p_{t,i}(\langle q_t^{(i)}, \ell_t \rangle - \ell_{t,\varphi(i)}) = \sum_t \langle p_t, \ell_t \rangle - \sum_t \sum_i p_{t,i} \ell_{t,\varphi(i)}$, which is the swap regret for φ . Each copy has regret $O(\sqrt{T \ln N})$ on losses in $[0, 1]$ (scaled by $p_{t,i} \leq 1$), and a sharper analysis using $\sum_i \sqrt{\sum_t p_{t,i}} \leq \sqrt{NT}$ gives $O(\sqrt{NT \ln N})$.

Part VII

Toolkit, Reading and Projects

*The probabilistic and convex-analytic facts used throughout,
and where to go next.*

Mathematical Toolkit

A.1 Hoeffding's lemma and inequality

Lemma A.1 (Hoeffding's lemma)

If $X \in [a, b]$ almost surely, then for all $s \in \mathbb{R}$, $\ln \mathbb{E} e^{s(X - \mathbb{E}X)} \leq \frac{s^2(b-a)^2}{8}$.

Proof. Shift so that $\mathbb{E}X = 0$ (then $a \leq 0 \leq b$). By convexity of e^{sx} on $[a, b]$, $e^{sX} \leq \frac{b-X}{b-a} e^{sa} + \frac{X-a}{b-a} e^{sb}$. Taking expectations, $\mathbb{E} e^{sX} \leq \frac{b}{b-a} e^{sa} - \frac{a}{b-a} e^{sb} = e^{\psi(u)}$, where $u = s(b-a)$, $\theta = \frac{-a}{b-a} \in [0, 1]$ and $\psi(u) = -\theta u + \ln(1 - \theta + \theta e^u)$. Now $\psi(0) = \psi'(0) = 0$ and $\psi''(u) = \rho(1 - \rho) \leq \frac{1}{4}$ with $\rho = \frac{\theta e^u}{1 - \theta + \theta e^u}$. By Taylor's theorem, $\psi(u) \leq \frac{u^2}{8}$. $\square$

Theorem A.2 (Hoeffding's inequality)

If $Z_1, \dots, Z_m$ are independent with $Z_j \in [0, 1]$ and $\bar{Z} = \frac{1}{m} \sum_j Z_j$, then $\Pr[\bar{Z} - \mathbb{E}\bar{Z} \geq \varepsilon] \leq e^{-2m\varepsilon^2}$, and the same bound holds for the lower tail.

Proof. By Markov's inequality and independence, for $s > 0$, $\Pr[\bar{Z} - \mathbb{E}\bar{Z} \geq \varepsilon] \leq e^{-sm\varepsilon} \prod_j \mathbb{E} e^{s(Z_j - \mathbb{E}Z_j)} \leq e^{-sm\varepsilon + ms^2/8}$. Choose $s = 4\varepsilon$. $\square$

The same proof gives the Azuma–Hoeffding inequality for martingale differences bounded in $[a_t, b_t]$, which is what one uses to turn the expected-regret bounds of this book into high-probability bounds.

A.2 Norms, duality and strong convexity

- **Dual norm.** $\|g\|_* = \sup_{\|x\| \leq 1} \langle g, x \rangle$. Examples: ℓ_2 is self-dual; the dual of ℓ_1 is ℓ_∞ ; the dual of ℓ_p is ℓ_q with $\frac{1}{p} + \frac{1}{q} = 1$. Hölder's inequality: $\langle g, x \rangle \leq \|g\|_* \|x\|$.
- **Strong convexity.** R is α -strongly convex w.r.t. $\|\cdot\|$ if $R(y) \geq R(x) + \langle \nabla R(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$.
- **Pinsker's inequality.** $\text{KL}(p \| q) \geq \frac{1}{2} \|p - q\|_1^2$ for distributions p, q . Equivalently, negative entropy is 1-strongly convex w.r.t. $\|\cdot\|_1$ on the simplex.
- **Fenchel conjugate.** $R^*(\theta) = \sup_x \{\langle \theta, x \rangle - R(x)\}$. The conjugate of negative entropy restricted to the simplex is $\ln \sum_i e^{\theta_i}$ (log-sum-exp), and ∇R^* is softmax. This is why FTRL with entropy is Hedge: $x_t = \nabla R^*(-\eta L_{t-1})$.
- **Exp-concavity.** f is α -exp-concave iff $\nabla^2 f \succeq \alpha \nabla f \nabla f^\top$.

A.3 Entropy and KL divergence

For distributions on a finite set, $H(p) = -\sum_i p_i \ln p_i \in [0, \ln N]$ and $\text{KL}(p \parallel q) = \sum_i p_i \ln \frac{p_i}{q_i} \geq 0$ (Gibbs' inequality, from Jensen). For Bernoulli distributions, $\text{KL}(\text{Ber}(a) \parallel \text{Ber}(b)) \leq \frac{(a-b)^2}{b(1-b)}$, which is the fact behind bandit lower bounds: distinguishing $\frac{1}{2}$ from $\frac{1}{2} + \varepsilon$ needs about $1/\varepsilon^2$ samples.

A.4 Regret bounds at a glance

Setting	Algorithm	Regret	Chapter
Realizable experts, mistakes	Halving	$\log_2 N$	1
Experts, mistakes	Weighted Majority	$2(1 + \varepsilon)M^* + \frac{2 \ln N}{\varepsilon}$	2
Experts, convex loss	EWA	$\sqrt{2L^* \ln N} + \ln N$	3
Hedge, $[0, 1]$ losses	Hedge	$\sqrt{(T/2) \ln N}$ (optimal)	3, 5
Log loss / mixable	EWA, $\eta = 1$	$\ln N$	3
k -switch tracking	Fixed Share	$O(\sqrt{Tk \ln(NT)})$	4
Separable classification	Perceptron	$1/\gamma^2$ mistakes	8
OCO, Lipschitz	OGD	$DG\sqrt{T}$	9
OCO, α -strongly convex	OGD, $\eta_t = \frac{1}{\alpha t}$	$\frac{G^2}{2\alpha}(1 + \ln T)$	9
OCO, general norm	FTRL / mirror descent	$2BG\sqrt{T}$	10
Portfolios vs best CRP	Universal portfolio	$(n-1) \ln T + O(1)$	11
Stochastic bandit	UCB1	$\sum_j \frac{8 \ln T}{\Delta_j} + O(\sum_j \Delta_j)$	13
Adversarial bandit	EXP3	$\sqrt{2NT \ln N}$	14
Bandit linear optimization	SCRiBLE	$O(d\sqrt{\theta T \ln T})$	14
Approachability	Blackwell	distance $B/\sqrt{T}$	15
Swap regret	Blum–Mansour	$O(\sqrt{NT \ln N})$	16

Further Reading and Projects

B.1 Books

- N. Cesa-Bianchi and G. Lugosi, *Prediction, Learning, and Games*, Cambridge University Press, 2006. The course's main text. Chapters 2–4 (experts), 7 (games), 9–10 (log loss, portfolios), 6 (bandits).
- G. Shafer and V. Vovk, *Probability and Finance: It's Only a Game!*, Wiley, 2001; expanded as *Game-Theoretic Foundations for Probability and Finance*, 2019. The source for Chapter 12.
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- F. Orabona, *A Modern Introduction to Online Learning*, 2019– (arXiv:1912.13213). Clean treatment of FTRL and mirror descent, parameter-free methods.
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B.2 Key papers by chapter

- **Ch. 2–3.** Littlestone and Warmuth, "The weighted majority algorithm", *Inf. Comput.* 1994. Freund and Schapire, "A decision-theoretic generalization of on-line learning and an application to boosting", *JCSS* 1997.
- **Ch. 4.** Herbster and Warmuth, "Tracking the best expert", *Machine Learning* 1998.
- **Ch. 6–7.** Freund and Schapire, "Adaptive game playing using multiplicative weights", *GEB* 1999. Arora, Hazan and Kale, "The multiplicative weights update method: a meta-algorithm and applications", *Theory of Computing* 2012.
- **Ch. 9–10.** Zinkevich, "Online convex programming and generalized infinitesimal gradient ascent", *ICML* 2003. Hazan, Agarwal and Kale, "Logarithmic regret algorithms for online convex optimization", *Machine Learning* 2007. Kalai and Vempala, "Efficient algorithms for online decision problems", *JCSS* 2005.
- **Ch. 11.** Cover, "Universal portfolios", *Mathematical Finance* 1991. Helmbold, Schapire, Singer and Warmuth, "On-line portfolio selection using multiplicative updates", *Math. Finance* 1998. Kalai and Vempala, "Efficient algorithms for universal portfolios", *JMLR*

2002. Blum and Kalai, “Universal portfolios with and without transaction costs”, *Machine Learning* 1999.
- **Ch. 12.** DeMarzo, Kremer and Mansour, “Online trading algorithms and robust option pricing”, STOC 2006. Abernethy, Frongillo and Wibisono, “Minimax option pricing meets Black–Scholes in the limit”, STOC 2012. Abernethy, Bartlett, Frongillo and Wibisono, “How to hedge an option against an adversary: Black–Scholes pricing is minimax optimal”, NeurIPS 2013.
 - **Ch. 13–14.** Auer, Cesa-Bianchi and Fischer, “Finite-time analysis of the multiarmed bandit problem”, *Machine Learning* 2002. Auer, Cesa-Bianchi, Freund and Schapire, “The nonstochastic multiarmed bandit problem”, *SIAM J. Comput.* 2002. Abernethy, Hazan and Rakhlin, “Competing in the dark: an efficient algorithm for bandit linear optimization”, COLT 2008.
 - **Ch. 15–16.** Blackwell, “An analog of the minimax theorem for vector payoffs”, *Pacific J. Math.* 1956. Abernethy, Bartlett and Hazan, “Blackwell approachability and no-regret learning are equivalent”, COLT 2011. Foster and Vohra, “Asymptotic calibration”, *Biometrika* 1998. Hart and Mas-Colell, “A simple adaptive procedure leading to correlated equilibrium”, *Econometrica* 2000. Blum and Mansour, “From external to internal regret”, *JMLR* 2007.

B.3 Project ideas with a finance angle

The course weighted its final project at 50%. These ideas extend the book toward finance and would make good self-study projects.

1. **Universal portfolios on real data.** Implement Cover’s UP (by grid or Kalai–Vempala sampling), EG and ONS. Backtest on a few ETFs with daily rebalancing and proportional transaction costs. Measure regret to the best CRP and to buy-and-hold, and study how costs change the picture.
2. **Fixed Share for regime switching.** Run Hedge and Fixed Share over a library of factor strategies (momentum, value, carry, low volatility). Plot weights over time and compare tracking regret against the best k -switch sequence computed by dynamic programming.
3. **Model-free option bounds.** Reproduce the DeMarzo–Kremer–Mansour bounds for at-the-money calls as a function of realised quadratic variation, and compare with Black–Scholes prices at the corresponding volatility.
4. **Bandit order routing.** Simulate venues with fill rates and price impact that drift over time. Compare UCB, Thompson sampling, EXP3 and a sliding-window UCB, and discuss which assumptions each needs.
5. **Calibration of default forecasts.** Take a public probability-of-default or recession-probability series. Compute reliability diagrams and the ℓ_1 calibration score over time, and implement a randomised recalibration layer using the approachability forecaster of Chapter 16.
6. **Equilibrium computation.** Solve Homework 2’s 4×5 game with the four dynamics of Exercise 6.4, then a large random game, and compare convergence of the averages

and of the last iterates, including optimistic Hedge.

B.4 About the source course

EECS 598-006, *Prediction, Learning, and Games*, University of Michigan, Fall 2013, taught by Jacob Abernethy (with a guest lecture by Ambuj Tewari and a co-taught lecture with Rafael Frongillo). Prerequisites were algorithm analysis, probability, and comfort with calculus and convex analysis; the course described itself as about 90% theory. Grading: three homeworks (35%), a final project (50%) and participation, including scribing lectures (15%). The homeworks' problems appear throughout this book as exercises: Homework 1 in Chapters 2–3, Homework 2 in Chapters 6, 8 and 10, Homework 3 in Chapters 4, 6 and 14. The [course website](#) links the scribe notes (`notes/lecN_MMDDYY.pdf`) and homeworks (`homeworks/hw1.pdf` to `hw3.pdf`).

This book was written from the course website, the 24 scribed lecture notes and the three homeworks. Where the scribe notes had gaps or typos, standard statements from the literature were used and the proofs were checked; any remaining errors are ours, not the course's.

